NOTES OF STELLAR PHYSICS

version 1.0

Giacomo Pannocchia

a.y. 2025-2026

For those who come after

PREFACE

In a sense, you might say I'm getting quite fond of writing these kind of notes.

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor Scilla Degl'Innocenti during the academic year 2025-2026. However, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer[1].

You can find all the notes I've redacted here.

Without further ado, we'd better not lose much more time on a preface and get started with it.

> *There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur* [...] *But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly.*
> *Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.*

Ainulindalë, "The music of the Ainur",
Silmarillion, J. R. R. Tolkien

1 I'd like, however, not to see my house stormed by homing pidgeons.

CONTENTS

1 ASTROPHYSICS' BUILDING BLOCKS

1.1 INTRODUCTION

As per the name of this chapter, we're going to introduce a series of important quantities and concepts that are going to be particularly useful later on, for this first introductory part I'm going to follow [5], §1.

For expressing astrophysical measurements, it should be clear that the units that we customarily use for our everyday life wouldn't be doing a really good job. For example, most of the times, it isn't particularly convient to have masses expressed in grams or kilograms, for the masses we'll be usually concerned with are so large that saying that a star weighs 10^{33} g isn't particularly insightful, since we don't have an everyday-life object to compare it with.

What often happens in astrophysics is that masses are usually expressed in terms of *solar masses* $M_\odot$, which has value

$$M_\odot = 1.99 \cdot 10^{33}\,\mathrm{g} \tag{1}$$

For example Vega, a bright white star in the Lyra constellation, has a mass roughly estimated to $M_{\mathrm{Vega}} = 2.15^{+0.10}_{-0.15}\,M_\odot$. The masses of most stars lie within a relatively narrow range from $0.1 - 20\,M_\odot$.

Stars are formed in clusters, which are predominantely composed of neutral Hydrogen. Thanks to a whole series of mechanisms interplaying in said clusters, it's often possible to assume that stars born within a given cluster are approximately coeval and have a similar chemical composition. What we can't assume, however, is them having the same masses.

To describe how the masses of stars are distributed within a cluster, it is customary to invoke a IMF, an *initial mass function*, which is essentially an empirical function that describes the mass distribution of stars for a given cloud.

Similarly, it isn't particularly convenient to express distances with meters, centimeters, kilometers or, God forbids, miles and feet. For relatively "close" objects, it is often used the *astronomical unit* (AU)

$$1\,\mathrm{AU} = 1.50 \cdot 10^{13}\,\mathrm{cm} \tag{2}$$

which is the average distance of the Earth from the Sun. Although this is a rather useful quantity, we're not still up there. We have to kick it up a notch once more. As the Earth moves around the Sun, the "nearby" stars seem to change their position just slightly with respect to the faraway stars, which appear to be fixed in place.

This phenomenon is called *parallax*; it is not particularly different from what you experience when observing your own finger first with one eye and then with the other.

Consider the construction shown in Fig.1.

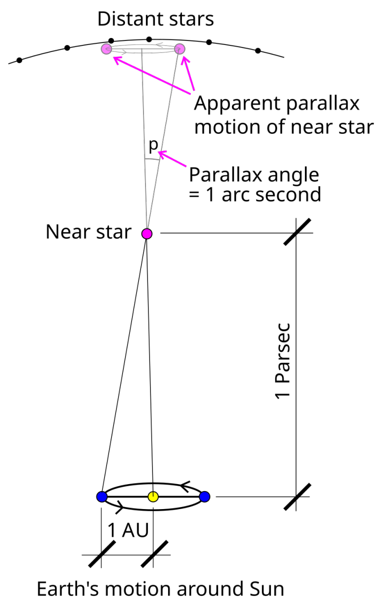

Figure 1: Defintion of parsec. Credits: Wikipedia.

Let us consider a star on the polar axis of the Earth's orbit at a distance d from the center of the orbit. The angle p in the picture is half of the angle by which this star appears to shift with the annual motion of the Earth on the distant stars' plane.

With simple geometrical considerations, if we assume Earth's orbit to be circular[1], then we can write the following relation

$$\tan(p) = \frac{1\,\mathrm{AU}}{d} \approx p$$

if we assume d to be much larger than an astronomic unit (which for all relevant scenarios is a condition that is well satisfied).

We can then define the *parsec* (pc) as the distance the star has to be so that its geometrical parallax turns out to be 1 arcsec. Hence

$$d = \frac{3.09 \cdot 10^{18}\,\mathrm{cm}}{p\,[\mathrm{arcsec}]} \implies 1\,\mathrm{pc} = 3.09 \cdot 10^{18}\,\mathrm{cm} \tag{3}$$

Obviously, for even larger distances the standard units are the *kiloparsec* (roughly the measure of galactic sizes), the *megaparsec* (the typical measure of galactic distances) and the *gigaparsec* (the typical measure of the observable Universe).

1.2 RELEVANT QUANTITIES FOR RADIATIVE TRANSFER

Although some books often start their description of radiatively relevant quantities from the definition of *monochromatic energy* and *monochromatic intensity*, I found that it is most misleading, since, in all but a few cases, what we experimentally measure are fundamentally *fluxes*.

1 Actually, it is not. Earth's orbital eccentricity is roughly $e = 0.017$, which is sufficiently small to let us approximate it as circular.

We shall then consider the *monochromatic flux* F_ν ($\mathrm{erg\,s^{-1}\,Hz^{-1}\,cm^{-2}}$) produced by some source passing through a small area $\mathrm{d}A$ located somewhere in space.

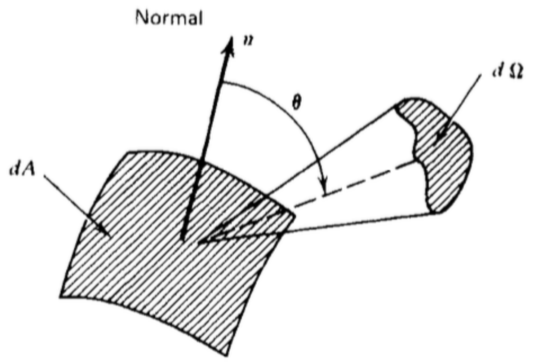

Figure 2: Schematic geometrical representation of the system.
Credits: G. Rybicki, A. Lightman [14].

If we call $\hat{k}$ the propagation direction of the flux and $\hat{n}$ the unit vector emerging from the surface $\mathrm{d}A$, it's easy to get convinced that what is actually passing through the surface is something proportional to $F_\nu(\hat{k}\cdot\hat{n})$.

From the monochromatic flux we can define the *bolometric flux*, which is just the monochromatic flux integrated over all frequencies (or wavelengths)

$$F = \int_0^{+\infty} F_\nu \,\mathrm{d}\nu = \int_0^{+\infty} F_\lambda \,\mathrm{d}\lambda \tag{4}$$

This also tells us how to convert a flux per unit frequency to a flux per unit wavelength

$$F_\nu \,\mathrm{d}\nu = F_\lambda \,\mathrm{d}\lambda$$

It should be clear that, despite being experimentally sensible for us to use the flux, we're losing much information sticking with it, namely directional information.

We consider then the amount of radiation $E_\nu \,\mathrm{d}\nu$ passing through the same area in time $\mathrm{d}t$ and solid angle $\mathrm{d}\Omega$. Hence we can write

$$\mathrm{d}E_\nu \,\mathrm{d}\nu = I_\nu(\mathbf{r}, t, \hat{k})(\hat{k}\cdot\hat{n})\,\mathrm{d}t\,\mathrm{d}\Omega\,\mathrm{d}A\,\mathrm{d}\nu \tag{5}$$

where the quantity $I_\nu(\mathbf{r}, t, \hat{k})$ is called the *specific monochromatic intensity*. If $I_\nu(\mathbf{r}, t, \hat{k})$ is specified for all directions at every point in a certain region of spacetime, then we'd have a complete prescription of the radiation field we intend on studying.

Capitalizing on the blatant similarities with distribution functions, we can evaluate the moments of the monochromatic intensity.

Definition 1.2.1. *Monochromatic mean intensity J_ν*

$$J_\nu = \frac{1}{4\pi}\int_\Omega I_\nu \, d\Omega = \frac{c}{4\pi} U_\nu$$

with U_ν the total energy density of radiation. Note that J_ν is essentially just an average of the monochromatic intensity over all solid angles.

Definition 1.2.2. *Monochromatic flux $\vec{F}_\nu$*

$$\vec{H}_\nu = \frac{1}{4\pi}\int_\Omega I_\nu(\hat{k})\hat{k}\, d\Omega \implies \frac{1}{4\pi}F_\nu = \vec{H}_\nu \cdot \hat{n}$$

I haven't explicitly proved the last equality, but it shouldn't be hard for you to convince yourself (or prove it yourself) that it is indeed true.

Definition 1.2.3. *Monochromatic radiation pressure p_ν*
The monochromatic pressure is defined starting from the different directions correlations of the monochromatic intensity

$$K_\nu^{ij} = \frac{1}{4\pi}\int_\Omega I_\nu(\hat{k})k^i k^j\, d\Omega$$

The pressure in particular is usually expressed as

$$P_\nu = \frac{1}{c}\int_\Omega I_\nu(\hat{k})\cos^2\theta\, d\Omega$$

where $\cos^2\theta = (\hat{k}\cdot\hat{n})^2$.

1.2.1 Blackbody radiation

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu)\,\mathrm{d}\nu = \frac{4\pi h g \nu^3}{c^3}\,\mathrm{d}\nu$$

from which deducing the expression from internal energy is straightforward. Remembering $g = 2$ is the quantum degeneracy of photons, a simple moltiplication of the previous expressions yields

$$U_\nu\,\mathrm{d}\nu = \frac{8\pi h}{c^3}\frac{\nu^3}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT) - 1}} \tag{6}$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but we'll soon see how to generalize this formulation for less "restrictive" environments.

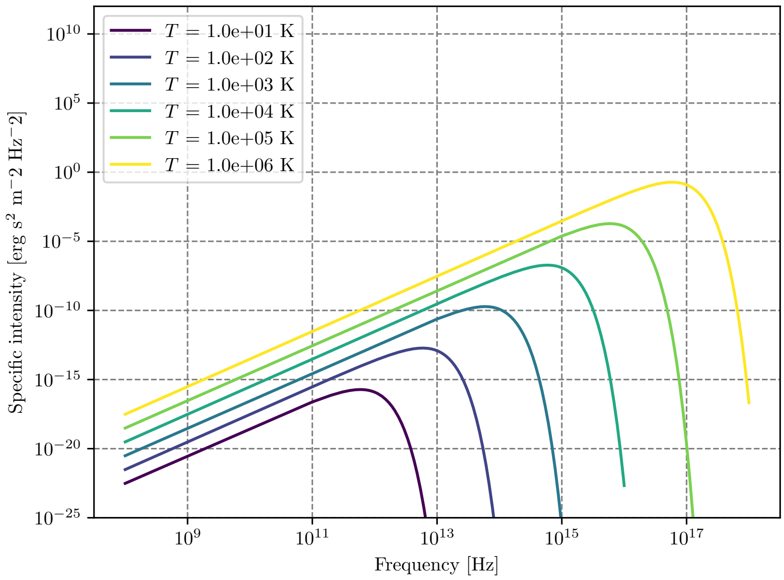

Figure 3: Blackbody frequency spectrum.

An incredible number of important results descends from (6), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu = \frac{2h}{c^2}\frac{\pi^4}{15}\left(\frac{kT}{h}\right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example[2]

$$F(T) = \pi \int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2h^3}T^4 \tag{7}$$

is what is usually known as the *Stefan-Boltzmann law*.

2 The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.6: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2}kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^{W} = \frac{2h\nu^3}{c^2}\exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

1.2.2 Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars.

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k}I_\nu \quad h\nu/kT \ll 1 \tag{8}$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν.

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν, from an unresolved source we cannot find I_ν, unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c, is obtained. Often the "fitting" procedure is nothing more than estimating the peak of the spectrum and applying Wien's displacement law to find a temperature.

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos\theta \, \mathrm{d}\nu \, \mathrm{d}\Omega := \sigma T_{eff}^4 \tag{9}$$

1.3 MAGNITUDE SCALES

When coming down to determining the properties of a star, there are two quantities that shine a little brighter than the others: The luminosity and the flux.

The luminosity of a star is generally defined as the power emitted by a given star over time (potentially over a given wavelength) and its expression is given by the relation

$$L = -\frac{\mathrm{d}E}{\mathrm{d}t} \quad \mathrm{erg} \cdot \mathrm{s}^{-1} \tag{10}$$

As we'll see, stars often emit isotropically in space. It goes without saying, however, that an observer at distance d from that star won't be able to collect all the light that is emitted. What we can measure is a flux, the fraction of energy that impinges on our detector. The flux is related to the luminosity through the following relation

$$\phi = \frac{L}{4\pi r^2} \quad \mathrm{erg} \cdot \mathrm{s}^{-1} \cdot \mathrm{cm}^{-2} \tag{11}$$

The flux, however, doesn't tel us much about the characteristics of the star, differently from the luminosity. But since what we can measure are fluxes, it is then crucial that we find a way to accurately measure distances so to infer the luminosity of an object.

Since ancient times, mankind has always tried to measure the brightness of stars. The efforts of those that came before us are nowadays crystallized in a quantities that is still in use: The *magnitude*.

The concept of magnitude is strictly related to the way the human eye was believed to perceive the differences in intensities.

Hypparchus, back in ancient Greece, classified all stars into six classes according to their apparent brightness. This classes went from class I, the brightest stars, to class VI, the less bright. According to Hypparchus, the brightest and faintest stars had an apparent brightness that is in the ratio of 100.

Let us consider a star with flux F_* and some reference flux F_0. We define the *apparent magnitude* as

$$m_* = x \log_{10}\left(\frac{F_*}{F_0}\right)$$

Calculating the difference in apparent magnitude between the brigthest and faintest stars and making use of the aforementioned characteristics, we can find the correct values for the constant x and define the apparent magnitude as

$$m_* = -2.5 \log_{10} \left(\frac{F_*}{F_0} \right) \tag{12}$$

Obviously, unless a bolometer is used, you can't perform a measurement covering all wavelengths. What you're going to measure is actually a convolution of the flux emitted by the star (as a function of the wavelength) and the transmission curve of the detector you're using

$$S_{det} = \int_0^\infty \mathrm{d}\lambda \, F(\lambda) T(\lambda)$$

More often than not, it is customary to measure magnitudes in given wavelengths' ranges, from which a "bolometric" magnitude can be inferred by adding a *bolometric correction* that is given from stellar atmosphere models.

Similarly, we can define the *absolute magnitude* as the magnitude a star would have if it was located 10 pc away.

Using the fact that $F \propto r^{-2}$, we find a relation for the absolute magnitude M

$$m - M = 5 \log_{10} \left(\frac{d}{10\,[\mathrm{pc}]} \right) \tag{13}$$

The difference $m - M$ is often called *distance module* (DM) and is sometimes used as an indirect way to express a distance.

For classifying stars according to their color it's often used the so-called UBV photometric system (from *Ultraviolet*, *Blue*, *Visual*), also called the Johnson system (or Johnson-Morgan system), which makes use of filters so that the mean wavelengths of response functions (at which magnitudes are measured to mean precision) are 364 nm for U, 442 nm for B, 540 nm for V.

Given this setup, we can easily calculate the magnitudes, say, in the Blue and Visible light-bands, so that the relative magnitudes are

$$m_B - m_V = -2.5 \log_{10} \left(\frac{F_B}{F_V} \right) := B - V$$

where $B - V$ is said *color index*.

When passing through interstellar dust[3] the intensity of radiation gets modified. The dust that makes the ISM absorbs light in the V-UV bands to later re-emit it as InfraRed radiation or scatters the "short" wavelength, producing a net-effect that essentially "increases" the wavelength of the incoming radiation and decreases its intensity.

These two phenomena are known as *reddening* (do not confuse it with *redshift*) and *extinction*.

The absolute magnitude in the, say, blue and visible bands, will thus be modified

$$M_V = M_V^0 + A_V$$
$$M_B = M_B^0 + A_B$$

3 The interstellar dust, which is usually referred to as ISM, consists in small particles of silicates, graphite, amorphous carbon, ices and organical mixtures of size $10^{-2} - 1\,\mu$m

where the quantities with the "o" superscript are the un-reddened magnitudes. We can define the new color index

$$B - V = (B - V)^0 + E[B - V]$$

where $E[B - V] = A_B - A_V$ is the extinction. Within our galaxy, the ratio of reddening to the extinction is a well-known quantity

$$R = \frac{A}{E[B - V]} \approx 3.0$$

1.4 ELEMENTS ABUNDANCES

Observing the spectrum emitted by stars we may expect, a bit naively, to observe a blackbody spectrum, but that isn't quite the case. As we'll have the chance to discuss later on, the radiation field produced by stars is only approximately equal to a blackbody function in the inner regions of stars.

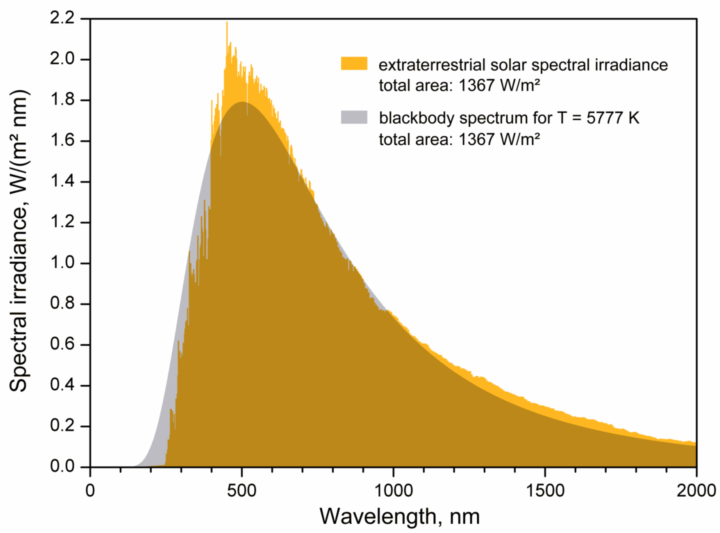

Figure 4: Deviation from blackbody behavior for the solar spectrum. Credits: Unknown source.

Although a tad annoying for the characterization of the radiation field, this deviation from ideal behavior allows us to infer the chemical composition of stars!

Deviation from blackbody behavior in fact is usual to take the form of *emission* and *absorption* lines in the spectrum, which tend to stick out like sore thumbs. Clearly, different elements (and different species of the same element) will produce very different lines, allowing us to reconstruct the chemical composition of the photosphere, which is the region where the photons we observe are coming from.

For our purposes, we'll be interested in the *original* chemical composition of a star rather than its actual, or future, value, so that that is shed of the modifications induced by ongoing nuclear reactions.

At this point, you may be objecting: Isn't the effect of nuclear reactions relevant only in the inner regions of stars?

As often happens in Physics, the correct answer is "It depends." Can we take for granted that the chemical composition we see in the photosphere is the same as the original one?

Generally, no. As we'll see, stars in a certain range of masses and in some evolutionary stage of their life may develope convective or radiative envelopes, and the answer to our question we can find once we determine which of the two is going to win.

1.4.1 Universal Curve of abundances

A matter we're going to be most interested in discussing is the provenience of the various (chemical) elements we observe on Earth and all around the Universe and *how* they are produced. For example, we now know that the almost entirety of Hydrogen was produced during the *Primordial Nucleosynthesis* along with a bit of Helium.

Metals, on the other hand, are entirely produced by *Stellar Nucleosynthesis*. That's because in the early stages of the formation of the Universe, temperature dropped too fast below the critical threshold that allows the relevant nuclear reactions to produce efficiently Helium and metals. On top of that, light metals (Lithium, Beriullium and Boron) are generally destroyed for the sake of producing lighter elements like Helium (see Chapter 4). Fig.5 presents a comparison of chemical composition of the solar photosphere with that of CI chondrites.

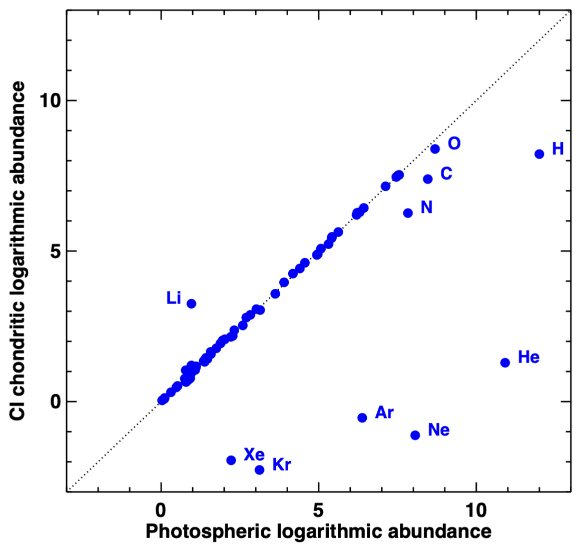

Figure 5: Comparison bewteen the recommended abundances by [1] in present-day solar photosphere and CI chondrites. Most elements agree well, with the exception of the highly volatile elements, which are depleted in meteorites, and Li, which has been largely destroyed in the Sun due to nuclear burning. Credits: [1].

Since elements are (on avergae) produced essentially via the same producedures, we may expect to measure a constant ratio between the various abundances of elements. Consider for example Fig.6.

First of all: As anticipated, light metals have a relatively low abundance, which is consistent with them being burned for the sake of producing lighter elements (Hydrogen and Helium) which have above-average abundance.

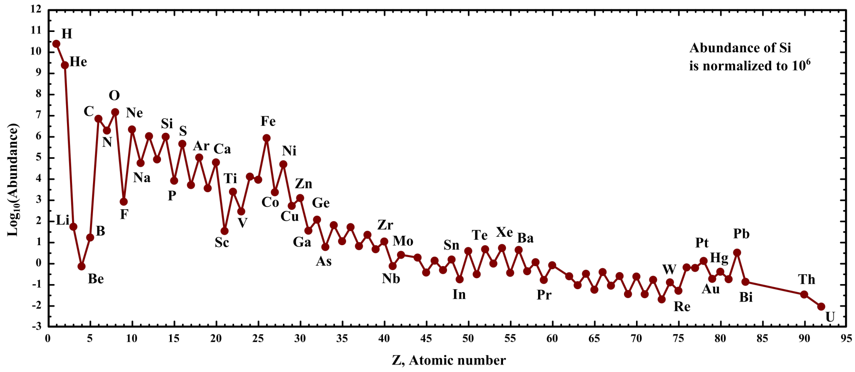

Figure 6: Universal curve of abundances, normalized on the relative abundance of Silicon. It represents the average value in the galactic disk of chemical abundances in stars. Credits: Wikipedia, but see also [1].

Similarly, the heavier elements which are produced by α-particles capture have roughly the same relative abundance, dropping off only when crossing the peak of Iron. The formation of elements with $Z > 26$ can happen only with proton/neutron capture on the nucleus, which is a fairly rare process that can occur only in the late evolutionary stages of massive stars.

As we have anticipated somewhere above, to model the evolution of a star, it's important to know the original chemical composition of the cluster the star was born from. You can think of it as "setting" the starting conditions for your system. In particular, it is of crucial importance to determine the primordial abundance of Helium.

For this purpose, blue galaxies hosting very bright, massive stars have been observed, where it's possible to study the recombination lines of Helium in metal-poor H_{II} regions. A very recent study (Aver et al., 2026 [2]) determined up to a precision of 0.5% the primordial helium abundance

$$Y_p = 0.2458 \pm 0.0013$$

If, for whatever reason, we want to determine the abundance of non-primordial Helium, we have to consider stellar models. That is because, usually, stars presenting He-lines aren't particularly trustworthy due to sedimentation of Helium, which essentially causes the superficial abundance not to reflect the original one.

To create a stellar model for this purpose, we consider solar-mass stars, whose original metal abundance has been corrected for diffusion[4], and an unknown abundance of Helium. Then you let the system evolve until the present-day age of the Sun and you cross-check if you've obtained the same

4 Don't worry, we're going to see this later on much more in detail.

luminosity $L_\odot$ and radius $R_\odot$. In particular the luminosity is going to be closely related to the Helium abundance.

Metallicity and Enrichment ratio

Usually, the total abundance of Helium is expressed in terms of the metallicity of a given star

$$Y_X = Y_p + \frac{\Delta Y}{\Delta Z} Z_* \tag{14}$$

We can also define the [Fe | H] as

$$[\mathrm{Fe} \,|\, \mathrm{H}] = \log_{10}\left[\frac{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_*}{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_\odot}\right] = \log_{10}\left[\frac{(Z/X)_*}{(Z/X)_\odot}\right] \tag{15}$$

where we've defined $N_i = X_i/A_i m_{\mathrm{H}}$. From this definition is obvious then that if $[\mathrm{Fe} \,|\, \mathrm{H}] = 0$, the abundance of Iron is equal to that of the Sun. Doing a slight manipulation[5] we can also write

$$[\mathrm{Fe} \,|\, \mathrm{H}] \approx \log_{10}\left[\frac{(\frac{Z}{1-Y})_*}{(\frac{Z}{X})_\odot}\right] = \log_{10}(Z_*) - \log_{10}(1 - Y_*) - \log_{10}\left(\frac{Z}{X}\right)_\odot \tag{16}$$

which can be inverted to obtain an expression of the form $\log_{10}(Z_*) \sim [\mathrm{Fe} \,|\, \mathrm{H}]^{-1.85}_{-1.77}$, where we're assuming $Y \sim Y_\odot \approx 0.24$. The subscript comes from the estimate of [12] (Magg et al, 2022), while the superscript comes from the estimate of [1] (Asplund et al, 2021) of the solar metallicity.

Similarly, we can define the *alpha-enhancement* as

$$[\alpha \,|\, \mathrm{Fe}] = \log_{10}\left[\frac{(N_\alpha/N_{\mathrm{Fe}})_*}{(N_\alpha/N_{\mathrm{Fe}})_\odot}\right] \tag{17}$$

which, on average, evaluates to $[\alpha \,|\, \mathrm{Fe}] \sim 0.3$ for stars in galactic halos.

1.4.2 Initial Mass Function (IMF)

As we mentioned somewhere above, stars are born in clusters. During the arduous process that is stacking bit after bit of Hydrogen (along with Helium and whatever the star my found in the cluster) to build its own body, it's clear that we cannot reasonably expect all stars to be born with the same masses, just like you wouldn't expect your brother/sister to weigh just as much as you did when you were born.

What we experimentally observe is a *distribution of mass* in the cluster, influenced by every little fluctuation occuring in a cluster.

It is worth to notice, however, that the distribution we usually refer to as IMF (*Initial Mass Function*) is not the *Present Day Mass Function*, which gives the statistic of the masses of stars as seen at present day. That the two distributions must be different is evident if you consider more massive stars: These, that would appear in the IMF, *would not* appear in the PDMF, because they are most likely dead. Using stellar evolution models, however, it is possible to pass from a given IMF to the PDMF.

5 We're making use of $X + Y + Z = 1$.

Evaluating the mass distribution of stars right away is a burdensome task, that, for all but a few rare cases[6], proves to be quite impossible. For that reason, it is customary to study the distribution in luminosity, or, equivalently, the distribution in given magnitude bands.

In fact, with a basic application of the chain rule, you can write

$$\frac{\mathrm{d}N}{\mathrm{d}M_\nu} = \frac{\mathrm{d}N}{\mathrm{d}m} \cdot \frac{\mathrm{d}m}{\mathrm{d}M_\nu} \tag{18}$$

Unable to directly measure our quantity of interest $\mathrm{d}N/\mathrm{d}m$, we can circumvent the problem by calculating the number of stars in a given luminosity range and calculating the relation between mass and luminosity, which is going to be one of the many jobs of stellar models.

Once you're finally done, you'll probably find a relation of the form

$$\frac{\mathrm{d}N}{\mathrm{d}m} \propto m^{-\alpha} \quad (\alpha > 0) \tag{19}$$

so less massive stars will be favored with respect to heavier stars.

For example, at present day, the IMF that is customarily into use is that found by [11]

$$M \in (0.08, 0.5)\, M_\odot \to \alpha = 1.3 \pm 0.3 \tag{20}$$

$$M > 0.5\, M_\odot \to \alpha \approx 2.3 \tag{21}$$

Note that Kroupa's IMF is in good accord with Salpeter's IMF [16], which was one of the first proposals for a IMF.

For the moment, although we know that it shouldn't actually be the case, we're observing a *constant* IMF in galaxies, clusters and so on. Just recently, we're discovering that, luckily for us, our prediction of non-constancy is verified when you take into account regions with a high star formation rate and not too low metallicity.

1.4.3 Stellar Trivia

In this subsection I'm going to anticipate some of the subjects that are going to be the main interest of many among the following chapters.

Starting from dimensional considerations only (which, however, requires that you know the equations describing stellar structures), you can easily show that more massive stars are hotter and have a higher luminosity, hence them having a shorter lifespan. You may argue that more massive stars have more "fuel" to burn, thus "cancelling out" the effect of a higher luminosity.

Although that is technically true, it doesn't take into account the fact that (effective) temperature is positively correlated with the mass of a star. That means that more massive stars are generally hotter, which in turn pumps up the efficiency of nuclear reactions[7]. This means that more fuel gets burned to balance the collapse caused by gravitational pulls.

6 For example, one of the few systems we're able to "accurately" determine the mass of is that of spectroscopic binaries.

7 Nuclear reactions in stars can be shown to present a power-law dependency on the temperature, scaling at least as T^4.

This phenomenon can be generally summarized under the name of *stellar thermostate*: When a star expands, its density decreases, and so its temperature; the efficiency of nuclear reactions thus decreases until its evenly balanced by gravitational collapse. Conversely, if the gravitational pull is stronger, the star contracts and gets hotter, the efficiency of the nuclear reactions ramps up along with the luminosity, halting the collapse.

This inspires our first, important deduction: If in a cluster stars are predominantely blue, the cluster must be young! Since more massive stars have a shorter lifespan, if the cluster was older, then we wouldn't be able to observe such stars.

It's important to distinguish to kinds of clusters: *Globular clusters* are generally old clusters found in galactic halos; they have little gas, no (recent) star formation and are generally arranged spherically symmetrically around the GC. In globular clusters would be logical to expect a lower metallicity, for a lower number of generations of stars might have followed one another in that region.

On the other hand *open clusters* are now found in the galactic plane[8]; they have much more gas available and there's continuous star formation. Generally, open clusters house a smaller number of stars, therefore allowing us to see only stars still in the *main sequence* stage[9]. In open clusters, it's sensible to expect a higher metallicity (with respect to globular clusters).

As a further classification, astrophysicists are used to distinguish two (three) populations of stars, depending on their characteristics and the cluster they're found in:

- *PopI Stars*: These are typically young, blue stars, which are very bright and with a high metallicity content. They're found in open clusters.
- *PopII Stars*: These are typically old, reddish stars, which aren't particularly bright and have a low metallicity content. They're found in globular clusters.
- *PopIII Stars*: Allegedly, this population of star should be the first one to form after the Big Bang. As so, its metallicity content should be virtually zero. This would imply that they should have large masses[10] and be very bright. As you might imagine, PopIII star wouldn't live for long.

Thick Disk

Along with the halo, the buldge and the *thin* disk, there's another structure in galaxies that is worth mentioning: The *thick disk*.

As the name suggests, the thick disk has a scale height that is greater than that of its thinner counterpart ($H_{\text{thick}} \sim 1\,\text{kpc}$ opposed to $H_{\text{thin}} \sim 0.1\,\text{kpc}$) but

8 Originally, they are likely to have formed in the halo, but their elliptic motion intersecting the GP have experienced numerous interactions with other stars and objects that have brought their trajectory to ultimately lie on the GP.

9 The main sequence stage is an evolutionary stage of stars during which Hydrogen is burned in the core.

10 In fact, they should be massive enough to leave behind, after their death, an intermediate mass black hole.

a shorter scale length ($l_{\text{thick}} \sim 3 - 4\,\text{kpc}$ opposed to $l_{\text{thin}} \sim 8 - 15\,\text{kpc}$). Like the halo, it is one of the more ancient structures, and presents characteristics that are hybrid between the halo and the thin disk, especially if you consider its rotational velocity and its metallicity content.

It is difficult to trace a sharp boundary between the thick disk and the halo/thin disk; usually considerations are made taking into account the kinematic of the disk.

Dwarf Galaxies

Dwarf galaxies are, unexpectedly, low-mass galaxies, in a range that can span from the typical mass of globular clusters ($10^5\,M_\odot$) to that of more "standard" galaxies ($10^9 - 10^{11}\,M_\odot$). An example of dwarf galaxies are the Large and Small Magellanic Clouds, that are both satellites of the MW.

One way to indentify dwarf galaxies is by observing the *streams* they leave in their wake as they move around their host galaxy, which are essentially groups of stars that get left behind when the smaller cloud gets gravitationally captured. The presence of streams is usually reflected into zones of (sensibly) different metallicity.

Moreover, it has been observed that the α-enhancement in dwarf galaxies is smaller than that of the MW's halo, making it quite improbable that they can make a relevant contribution.

On a concluding note, I think it's worth to mention at least the existence of Ultra-Faint Dwarf Galaxies (UFDs), which are a class of galaxies that contain from a few hundred to one hundred thousand stars, making them the faintest galaxies in the Universe.

UFDs resemble globular clusters in appearance but have very different properties. For once, UFDs contain a significant amount of dark matter and are more (spatially) extended.

UFDs are the most dark matter-dominated systems known to present day.

2 STELLAR STRUCTURES

2.1 STELLAR EQUILIBRIUM EQUATIONS

In this chapter we're going to introduce and describe the so-called *stellar equilibrium equations*, which account for the physical mechanisms that are interplaying within stellar structures, and compare it with what real-life observations tell us.

The goal of stellar structure equations is making possible to predict instant after instant in every point of the star the value of each physical and chemical properties, so that once an initial condition is set, we're going to be virtually able to accurately determine the evoluition of those properties.

On a first approximation, we can consider stars to be autogravitating mass conglomerates, for which the gravitational pull is evenly balanced by the pressure gradients (and possibly other forces) that set in within the star. Since gravity is a *central force*, we can well expect the equilibrium configuration to be that of a sphere. This greatly simplifies the problem at hand, since from a three-dimensional configuration is reduced to a one-dimensional configuration by means of symmetry.

The approximation of spherical symmetry allows us to formally decompose the star in a series of *shells* of infinitesimal width $\mathrm{d}r$, so that the physical properties of the star are approximately constant withing said shell. The quantities we'll be interested in considering are the pressure $p(r)$, the density $\rho(r)$, the temperature $T(r)$, the various relative abundances $X_i(r)$, the mass $M(r)$ (that is considered to be the mass contained in the sphere of radius r) and the luminosity $L(r)$ (the energy per unit time entering the shell at distance r from the center of the star).

Given this notation, the mass found inside radius $r + \mathrm{d}r$ should be $M(r) + \mathrm{d}M$, where $\mathrm{d}M$ can be identified with the mass of the infinitesimal spherical shell. Said $\rho(r)$ the density, then the mass of this shell is

$$\mathrm{d}M = 4\pi r^2 \rho \, \mathrm{d}r$$

which yields our first stellar structure equation

$$\boxed{\frac{\mathrm{d}M}{\mathrm{d}r} = 4\pi r^2 \rho(r)} \tag{22}$$

Note that (22) is essentially a restating of the continuity equation of mass.

Let's stop for a moment to consider how sensible are the two approximations we're implementing in our description, that are the request of spherical symmetry and that only pressure gradients are balancing the gravitational pull. The former is generally met by the vast majority of stars, while for the latter it might be necessary to use some more care, especially when dealing with (highly) rotating stars, forcing us to add centrifugal forces to the picture[1].

Generally, if the centrifugal acceleration at the surface is too high with respect to the surface gravitational acceleration, the necessity of modifying our equations to keep track of centrifugal forces may arise.

For this purpose, we define the *critical angular velocity* Ω_c so that

$$\Omega_c^2 \sim \frac{GM_*}{R_*^3} \implies v_c \sim \sqrt{\frac{GM_*}{R_*}} \tag{23}$$

Note that depending on the mass and the radius of the star we're considering, critical velocities might get as high as $v_c \sim 300 - 600\,\mathrm{km/s}$; for comparison, the equatorial velocity of the Sun is $v_\odot^{\mathrm{eq}} \sim 2\,\mathrm{km/s}$.

To result in non-negligible effects, it is sufficient that the angular velocity gets as high as $\Omega_{\mathrm{sup}} \gtrsim 0.5\Omega_c$; if, instead, $\Omega_{\mathrm{sup}} \gtrsim 0.7\Omega_c$ you start observing deformations in the structure of the star, but this are fairly rare. As you may easily guess, these are problems that concern with the more massive stars, getting more and more relevant as the mass of the star gets bigger.

About 20% of the stars with mass in the range $10M_\odot \lesssim M \lesssim 19M_\odot$ (so typically stars of spectral class B or O) have velocities of order

$$150 < v_{\sin i} < 300 - 400\,\mathrm{km/s}$$

Note that through Doppler effect, we're able to measure the velocity of objects along the line of sight, but in principle, the rotation axis of the object we're observing may be tilted of an angle i with respect to the observer, hence the notation/factor $\sin i$.

Increasing the mass, we find that about 30% of the stars with mass in the range $20M_\odot \lesssim M \lesssim 40M_\odot$ have roughly the rotational velocities of the order of the ones we've written above.

To "detect" the presence and relevance of rotation it is customary to perform a spectroscopic analysis of the star. In fact, it is clear that the presence of rotation issues the arising of convection motion within the star, which in turn implies a "mixing" of the star's contents.

In principle, if we observe main sequence stars, the chemical composition of the surface should be roughly equal to the starting composition[2], for temperature isn't high enough to start the nuclear processing on the surface. But, in presence of strong mixing mechanisms, we may observe an excess of Helium and Nitrogen that are brought to the surface from the inner regions of the star (where nuclear reactions are at least partially active).

---

1 Note that in the presence of rotation, the equilibrium configuration is no longer spherically symmetric, since rather than the gravitational potential, we have now to consider a baroclyinic potential (gravitational+centrifugal).

2 If we neglect the effects of *sedimentation*, that is. The characteristic times are however too short for sedimentation to occur, so it should be probably safe neglecting it.

The Effect of Magnetic Fields (anticipation)

In principle, even particularly strong magnetic fields may influence the surface composition and structure of the star. This, however, requires

$$\beta = \frac{p_{\text{gas}}}{p_{\text{mag}}} \sim 1 \text{ with } \beta = \frac{8\pi p}{B^2}$$

that in turn would mean having magnetic fields as high as $B \approx 3000 - 5000\,$Gauss. In general they are not relevant, but there exist very rare cases such that of *magnetic stars* of class A_p or B_p, where the subscript "p" is to indicate particular lines in the spectrum of such stars. For comparison

$$\langle B \rangle_\odot \sim 0.5 - 4\,\text{Gauss vs } \langle B \rangle_{MS} \sim 1 - 30\,\text{kGauss}$$

Magnetic fields that strong are assured to influence the external layers of such stars, inducing mixings that lead to a modification of the surface abundances.

Another class of magnetic stars is that of *magnetic white dwarves* that, although very rare, can reach magnetic fields of order $10^6 - 10^9\,$Gauss. Similarly, in *magnetars* (magnetic neutron stars), magnetic fields can reach intensities of $10^{14} - 10^{15}\,$Gauss.

Note that in principle we're able to detect the presence of strong magnetic fields by studying the surface oscillation of the star, performing what's called *asteroseismology*.

2.1.1 Free Fall Time

Our initial claim was that for the majority of their life, stars are in *hydrostatic equilibrium*, in a configuration where pressure gradients are evenly matching the collapse forces. But can we be so sure that a star is, as a matter of fact, in equilibrium?

We know, for example, that the Sun has been stable on timescales of at least a couple hundred years. On top of that, looking at the Milky way we can find billions of stars with characteristics similar to those of the Sun but in different stages of their life, so that we might have an idea of the whole life cycle of our dearest yellow dwarf.

From those observations, we learn that the Sun must have been stable for times of the order of millions of years. Of course, to make sure it is indeed in hydrostatic equilibrium, we should compare with the *dynamical timescales* the observation timescales over which we can safely claim that its characteristics (e.g., the luminosity and the radius) haven't changed.

Let's consider a spherically symmetric star with constant density. In presence of gravity alone, we can write

$$\ddot{r} = -\frac{GM}{r^2}$$

which can be immediately integrated to yield

$$\begin{aligned}\frac{1}{2}\dot{r}^2 &= \frac{GM}{r} + \text{const.}\\ &= \frac{GM}{r} - \frac{GM}{R}\end{aligned}$$

where we have set the initial conditions for $t = 0$: $r = R$ and $\dot{r} = 0$.

The free-fall timescale can be thus obtained by committing a mathematical atrocity, but I (and, for extension, you too) don't care[3]

$$\begin{aligned}\tau_{ff} &= \int_0^{t_{ff}} \mathrm{d}t = \int_R^0 \frac{1}{\dot{r}}\,\mathrm{d}r \\ &= \int_0^R \frac{\mathrm{d}r}{\sqrt{2GM\left(\frac{1}{r} - \frac{1}{R}\right)}} = \frac{\pi}{2\sqrt{2}}\sqrt{\frac{R^3}{GM}}\end{aligned}$$

thus, if we use $R^3/GM = 3/4\pi G\rho$

$$\tau_{ff} = \sqrt{\frac{3\pi}{32G\rho}} \approx 1.63\,\mathrm{Myr} \cdot \left(\frac{10^3\,\mathrm{cm}^{-3}}{\mu n}\right)^{1/2} \tag{24}$$

For the Sun, the free-fall timescale amounts to roughly $\tau_{ff} \sim 30\,\mathrm{min} - 1\,\mathrm{h}$ depending on the calculation that has been used. So we can safely claim that the Sun is definitely not collapsing. What if, instead, it is exploding?[4]

To see that we can calculate the explosion time, sound time, or dynamical time as

$$\tau_{\mathrm{sound}} = \frac{R}{c_s} \approx 0.5\,\mathrm{Myr}\left(\frac{R}{0.1\,\mathrm{pc}}\right)\left(\frac{0.2\,\mathrm{km\,s^{-1}}}{c_s}\right) \tag{25}$$

If $\tau_{ff} > \tau_{\mathrm{sound}}$ the system returns to a stable equilibrium as the pressure forces can overcome gravity. Conversely, for $\tau_{ff} < \tau_{\mathrm{sound}}$ gravitational collapse takes place.

2.1.2 Hydrostatic Equilibrium

In the last section we showed that is fair for us to assume that stars are in *hydrostatic equilibrium* for most of their lives. We now turn to write an equation (the second of four) that will allow to characterize our star.

Consider a star in which the only relevant forces acting on a spherical shell between r and $r + \mathrm{d}r$ and transverse area $\mathrm{d}A$ are the gravitational pull and the pressure forces (both due to the gas and photons). It's not difficult to show that the balance condition for the mass of the small element is

$$-\mathrm{d}p\mathrm{d}A - \frac{GM_r}{r^2}\rho\mathrm{d}r\mathrm{d}A = 0$$

from which

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM_r}{r^2}} \tag{26}$$

A quick look at (22) and (26) will convince you that we have more free variables (functions of the radial coordinate) than equations. This is one of the many cases in which the *closure problem* pops out. To solve it we're gonna need to introduce an equation of state (EoS)

$$\boxed{p = p(\rho, T, X, Y, X_i)} \tag{27}$$

3 It is actually less atrocious than I'm making it sound like.
4 Read: expanding outwards.

where the various quantities have their obvious meaning. A particularly simple configuration (that we'll have the pleasure to look somewhere in the following) is the one where we're allowed to take a *politropic equation of state*

$$p = K\rho^{\gamma} \tag{28}$$

with K some constant and γ some power that depends on the particular gas and its degenerate state.

Let us now take a look at another particularly simple EoS, that is the equation of state of perfect (non degenerate) gas

$$p = nkT = \frac{\rho(r)}{\mu(r) m_H} kT(r) \tag{29}$$

with $\mu(r)$ the *mean molecular weight*, which clearly depends on the elements present in the star and their ionization state, which can (hopefully) inferred from considering Saha's equation (118). In general it's not easy to determine the exact form of $\mu(r)$ so it is customary to consider the easy scenario of a *completely ionized* star[5]

If we consier X the relative abundance of Hydrogen, Y the relative abundance of Helium and X_i the relative abundance of the metal with i so that $A > 4$[6]. If we consider the number of ions and electrons that complete ionization generates:

- Hydrogen: #ions X/m_H, #electrons X/m_H;
- Helium: #ions $Y/4m_H$, #electrons $2Y/4m_H$;
- Metal i-th: #ions $X_i/A_i m_H$, #electrons $Z_i X_i/A_i m_H$

so that the total number of particles (per unit volume) is just

$$n = \left[2X + \frac{3Y}{4} + \sum_{i=4}^{N} \left(\frac{X_i}{A_i} + \frac{Z_i X_i}{A_i} \right) \right] \frac{\rho}{m_H} \tag{30}$$

If we consider the following approximation for (30)

$$\frac{X_i}{A_i} \ll 1 \quad \frac{Z_i}{A_i} \sim \frac{1}{2}$$

we find

$$\mu^{-1} \approx 2X + \frac{3Y}{4} + \frac{1}{2} \sum_{i=4}^{N} X_i = 2X + \frac{3Y}{4} + \frac{Z}{2} \tag{31}$$

where we've introduced the *metallicity* Z.

We can also consider the molecular weight restricted to ions only

$$\mu_i^{-1} = 2X + \frac{3Y}{4} + \sum_{i=4}^{N} \frac{X_i}{A_i}$$

5 People tend to say that a star is completely ionized when its temperature is at least $T > 10^{6\,\mathrm{K}}$, although that's a completely arbitrary lower limit. For instance, heavier elements like Iron, which have electrons to spare, are completely ionized at much higher temperatures.

6 Sorry $^4_3$He, you're too less abundant to be considered. But your time will come soon.

and consider an average charge for the mixture of elements $\bar{Z}$, so that $n_e = \bar{Z}n_i \approx \bar{Z}\rho/\mu_i m_H$. From this we can define the mean molecular weight of electrons as

$$\frac{\mu_i}{\bar{Z}} \to n_e = \frac{\rho}{\mu_e m_H}$$

If we plug this all back in the EoS for the perfect gas

$$p_{\text{gas}} = nkT = (n_e + n_i)kT = \frac{\rho kT}{m_h}\left(\frac{1}{\mu_i} + \frac{1}{\mu_e}\right)$$

we conclude

$$\frac{1}{\mu} = \frac{1}{\mu_i} + \frac{1}{\mu_e} \tag{32}$$

2.1.3 Temperature Gradients

As you probably know, in the inner regions of stars holds the so called *Local Thermodynamic Equilibrium* (for friends, LTE): In these regions there is, as the name suggests, punctual and *local* thermodynamic equilibrium between matter, which will be then well described by a Maxwell-Boltzmann distribution, and radiation, which can be described by (6).

This fact should also suggest you that, since we're able to observe temperature gradients within stars, it is not possible for stars to be in *stric thermodynamic equilibrium* (STE) and thus emit like perfect blackbodies with fixed temperature.

Let us try to give an order of magnitude estimate of the temperature gradient within, say, the Sun

$$\langle \frac{\mathrm{d}T}{\mathrm{d}r} \rangle = \frac{T_c - T_S}{R_\odot} \approx 10^{-4}\,\mathrm{K\,cm^{-1}}$$

which means that over the average mean free path of a photon in the inner regions of stars ($\lambda \sim 0.1 - 1\,\mathrm{cm}$), the difference in temperature will be

$$\Delta T = \langle \frac{\mathrm{d}T}{\mathrm{d}r} \rangle \cdot \lambda \approx 10^{-5} - 10^{-4}\,\mathrm{K}$$

which is very small. But to actually check if the assumption of LTE is valid (if at all sensible) we shall consider the variation in internal energy of the photons. We know that in thermal equilibrium $U = aT^4$, so

$$\frac{\mathrm{d}U}{U} = 4\frac{\mathrm{d}T}{T} \sim 10^{-12} - 10^{-11}$$

which means the deviation from isotropic (blackbody) behavior is negligibly small over the mean free path of photons, therefore allowing us to assume the LTE configuration.

We know, either through direct observation or because you've read the upcoming section about radiative transfer, that wherever is a temperature gradient there is also a radiative flux (along with a convective motion and electronic conduction)!

On a first approximation we may write

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{rad}} \propto F(r)\bar{\kappa}$$

wnere F is the flux we've defined in (4) and $\bar{\kappa}$ is some average *opacity*[7] that sums up the various interaction channels between matter and radiation. Note that in general it is frequency dependent[8].

In the equation we've written above we're assuming that the transport of energy is carried out by radiation alone, which is essentially the temperature gradient that we'd observe if all the flux was transported by radiation.

Taking a closer look to the micro-physics involved in absorption phenomena we learn that is possible to link the opacity to the mean free path of electrons

$$\lambda_\sigma = (n\langle\sigma\rangle)^{-1} = (\rho\bar{\kappa})^{-1}$$

If, on the other hand, there were another channel available for transporting energy on top of radiation, we may be writing

$$F(r) = \frac{L(r)}{4\pi r^2} = F_{\text{rad}} + F_{\text{conv}}$$

where we've assumed the presence of, say, relevant convective motions within the star. It should go without saying that the temperature gradient will no longer be the gradient induced by radiative transport, but will rather be an *ambient gradient*, which we might express as

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{amb}} \propto \bar{\kappa}F_{\text{rad}}$$

Clearly, since now radiation is transporting only a part of the total flux

$$\left|\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{amb}} < \left|\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{rad}}$$

If we define with ϕ the flux of photons per unit time, and approximate the number of interactions in an infinitesimal distance $\mathrm{d}r$ as $\mathrm{d}r/\lambda$, with λ the mean free path, we can calculate the transferred momentum

$$\delta p = \phi\frac{\mathrm{d}r}{\lambda}\frac{h\nu}{c} = \frac{F(r)}{\lambda c}\mathrm{d}r$$

But the variation in (linear) momentum is equal to the variation of pressure $\delta p = \delta P_{\text{rad}}$

$$\frac{\mathrm{d}P_{\text{rad}}}{\mathrm{d}r} = \frac{F(r)}{\lambda c}$$

If we recall that $P_{\text{rad}} = U/3 = aT^4/3$ and correct the sign of the expression (to address the mismatch of signs between the flux and temperature gradient), we finally yield

$$F(r) = -\lambda ca\frac{4}{3}T^3\frac{\mathrm{d}T}{\mathrm{d}r}$$

Recalling that $\lambda = (\rho\bar{\kappa})^{-1}$ and that $\sigma_B = ac/4$ we can write our third stellar structure equation

$$\boxed{F(r) = -\frac{16}{3}\frac{\sigma_B T^3}{\rho\bar{\kappa}}\frac{\mathrm{d}T}{\mathrm{d}r}} \tag{33}$$

7 Opacity is measured in $\mathrm{cm}^2\,\mathrm{g}^{-1}$.
8 For a more rigorous definition see either the next chapter or [14].

It is often useful to flip the equation

$$\frac{dT(r)}{dr}\bigg|_{\rm rad} = -\frac{3\bar{\kappa}\rho}{4ac}\frac{1}{T^3}\frac{L(r)}{4\pi r^2}$$

so that in general, if $F_{\rm rad} \neq F_{\rm tot}$, we can make a slight manipulation and write

$$\frac{dT(r)}{dr}\bigg|_{\rm amb} = -\frac{3\bar{\kappa}\rho}{4ac}\frac{1}{T^3}F_{\rm rad}$$

If you want to see a slight more rigorous derivation of (33), you may check the last part of the "Radiative Transfer" section.

2.1.4 Energy Production

Let us consider what happens if the stellar atmosphere was not constantly replenished with energy from the inner layers. How long would it take for the star to deplete completely the energy stored in the atmosphere.

The average density of the outer layers of the atmosphere is $\rho^{\rm ext_\odot} \sim 10^{-7}\,{\rm g\,cm^{-3}}$. We can thus give an estimate of the mean free path $\lambda_j \sim (\rho\bar{\kappa})^{-1} \sim 100 - 300\,{\rm km}$. We've made the (usually decent) approximation of opacity being dominated by photoelectric effect on H^- ions, $\bar{\kappa} \approx 0.3\,{\rm cm^2\,g^{-1}}$. Note that in these conditions, the width of the atmosphere is of the same order of the mean free path of photons.

We know the solar flux $F_\odot \sim 6 \cdot 10^{10}\,{\rm erg\,cm^{-2}\,s^{-1}}$, as well as its temperature and pressure $T_{eff} \sim 6000\,{\rm K}$, $p \approx 10^5\,{\rm dyne\,cm^{-2}}$. The average number of particles is $n \sim 10^{17}\,{\rm cm^{-3}}$, which can be integrated over the vertical direction to find an average number of photons per unit surface.

The mean energy of photons can be expressed as $E_p = 3kT/2 \sim 10^{-12}\,{\rm erg}$. The energy per unit surface is thus $E_s \approx 10^{12}\,{\rm erg\,cm^{-2}}$. Finally, we find that the energy stored in the atmosphere would be depleted in roughly

$$\tau = \frac{E_s}{F} \approx 15\,{\rm s}$$

In such a short time, we would be able to discern immediately the changes in color in the atmosphere. Similarly, through the random-walk approximation, we may find an estimate of the time it takes photons to escape the inner regions of stars.

Random Walks

Consider a photon emitted in an infinite, homogeneous scattering region. It travels a displacement $\vec{r}_1$ before being scattered, then travels in a new direction over a displacement $\vec{r}_2$ before being scattered, and so on. After N free paths, the total displacement will look like

$$\vec{R} = \sum_{i=1}^{N} \vec{r}_i$$

The average total displacement will be identically null. Therefore, we must evaluate the mean square of the total displacement, which, in the assumption of independent and isotropic scattering, must be expressed as

$$l_*^2 = \langle \vec{R}^2 \rangle = \sum_{i=1}^{N} \langle \vec{r}_i^{\,2} \rangle = \sum_{i=1}^{N} l_i^2$$

The quantity l_* is the root-mean-square net displacement of the photon, while l^2 denotes the mean square of the free path of a photon, which within a factor of order unity, it's simply the mean free path of a photon.

Since the mean square displacements of each scattering iteration have no reason to be different, we'll just write

$$l_* = \sqrt{N}l$$

Said L the linear dimension of the object photons are trying to "escape" from, the number of scatterings events required to actually escape from regions of large optical depth is roughly determined by setting $l_* \approx L$. Then $N \sim L^2/l^2$.

This way, we can find the number of interactions needed to travel through a solar radius $R_\odot$ if the average displacement is a mean free path λ

$$\Delta x = \sqrt{z}\lambda \implies z \approx \left(\frac{R_\odot}{\lambda}\right)^2 = 1.96 \cdot 10^{22}$$

If an interaction takes, on average, $\tau = 10^{-8}$ s, the time needed to perform all those interactions is $\Delta t \approx 10\,\mathrm{Myr}$. This usually goes by the name of *thermal timescale*.

If we consider the average density of a star and a mean opacity independent of temperature and density, we can write

$$\bar{\rho} = \frac{M}{\frac{4\pi}{3}R^3}$$

but for stars with mass in the range $M_\odot \lesssim M \lesssim 10M_\odot$, mass scales linearly with the radius $M \propto R$. So

$$\lambda = (\bar{\kappa}\rho)^{-1} \propto R^2 \to \sqrt{z} \propto R^{-1} \implies \Delta t_{th} \propto R^{-2}$$

This is telling us that increasing the mass of a star (and thus the radius) has the surprising consequence of a (very) reduced lifespan.

We now present the last equation of stellar structure, that relates the luminosity gradient to the rate of production of energy

$$\boxed{\frac{\mathrm{d}L(r)}{\mathrm{d}r} = 4\pi r^2 \rho(r)\epsilon(r)} \tag{34}$$

where $\epsilon(r)$ is the "energy production rate", that is usually decomposed in three contributions

$$\epsilon(r) = \epsilon_g(r) + \epsilon_N(r) - \epsilon_\nu(r)$$

The easier to address is the term $-\epsilon_\nu$, which expresses the energy lost to neutrinos that are produced where there's no nuclear reactions. These neutrinos, produced via various channels, are often called *thermal neutrinos*. The main channels of production are summed up below:

$$\begin{aligned}
e^- + (Z,A) \text{ or } e^- \to e^- + (Z,A) \text{ or } e^- + \nu_e + \bar{\nu}_e \quad &\text{Bremsstrahlung} \\
\gamma + e^- \to e^- + \nu_e + \bar{\nu}_e \quad &\text{Photoproduction} \\
\gamma \to e^- + e^- \to +\nu_e + \bar{\nu}_e \quad &\text{Pair annihilation} \\
\gamma_{\text{plasmon}} \to \nu_x + \bar{\nu}_x \quad &\text{Neutrinic oscillations} \\
(Z,A) + e^- \to (Z-1,A) + \nu_e \quad &\text{Neutralization}
\end{aligned}$$

Please note that the subject of neutrinic oscillations is rather technical and I know nothing of it to make this surely intriguing subject justice. The only thing I know is that the collective oscillations in a plasma, *plasmons*[9], may decay into one of the three leptonic pairs of neutrinos.

The next one to address is the contribution given by *gravitational energy* ϵ_g. Consider the 1st principle of thermodynamics

$$\mathrm{d}Q = \mathrm{d}U + p\mathrm{d}V \quad \mathrm{d}Q_m = \mathrm{d}U_m + p\mathrm{d}\left(\frac{1}{\rho}\right) \tag{35}$$

where the subscript m in the second equation means "per unit mass". We may then define $\epsilon_g = -\mathrm{d}Q_m/\mathrm{d}t$. Invoking entropy, this becomes

$$\epsilon_g = -\frac{\mathrm{d}Q_m}{\mathrm{d}t} = -T\frac{\mathrm{d}S_m}{\mathrm{d}t} = -T\left[\partial_p S)_T\frac{\mathrm{d}p}{\mathrm{d}t} + \partial_T S_m)_p\frac{\mathrm{d}T}{\mathrm{d}t}\right]$$

which, once we introduce the *specific heat capacity at constant pressure* $c_p = T\partial_T S_m)_p$ and the heat exchanged during an isothermal transformation $E_T = -T\partial_p S)_T$, we may finally write

$$\epsilon_g = -\frac{\mathrm{d}Q_m}{\mathrm{d}t} = E_T\dot{p} - c_p\dot{T} \tag{36}$$

Since time derivatives are involved, it should be clear that we are in the need of having two different models to describe the structure of a star in different times. To do so, it is necessary to solve *mesh by mesh* the stellar structure equations. It is then of crucial importance knowing how abundances change within a given mesh. Those who work with this stuff usually linearize the function describing the evolution of the abundance of the i-th element as such

$$X_{i,k}^{\mathrm{new}} = X_{i,k}^{\mathrm{old}} + \frac{\mathrm{d}X_i}{\mathrm{d}t}\Delta t \tag{37}$$

where the subscript k refers to a given mesh. The time interval Δt is often chosen so that in each mesh the physical variables and/or the abundances do not vary more than a predetermined value.

2.2 VIRIAL THEOREM

For stellar structures in hydrostatic equilibrium, for which only pressure and gravitational forces are present, the following *virial theorem* holds

$$2\langle K\rangle_{\mathrm{trasl}} + \langle\Omega\rangle = \frac{1}{2}\ddot{I} \approx 0 \tag{38}$$

This strikingly simple equation allows us to find some enlightning relations. Let us then consider a monoatomic perfect gas for which the equaipartition theorem grants us $\langle K\rangle = 3kT/2 = E_{th}$. The virial theorem thus implies

$$E_{th} = -\frac{\Omega}{2} > 0 \implies E_{tot} = \frac{\Omega}{2}$$

9 Not to be confused with the omonimous biscuits.

Note that under the assumption of spherical symmetry, the gravitational energy can be found evaluating the following integral

$$\Omega = -\int_0^{R_*} \frac{GM_r}{r} 4\pi\rho r^2 \,\mathrm{d}r < 0$$

From the former expression, we can calculate the luminosity

$$L = -\frac{\mathrm{d}E_{tot}}{\mathrm{d}t} = -\frac{1}{2}\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{\mathrm{d}E_{th}}{\mathrm{d}t} > 0$$

From this we're learning that from the gravitational contraction, half the energy is dissipated through radiation, the other half is used to "fuel" the thermal content of the star (i.e., the star gets hotter).

What if we relax the condition of the gas being monoatomic? The virial theorem (38) still holds, but in general we can't say $\langle K \rangle = E_{th}$. The traslational (average) kinetic energy is still $\langle K \rangle = 3kT/2$, but now $E_{th} = lkT/2$, where l is the number of degrees of freedom of the single gas element. It isn't particularly convenient to work with the degrees of freedom, so we're going to convert it to a more thermodynamically acceptable quantity, the adiabatix index γ. It is possible to show that

$$\gamma = \frac{C_p}{C_V} = 1 + \frac{2}{l}$$

This way we can cast the virial theorem in an alternative form

$$3(\gamma - 1)E_{th} + \Omega \approx 0$$

The total energy can be calculated with a direct substitution

$$E_{tot} = E_{th} + \Omega = \frac{3\gamma - 4}{3(\gamma - 1)}\Omega$$

from which

$$\Delta E_{tot} = \frac{3\gamma - 4}{3(\gamma - 1)}\Delta\Omega$$
$$\Delta E_{th} = -\frac{\Delta\Omega}{3(\gamma - 1)}$$

Since $\Delta\Omega$ is necessarily smaller than 0, we thus obtain a stability criterion for the star: A star can be stable if $\gamma > 4/3$. If such condition is not met, all the energy released by gravitational collapse is converted in thermal energy, thus resulting in instabilities.

We can also extend the virial theorem to different equations of state if we allow for the translational kinetic energy to be written as $\langle K \rangle = \xi E_{th}$. Clearly, we have to assume that the coefficient ξ is constant over the whole stellar structure. Repeating the same calculations as before (*mutatis mutandis*, that is), virial theorem yields

$$E_{tot} = \frac{\xi - 1}{\xi}\Omega$$

and the stability condition therefore becomes $\xi > 1$.

2.2.1 Thermal Timescale Revisited

A few pages ago we made use of the random walk approximation to find the time that it takes on average to a photon to scatter its way out of the inner region of a star. We called that time *thermal timescale*. It is the aim of this section to show that it is indeed possible to find an equivalent formulation for that timescale starting from the virial theorem.

Let us recall (38)

$$2\langle K\rangle + \langle \Omega\rangle \approx 0$$

For the sake of simplicity, we're going to assume a monoatomic gas, so that $\langle K\rangle = E_{th}$. We can assume the following expression for the gravitational potential

$$\Omega = -q\frac{GM^2}{R}$$

with q that depends on the specific density profile that is given. For a homogeneous star, for example, $q = 3/5$. We showed that the luminosity of the star will be such that

$$L = -\frac{1}{2}\dot{\Omega} = -\frac{q}{2}\frac{GM^2}{R^2}\dot{R}$$

Separating variables and integrating, we find

$$\tau_{KH} = \frac{q}{2}\frac{GM^2}{RL} \tag{39}$$

which is called *Kelvin-Helmoltz timescale*. In short, it represents the time during which the stellar structure is able to sustain itself radiating away the energy generated from the gravitational shrinking.

Note that we're implicitly assuming that the luminosity doesn't change while the star collapses, which is in general a very bad approximation.

Under the approximation of homogeneous density $q = 3/5$, and the Kelvin-Helmoltz timescale is

$$\tau_{KH} \approx 10\,\mathrm{Myr}\left(\frac{M}{M_\odot}\right)^2\left(\frac{R_\odot}{R}\right)\left(\frac{L_\odot}{L}\right)$$

which is much greater than the other dynamical times, but still not quite as large as it should be to match the estimated lifespan of the Sun (that it's also connected to the fairly long half-life of some radionuclides we observe here on Earth).

2.3 QUALITATIVE STELLAR STRUCTURE RELATIONS

Now that we have the virial theorem at our disposal, we migth think of looking for very approximated expressions that can give us an idea of how physical properties in a star are interconnected. All quantities will be interpreted as *averaged* quantities.

Let us start with density and temperature. If we know M and R, then

$$\Omega = -q\frac{GM^2}{R}$$

with q a constant given by the profile density. We can now use the virial theorem and invert the relation between mass, density and radius

$$E_{th} \sim \frac{3kT}{2}\frac{M}{\mu m_{\mathrm{H}}} \quad \rho = \frac{M}{\frac{4\pi}{3}R^3}$$

finding

$$\boxed{T \propto M^{2/3}\rho^{1/3}} \tag{40}$$

This means that two stars with different masses must have different densities to achieve the same temperature! This will then reflect onto smaller stars having the higher densities.

Upon a first approximation, the luminosity L does not depend on the energy production rate ϵ. If we butcher the equation of hydrostatic equilibrium a plug in the proportionalities of the density

$$p_C = \frac{M^2}{R^4}$$

where p_C is the central pressure. We can make the further assumption of a perfect gas and negligible radiation pressure, so that

$$\boxed{T_C \propto \frac{p_C\mu}{\rho} \propto \mu\frac{M}{R}} \tag{41}$$

If we make the assumption of exclusively radiative transport, we find that

$$\frac{\mathrm{d}T}{\mathrm{d}r} \propto -\mu M R^{-2} \to L \propto -\frac{T^3}{\bar{\kappa}\rho}\frac{\mathrm{d}T}{\mathrm{d}r}R^2$$

We then make the (very wrong, both conceptually and experimentally) approximation of constant opacity, thus finding

$$\boxed{L \propto \mu^4 M^3} \tag{42}$$

Finally, we can find a relation between luminosity and density

$$\frac{\mathrm{d}L}{\mathrm{d}r} = 4\pi r^2\rho\epsilon \to \frac{L}{R^3} \propto \rho\epsilon \to \rho \propto \frac{L}{R^3}\frac{1}{\epsilon}$$

but we showed that $R \propto M/T$, so

$$\rho \propto \frac{T^3}{\epsilon}$$

without some kind of prescription for the energy production rate, this is as far as we can get. Fortunately for us, for main sequence stars that still burn Hydrogen, it is given (as we'll see) the following expression

$$\epsilon \propto \rho^m T^n$$

with $m = 1$ both in the case of energy production through the ppI chain and the CN-NO chain. The two of them, however, have really different dependencies on the temperature: The ppI scales as $\epsilon \propto T^4$, while the CN-NO as $\epsilon \propto T^{15}$. Plugging in the aforementioned expression, we finally find

$$\boxed{\rho^{m+1} \propto T^{3-n}} \tag{43}$$

A simple way of solving stellar structure equations is by killing off one of the variables, namely temperature. This isn't always possible, but there exist some special cases in which pressure can be assumed to be dependent only on density. This is the case of the so-called *politropic structures*. Politropic structures are described by the following equation of state

$$p(\rho) = K\rho(r)^{1+\frac{1}{n}} = K\rho(r)^{\Gamma} \tag{44}$$

where n is the *index of the politropic*. Consider the equation for hydrostatic equilibrium (26) and twist it so that

$$\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho} = -GM_r$$

This we can derive with respect to the radial coordinate, finding

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho}\right) = -G\frac{\mathrm{d}M_r}{\mathrm{d}r} = -4\pi G\rho r^2$$

where we made use of (22). Dividing by r^2 we finally obtain

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho}\right) = -4\pi G\rho \tag{45}$$

which is still the same equation, but with a *crispier* form that will make a little easier the discussion we'll do in the next few paragraphs.

Lane-Emden equation

Assume for example a density profile of the form

$$\rho(r) = \rho_c\phi^n(r)$$

with ϕ some adimensional density profile with the following characteristics: $\phi(0) = 1$ and $\partial_r\phi|_{r=0} = 0$. Substituting in the politropic equation (44) we find that the pressure is simply

$$p(r) = K\rho_c^{1+\frac{1}{n}}\phi^{n+1} = p_C\phi^{n+1}$$

In terms of an adimensional radius $r = \lambda\xi$, with λ constant depending only on n

$$\lambda = \left[\frac{K\rho_c^{1/n-1}(n+1)}{4\pi G}\right]^{1/2}$$

eq.45 takes a much simpler and elegant form

$$\boxed{\frac{1}{\xi^2}\partial_\xi(\xi^2\partial_\xi\phi) = -\phi^n} \tag{46}$$

which is the general *Lane-Emden equation*. Solutions for this equation have been calculated numerically in Fig.7.

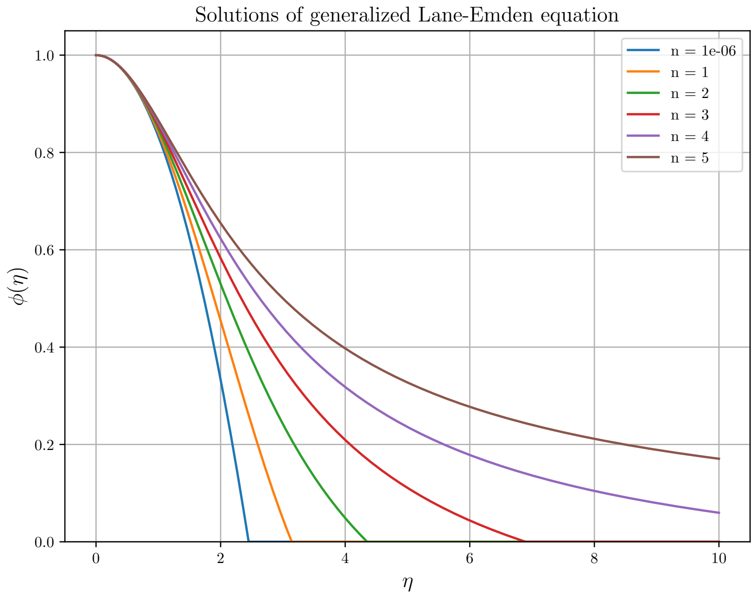

Figure 7: Numerical solutions to the Lane-Emden equation.

Note that solving (46) gives us two things: The profile density ϕ and the adimensional radius ξ_0 for which $\phi(\xi_0) = 0$, which is essentially the radius of the star.

Since we know quite a few things already, we may ask ourselves if it is possible to calculate the central density of the star as well. Assuming that we know from some other observations the total mass of the star, we can indeed calculate ρ_c. In fact, we can simply integrate the equation of continuity (22) up to the radius of the star

$$M = \int_0^R 4\pi r^2 \rho \, \mathrm{d}r = 4\pi\lambda^3 \rho_c \int_0^{\xi_0} \xi^2 \phi^n \, \mathrm{d}\xi \tag{47}$$

The value of that integral is actually fixed by the original equation itself to $-\xi_0^2 \partial_\xi \phi|_{\xi_0}$. So we can promptly invert the equation above and find the central density of the star. It goes without saying that once we've completely specified the density profile we'll have automatically determined the pressure profile as well by means of mere substitution.

Politropic Integration for the Solar Structure

In the following section we'll try to integrate analitically the stellar structure equations for the Sun assuming a politropic EoS (44). Before we begin, let us make the following approximation

$$\beta = \frac{p_{\text{gas}}}{p_{\text{tot}}} \sim \text{const.}$$

where $p_{\text{tot}} = p_{\text{gas}} + p_{\text{rad}}$. This assumption follows from the conditions of hydrostatic equilibrium and radiative transfer, by means of which it's possible to prove that

$$1 - \beta \sim \bar{\kappa}(r)\epsilon$$

This is the so called *Eddington Approximation*: Since $\epsilon \propto \rho T^4$ (at least as far as Hydrogen burning in main sequence stars is concerned) and $\bar{\kappa} \propto \rho^\alpha T^{-3.5}$ (Kramer's law), their product will be approximately constant[10].

Moreover

$$p_{\text{tot}} = p = p_{\text{gas}} + p_{\text{rad}} = \frac{\rho k T}{\mu m_{\text{H}}} + \frac{1}{3} a T^4$$

Let us use the expression for β to remove the temperature, so to write a politropic equation

$$\frac{p_{\text{rad}}}{p_{\text{gas}}} = \frac{1-\beta}{\beta} = \frac{1}{3}\frac{aT^3}{\rho}\mu m_{\text{H}}$$

so that

$$T = \left[\frac{1-\beta}{\beta}\frac{3\rho}{\mu m_{\text{H}} a}\right]^{1/3}$$

This way, if we substitute in the equation for the pressure, we find

$$p = \left[\frac{3}{a}\left(\frac{k}{\mu m_{\text{H}}}\right)^4 \frac{1-\beta}{\beta^4}\right]^{1/3} \rho^{4/3} \tag{48}$$

which is a politropic equation as requested. With reference to (44), this relation would have a politropic index $n = 3$. Note that in this case, since we don't know β, we don't know the multiplicative factor, nor we have the absolute certainty of it being constant. For the moment, we'll consider an averaged molecular weight over the whole stellar structure[11].

We still don't know the multiplicative constant, but we know both the mass and the radius of the Sun. We've shown in (47) that if we know both the mass and the radius of a star we're able to infer its central density

$$M = M(\beta, \rho_C) = 4\pi\lambda^3 \rho_c (-\xi^2 \partial_\xi \phi)_{\xi_0} = M_\odot$$

with

$$\lambda = \left[\frac{K\rho_c^{1/n-1}(n+1)}{4\pi G}\right]^{1/2}$$

Moreover, we know that $R_\odot = R(\beta, \rho_C)\lambda\xi_0$, thus we have enough equations to solve for. Behold

$$\rho_C = \frac{M_\odot}{4\pi R_\odot^3} \frac{\xi_0^3}{(-\xi^2 \partial_\xi \phi)_{\xi_0}}$$

So if we plug in the necessary solar parameters and look up to some table to find the values for $\xi_0 \approx 6.9$ and $-\xi^2 \partial_\xi \phi|_{\xi_0} \approx 2.02$ for a politropic index $n = 3$, we'd find

$$\rho_C \sim 75\,\text{g}\,\text{cm}^{-3}$$

which is about half of the best estimate we'd find using less approximate models, so everything considered we've done a good job.

---

10 Note that this is only approximately true for Sun-like stars which are burning Hydrogen in the Main Sequence stage.

11 Note that generally the solar structure is divided into two regions, one where Hydrogen burning has happened (*solar interior*: $X \sim 0.34$, $Y \sim 0.64$, $Z \sim 0.02$ and $\mu \approx 0.85$, assuming complete ionization) and one where it has not (*external envelope*: $X \sim 0.70$, $Y \sim 0.28$, $Z \sim 0.02$ and $\mu \approx 0.62$, assuming complete ionization).

At this point we can equate the expression for K coming from the definition of the parameter λ and that coming from the multiplicative factor of (48)

$$K = \frac{\lambda^2 4\pi G}{(1+n)\rho_c^{1/n-1}} \qquad K_\odot = \left[\frac{3}{a}\left(\frac{k}{\mu m_\mathrm{H}}\right)^4 \frac{1-\beta}{\beta^4}\right]^{1/3}$$

thus allowing us to solve for β (since we've calculated ρ_C and we can deduce the value of $\lambda = R_\odot/\xi_0$). The solution is not exactly trivial, and is shown in (49)

$$\frac{1-\beta}{\mu\beta^4} = \frac{a\pi}{48}\frac{G^3 M_\odot^2 m_{\mathrm{H}^4}}{k^4(-\xi^2\partial_\xi\phi)_{\xi_0}} = A \tag{49}$$

which you can use also for sun-like stars. Given what we've found earlier we can substitute for $\beta \approx 0.998$, meaning that most of the pressure is due to raw gas pressure (and not from radiation); the value of β coming from evolution models is $\beta \sim 0.999$ which is in great agreement with our present approximated result.

We can also calculate the central pressure $p_C \sim 1.2 \cdot 10^{17}\,\mathrm{dyne/cm^2}$ and the central temperature $T_C \sim 1.6 \cdot 10^7\,\mathrm{K}$.

Similarly to what we've done in (47), we can calculate the mass within a sphere of radius r. The expression is almost identical

$$M = \int_0^r 4\pi r^2 \rho\,\mathrm{d}r = 4\pi\lambda^3\rho_c \int_0^\xi \xi^2\phi^n\,\mathrm{d}\xi = 4\pi\lambda^3\rho_c(-\xi^2\partial_\xi\phi) \tag{50}$$

2.5 HOMOLOGOUS MODELS: SELF-SIMILARITY

Let's start with a simple consideration: Different politropic structures, *with the same politropic index n*, differ only for a scaling factor. This is not particularly difficult to show. Consider

$$\frac{\rho(\xi)}{\rho'(\xi)} = \frac{\rho_C}{\rho'_C}$$

and the same thing holds for the pressure, mass and so on

$$\frac{m(\xi)}{m'(\xi)} = \frac{M}{M'} \qquad \frac{r(\xi)}{r'(\xi)} = \frac{R}{R'}$$

All we have done was writing down the various politropic equations (and what descends from them) and *scale* them, finding out that you can jump from one another by simply rescaling an already existent solution. That is a notable simplification if you think about it.

What are, then, in general the properties a star must have to be self-similar? The answer you can find in the following Table.

Self-Similarity

Suppose to have a quantity dependent on two independent variables $Q(x, y)$. Characteristic scales $Q_0(x)$ and $\delta(x)$ are defined for Q, y respectively. Then the scaled variables are just

$$\xi = \frac{y}{\delta(x)} \quad \tilde{Q}(\xi, x) = \frac{Q(x, \xi)}{Q_0(x)}$$

If the scaled dependent variable is independent of x, i.e.

$$\exists \hat{Q}(\xi) : \tilde{Q}(\xi, x) = \hat{Q}(\xi)$$

then $Q(x, y)$ is self-similar. The interested reader can refer, for example, to [13].

Let's make a few other definitions, starting from $\eta = m(r)/M(R)$

$$\frac{r(\eta)}{r'(\eta)} = z \qquad \frac{P(\eta)}{P'(\eta)} = p$$
$$\frac{T(\eta)}{T'(\eta)} = t \qquad \frac{L(\eta)}{L'(\eta)} = s$$
$$\frac{\rho(\eta)}{\rho'(\eta)} = d$$

Given these definitions, we'll be interested in finding out how these scaling factors depend on the variations in mass and chemical composition. We'll then define

$$\frac{M(\eta)}{M'(\eta)} = x \quad \frac{\mu(\eta)}{\mu'(\eta)} = y$$

Tragically, we have to make another approximation: We have to assume that also the opacity and the energy production rate are self-similar

$$\frac{\epsilon(\eta)}{\epsilon'(\eta)} = \varepsilon \quad \frac{\bar{\kappa}(\eta)}{\bar{\kappa}'(\eta)} = k$$

This is a very restrictive limitation to the possible self-similar stellar structures: We've already seen that for the energy production rate the greater the mass, the greater the temperature and the greater gets the energy production rate, so we can't really compare much different masses. A similar thing holds for the opacity. This works kinda well for stars that have just entered the main sequence, when nuclear reactions haven't really started.

We can write all the structure equations in a self-similar fashion

$$\frac{\mathrm{d}p}{\mathrm{d}\eta} = \frac{\mathrm{d}p}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}\eta} = \frac{\mathrm{d}p}{\mathrm{d}r}\frac{M}{4\pi\rho r^2}$$

so that plugging this in the hydrostatic equilibrium condition yields

$$\frac{\mathrm{d}p}{\mathrm{d}\eta} = -\frac{G}{4\pi}\frac{M^2\eta}{r^4}$$

for both the structures we're considering (*mutatis mutandis*, that is). We can extend this calculation to all the other equations and start using all the self-similar relations we've written above. This way we can write

$$\frac{\mathrm{d}p'}{\mathrm{d}\eta} = -\frac{G}{4\pi}\frac{M'^2\eta}{r'^4}\left(\frac{x^2}{z^4 p}\right)$$
$$\frac{\mathrm{d}r'}{\mathrm{d}\eta} = \frac{M'}{4\pi\rho' r'^2}\left(\frac{x}{z^3 d}\right)$$
$$\frac{\mathrm{d}T'}{\mathrm{d}\eta} = -\frac{3}{64a\pi^2 c}\frac{\bar{\kappa}' M' L'(\eta)}{r'^4 T'^3}\left(\frac{kxs}{z^4 t^4}\right)$$
$$\frac{\mathrm{d}L'}{\mathrm{d}\eta} = M'\epsilon'\left(\frac{\varepsilon x}{s}\right)$$

but since the stellar structure equations must hold also for the second star (the one with the primed superscript), all the terms within parentheses must be equal to 1!

$$1 = \frac{x^2}{z^4 p} \qquad 1 = \frac{x}{z^3 d} \tag{51}$$
$$1 = \frac{kxs}{z^4 t^4} \qquad 1 = \frac{\varepsilon x}{s} \tag{52}$$

and we also have the equation of state, so we have enough equations to solve this[12]. It is possible to show that the system of equations can be solved with solutions given by

$$R: \ z = x^{z_1} y^{z_2} \qquad L: \ s = x^{s_1} y^{s_2}$$
$$P: \ p = x^{p_1} y^{p_2} \qquad T: \ t = x^{t_1} y^{t_2}$$

thus reducing the equations to equations for the powers of x and y, whose sum must necessary be equal to zero (by simple means of dimensional analysis). You can read an example of solution right below (or in any book that deals with self-similar solutions for stellar structures)

$$z_1 = \frac{1}{2}(1+A) \quad A = \left[\frac{4\delta(1+a+\lambda\alpha)}{\nu+b-4-\lambda\delta} + 4\alpha - 3\right]^{-1}$$

2.6 CONVECTIVE TRANSPORT

Matter inside the stars is never at rest, but usually the movements of the gas elements are small, random perturbations around their equilibrium positions. Under certain conditions, these small random perturbations can trigger large-scale motions that involve sizable fractions of the total stellar mass. These large-scale motions are called *convection*. Hot elements may rise to the top, thereby transporting energy from the hottest to the coolest regions, where they cool and then fall down as cold material.

12 Note that the EoS isn't enough to completely specify and solve the system. We have seven variables and only four equations, meaning we're in the need of three other relations. These can be found in the EoS $\rho \propto p^{\alpha} T^{-\delta} \mu^{-\phi}$, the energy production rate $\epsilon \propto \rho^{\lambda} T^{\nu}$ and the opacity $\bar{\kappa} \propto p^{a} T^{b}$.

In order to include convective transport in the stellar model computation we'll need to find a criterion for the onset of convection, and the expression of the temperature gradient in a convection zone. Note that the treatment of convections in the stellar interior is extremely complicated and needs the introduction of various approximations. This stems from the fact that the flow of gas in a stellar convective region is turbulent, in the sense that the velocity and all the other properties of the flow vary in a random and chaotic way.

The random nature of tyrbulence precludes computation based on a complete description of the motion of all the fluid particles based on a solution of the Navier-stokes equations for the stellar fluid.

In the following we're going to adapt an extremely simple model in which the gas flow is made of gas elements with a certain characteristic size, the same in all dimensions, that move by a certain mean free path before dissolving. All gas elements have the same physical properties at a given distance r from the center. We're going to assume that the "bubble" always remains

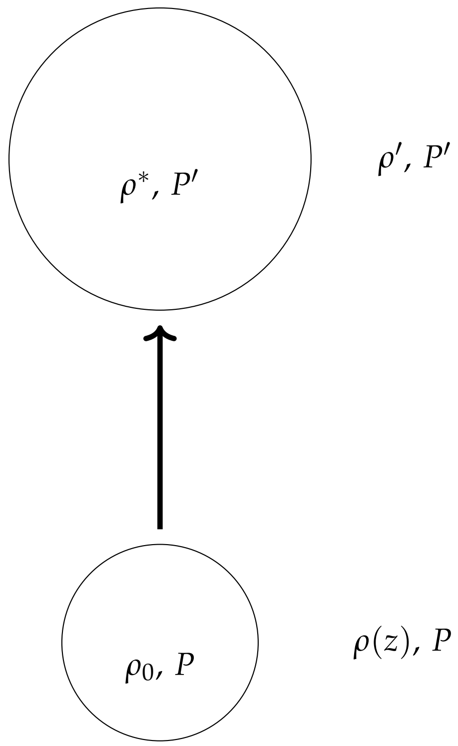

Figure 8: Vertical displacement of a blob of gas in a stratified atmosphere.

in *pressure equilibrium* as it moves, which is a condition that is generally met for subsonic flows for which

$$\tau_d \sim \frac{l}{c_s} \ll \frac{l}{v_{\text{bubble}}}$$

For example, in the Sun's convective regions, which extend approximately to 30% of the total solar radius, the sound speed is roughly $2 - 5 \cdot 10^2$ km/s, while the convective velocity is of order of a few meters per second. Note that only in the outskirts of the star the convective velocity gets high enough to give a little trouble to our initial assumption With some simple computations we are able to show that the time needed to roughly move across the convective region is of the order of a week, while the dynamic time is much smaller. Under these assumption, convective motions are unlikely to disturb too much the hydrostatic equilibrium condition.

Moreover, we're going to assume that the bubble moves *adiabatically*, so that there's no energy exchange with the surrounding fluid. As a further approximation we'all assume that the chemical composition of the star μ is constant along the bubble's motion.

With reference to Fig.8, we can expand to first order the temperature of the bubble and that of the surrounding fluid

$$T_b' = T_0 + \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{ad}} l \quad T_1 = T_0 + \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{rad}} l$$

we'all assume both the temperature gradients to be constant across a small displacement. To achieve the onset of covection, it's necessary that the bubble is *lighter* than the surrounding fluid, so that it will start floating upwards because of buyoancy. This is equivalent to requiring

$$(\rho_{amb} - \rho_{ad})g > 0$$

This condition we can in principle transform in a condition for temperature gradients, but to do so, we're in the need of some equation of state. Before taking a closer look at the various configurations, let us write down a few useful expansions

$$\begin{aligned} T_a &= T_0 + \partial_r T|_{\mathrm{rad}} l \\ T_b &= T_0 + \partial_r T|_{\mathrm{amb}} l \\ \mu_a &= \mu + \partial_r \mu|_{\mathrm{amb}} l \\ \mu_b &= \mu + \partial_r \mu|_{\mathrm{b}} l = \mu \end{aligned}$$

In the last equation we assumed that the chemical composition of the bubble remains constant as it moves[13].

Perfect Gas

Let us assume a perfect gas EoS, so that

$$\frac{\mathrm{d}\rho}{\rho} = \frac{\mathrm{d}p}{p} + \frac{\mathrm{d}T}{T} + \frac{\mathrm{d}\mu}{\mu}$$

If we assume that the chemical composition of the background fluid is constant, we're left with a relation that connects only the variation in temperature and the variation in density. Thus, if we request $\mathrm{d}\rho > 0$

$$-\frac{1}{T}\partial_r T|_{\mathrm{rad}} + \frac{1}{T}\partial_r T|_{\mathrm{b}} > 0$$

which we can convert into gradients with respect to the pressure simply multiplying by $\mathrm{d}r/\mathrm{d}p$. Taking the absolute values

$$\left|\frac{\mathrm{d}\ln T}{\mathrm{d}\ln p}\right|_{\mathrm{rad}} > \left|\frac{\mathrm{d}\ln T}{\mathrm{d}\ln p}\right|_{\mathrm{ad}} \to \nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} \tag{53}$$

13 This clearly is a possibly faulty approximation, because as the bubble moves, instabilities will surely arise, producing mixing between the bubble and the surrounding fluid.

which is known as *Schwartzschild criterion*. We have defined with ∇_x the gradient of temperature with respect to pressure. In the presence of convection we've seen that

$$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{amb}} = -\frac{3}{4ac}\frac{\bar{\kappa}\rho}{T^3}F_{\mathrm{rad}}(r)$$

but the total flux does not completely overlap with the radiative flux $F_{\mathrm{tot}} = F_{\mathrm{rad}} + F_{\mathrm{conv}}$.

Note that if we relax the perfect gas condition

$$\rho = \rho(p, T, \mu) \rightarrow \alpha = \frac{\partial \ln \rho}{\partial \ln p} \quad \delta = \frac{\partial \ln \rho}{\partial \ln T} \quad \phi = \frac{\partial \ln \rho}{\partial \ln \mu}$$

we obtain a condition like

$$\frac{\mathrm{d}\rho}{\rho} = -\delta\frac{\mathrm{d}T}{T} + \phi\frac{\mathrm{d}\mu}{\mu}$$

which if we assume $\mathrm{d}\mu = 0$ bears just the same result as before.

Perfect Gas with Non-Uniform Chemical Composition

As per our previous discussions, we can write

$$\frac{\Delta\rho}{\rho} = -\frac{(T_{\mathrm{amb}} - T_{\mathrm{b}})}{T} + \frac{(\mu_{\mathrm{amb}} - \mu_{\mathrm{b}})}{\mu} > 0$$

which in terms of the gradients we've defined earlier becomes

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{rad}} > \left|\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{amb}} + \frac{T}{\mu}\left|\frac{\mathrm{d}\mu}{\mathrm{d}r}\right|_{\mathrm{amb}}$$

Now we can make the same trick of multiplying for the spatial pressure gradient so to obtain

$$\nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} + \nabla_{\mu} \tag{54}$$

which is called *Ledoux Criterion*. Note that in this case it's much harder to prompt the onset of convection (the radiative gradient must be greater than that predicted by the Schwartzschild criterion). I think it won't surprise you that we can generalize this condition to a non-perfect gas, thus finding

$$\nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} + \frac{\phi}{\mu}\nabla_{\mu}$$

2.6.1 The Overshooting Problem

The *overshooting problem* can be quite a nuisance if we want to calculate the spatial extension of the convective region of the star. To put it simply, in the convective region, the blob of gas will raise with non-zero velocity which, in turn, won't be magically disappearing as soon as it crosses the non-convective region. This causes the convective region to formally "expand" outwards. This can potentially impact significantly on the permanency of a star in the Main Sequence stage. If we want to address the increasing in size of the convective region, it is customary to introduce a new parameter β_{ov} so that

$$l_{ov} = \beta_{ov} H_p$$

where H_p is the scale height of the pressure profile.

To find the ambient temperature gradient, we're in the need of recalling the 1st principle of thermodynamic

$$\delta Q = c_p \rho V (T_0' - T_1) = C_p \Delta T \tag{55}$$

The temperature difference can be expressed in terms of derivatives with respect to pressure

$$\Delta T = \left(\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{amb}} - \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{ad}} \right) \Delta p$$

where we can define the *superadiabaticity* δ_{ad}

$$\delta_{\text{ad}} = \left(\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{amb}} - \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{ad}} \right) \tag{56}$$

Since the thermal capacity is very large in the stellar interiors, we can assume the superadiabaticity to be vanishing

$$\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{amb}} \sim \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\text{ad}}$$

Note that in the stellar external envelopes, we can't assume the same principle to hold. But the problem is partially mitigated by the fact that in the most external layers there's roughly no energy production. This means that convection will influence the effective temperature, but not the luminosity.

To find the radiative gradient, we first have to find F_{rad} and, consequently, F_{conv}. In order to do so, we're going to implement somewhat "empirical" models, that tells us that $F_{\text{conv}} \propto l$. The mean free path l we'll parametrize as

$$l = \alpha_{ML} H_p$$

where α_{ML} is the *mixing length parameter*. We're not assured that such parameter is the same in different stars–or even in the same star but in different evolutionary stages.

We know that in order for convection to settle

$$\nabla_{\text{rad}} > \nabla_{\text{ad}}$$

and so that energy can be transported

$$\nabla_{\text{amb}} \geq \nabla_{\text{ad}}$$

In short

$$\boxed{\nabla_{\text{ad}} \leq \nabla_{\text{amb}} \leq \nabla_{\text{rad}}}$$

This tells us that the adiabatic gradient is the maximum gradient that can set into a given structure (assuming $\delta_{\text{ad}} \approx 0$). What we still have to do is find the convective gradient in the external regions.

2.6.2 Mixing Length Theory

In order to find what we're looking for (an approximate description of the convective gradient) we're going to need some simplifying assumptions:

1. The theory behind mixing length is going to be a *local*, so that this criterion can be applied on a "mesh perby mesh" basis to check its stability;

2. The mean free path of the bubble we can parametrize as $l = \alpha_{ML} H_p$;

3. Small variations in temperature and density;

4. The variations induced by different shapes and sizes of the bubbles are negligible (this is usually not true, but their contribution enters as a second order correction).

First and foremost, let us define the *pressure scale height* H_p as the distance the bubble has to travel so that pressure drops to e^{-1} of its original value, in formulas

$$\frac{1}{H_p} = -\frac{1}{p}\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{\mathrm{d}\ln p}{\mathrm{d}r}$$

If we assume that all contribution to the pressure comes from the gas (and so we neglect other contributions, like radiation pressure), we can write plug in the equation for hydrostatic equilibrium and find

$$H_p = \frac{p}{g\rho}$$

where g is the local acceleration of gravity.

Recall when we have first introduced the 1st Principle of Thermodynamics (55). We assumed tht the temperature difference was proportional to the difference between the adiabiatic gradient and the ambient temperature gradient. It's clear that if the two gradients were the same, no heat would be released and no energy transport is possible. As we've mentioned before, when convection involved layers closer to the stellar surface, we cannot make the the simplifications we've used in last section.

In this case we need to find a more complex expression for ∇_{conv}. We start by enforcing the confition that at a given layer within the convective region the total energy flux is simply the sum of the flux carried by radiation and convection

$$F_{\text{tot}} = F_{\text{rad}} + F_{\text{conv}} \quad F_{\text{rad}} = -\frac{4acT^3}{3\bar{\kappa}\rho}\frac{\mathrm{d}T}{\mathrm{d}r} \tag{57}$$

where $\mathrm{d}T/\mathrm{d}r$ is the the actual temperature gradient of the convecting region. So far so good, there's nothing new here.

Taken that we refer with the subscript "u" to rising matter and "d" to downwards moving matter, we can write an expression for the amount of stuff that is either rising or descending

$$\rho_u \sigma_u v_u \quad \rho_d \sigma_d v_d$$

where σ is the surface area of the convection column. The net transport of energy will then be

$$2F_{\text{conv}} = \rho_u v_u c_p T_u + \frac{1}{2}\rho_u v_u v_u^2 - \left[\rho_d v_d c_p T_d + \frac{1}{2}\rho_d v_d v_d^2\right]$$

Note that in order to preserve hydrostatic equilibrium, the amount of falling material must be the same as the rising one, otherwise the star would either dissolve or concentrate all the mass in the interior. To avoid this, the only solution is that after a distance l of order H_p the bubble stops and invert its motion. This reflects in requiring

$$\rho_u \sigma_u v_u = \rho_d \sigma_d v_d$$

so that if we assume little differences in density and equal cross-sections for the convective columns, the condition simply becomes $v_d \sim v_u = v$.

By neglecting the energy losses during the bubble's displacement (adiabatic assumption), the convective flux transported by bubbles that move *upwards* with a velocity v is then given by

$$F_{\text{conv}} = \frac{1}{2}\rho v c_p \left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\text{ad}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)\right] l$$

The factor $1/2$ accounts for the fact that at each layer approximately half of the matter is rising and half is moving downwards. We've already expanded T_u and T_d. Assuming an "average" temperature T_{amb} between the temperatures of the two columns, we can write

$$T_u - T_d = 2\Delta T = 2(T_u - T_{\text{amb}}) \rightarrow \Delta T(s) = \left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\text{ad}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)\right] s$$

where s is an average displacement which we can conveniently take as $s = l/2$, so that everything is consistent with the equation presented above. Performing the usual trick, we can transform all the gradient in logarithmic gradients (∇), so that

$$F_{\text{conv}} = \frac{1}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}\left[\nabla_{\text{amb}} - \nabla_{\text{ad}}\right] l$$

where $\mathrm{d}z = -\mathrm{d}r$ is the depth. We can substitute $\mathrm{d}p/\mathrm{d}z$ from the equation of hydrostatic equilibrium obtaining the following expression that is all nice and dandy (*said Lying Odysseus*)

$$\nabla_{\text{amb}} = \frac{F_{\text{tot}} + \frac{l}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}\nabla_{\text{ad}}}{\frac{l}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z} + \frac{T^3}{\bar{\kappa}\rho}\frac{4ac}{3}\frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}} \tag{58}$$

where we've used the decomposition of the total flux and the expression for the radiative flux (57).

What we are still lacking is a prescription for the velocity of the convective elements, which, however, we can conveniently obtain from the kinetic energy theorem ("work-energy principle")[14]: The net force per unit volume

14 The interested reader can refer to either [6] or [15] for a more detailed discussion.

f_r acting on the convective element at the layer r is given by $f_r = -g\Delta\rho(r) = -g\Delta T/T$, meaning the work done per unit volume in moving the bubble through a distance Δr is

$$W(\Delta r) = -\left(\frac{1}{2}\right) g\Delta\rho(\Delta r)\Delta r$$

which we can average over a distance $l/2$. Moreover, it is customary to express $\langle W(\Delta r)\rangle$ in terms of the average speed of the convective elements as

$$\langle W(\Delta r)\rangle = \frac{1}{2}\rho v^2$$

Equating the two expressions we finally obtain

$$\boxed{v^2 = \left(\frac{1}{2}\right)\frac{g}{T}\left(\frac{l}{2}\right)^2\left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{amb}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{ad}\right]} \tag{59}$$

Note that the factor 1/2 within parentheses was introduced by [15] when performing the integral of the force per unit volume. It is still not clear to me where that comes from, but I think it may be to address for some sort of average value for the density.

2.7 OPACITY

As we might have mentioned somewhere in the above, the physical property known as *opacity* is the collection of all the mechanisms of interaction between radiation and matter. A few important mechanisms are listed below

- Scattering $e^-\gamma$: Since the average energy of particles in a star is generally such that $kT \ll m_e c^2 \approx 0.511\,\mathrm{MeV}$, the prevalent scattering channel is the *Thomson scattering*;
- *Inverse Bremsstrahlung*: also known as "braking radiation" is a channel of interaction that leaves an unbound particle unbound (Free-Free);
- *Photoionization*: it's a type of interaction between a photon and a bound electron, that gets freed from a given atom. So, it's a type of Bound-Free interaction;
- *Level Jump*: radiative excitation/de-excitation of an electron, that is made jump from an energetic level to another. In this sense, this is a Bound-Bound interaction.

It won't come as a surprise that the correct calculation of opacity is made all the more difficult when we're not assured a complete-ionization scenario or if we can't exclude the presence of dust grains (or grains in general) in the stellar structure.

In general, the total opacity is the sum of all the contributions from the various channels which, of course, will also be frequency dependent

$$\kappa_\nu = (\kappa_{bb} + \kappa_{bf} + \kappa_{bf})\left[1 - \exp\left(-\frac{h\nu}{kT}\right)\right] + \kappa_{sc} \tag{60}$$

If you recall, in our stellar structure equations enters an opacity averaged over all frequencies, that is generally known as *Rosseland Mean Opacity*. A possible way to obtain it is presented at the end of Appendix A (131).

We're going to dedicate the next few sections to obtaining analitical expressions for the most common sources of opacity, under the assumption of being the dominating contribution.

2.7.1 H^- opacity

The opacity due to Bound-Free or Free-Free interactions with the H^- ion is dominant, for example, in the solar photosphere. In order for this channel to grow relevant, it's clear that we're going to need plenty of Hydrogen and free electrons (which will be typically coming from ionized metals).

If we want to exclude the possibility of other sources of opacity being more relevant, we'll have to work with temperatures in the range $3000 - 6000\,\mathrm{K}$ and with a metallicity Z in the range $10^{-3} - 3 \cdot 10^{-3}$. Similarly, we have to require a low density $10^{-10} < \rho < 10^{-5}\,\mathrm{g/cm^3}$ to avoid what's called *collision induced absorption*, which becomes more and more relevant at low temperatures and high densities, as can happen in main sequence stars with masses $M \leq 0.5 M_\odot$ or in the atmosphere of some white dwarves.

It is possible to show that the opacity due to interactions with the H^- ion is given by

$$\bar{\kappa}_{\mathrm{H}^-} \sim 2.5 \cdot 10^{-31} \left(\frac{Z}{0.02}\right) \rho^{1/2} T^9 \,\mathrm{cm^2 g^{-1}} \tag{61}$$

while for the collision induced absorption

$$\bar{\kappa}_{\mathrm{CIA}} = k_2 \rho^2 \tag{62}$$

where k_2 is a constant.

Note that if temperature drops particularly low ($T \lesssim 1400 - 1800\,\mathrm{K}$) we can observe the formation of *grains*, which are another important source of opacity.

2.7.2 Thomson Scattering

We know that in your typical star temperature is such that

$$kT \ll m_e c^2 \approx 0.511\,\mathrm{MeV}$$

We have formerly defined the opacity as

$$\rho \kappa_\nu = n \sigma_\nu \tag{63}$$

where n is the numeric density of the particles that interact with photons. Particle physics (or your Physics II classes, if you prefer) tells us that scattering Thomson is a particular type of *isotropic* and *frequency independent* scattering channel, which is automatically translated into the expression for its cross-section

$$\sigma_T = \frac{8\pi}{3} \left(\frac{e^2}{m_e c^2}\right)^2 = 0.66\,\mathrm{b} \tag{64}$$

where the b stands for *Barns* ($10^{-24}\,\mathrm{cm}^2$) a unit that is particularly used in particle physics. The opacitive channel of scattering Thomson benefits from high temperatures (more electrons since matter's more likely to be ionized) and not too high densities (other channels become dominant). As a rule of thumb, you might want to remember that the more dominant opacitive channels are the level jump, closely followed by photoionization, and then the others.

A while back ago, we've defined the number density of electrons in the case of total ionization (30)

$$n_e = \frac{\rho}{m_H}\sum_{i=i}^{N}\frac{Z_iX_i}{A_i} \approx \frac{\rho}{m_H}\left(X+\frac{Y}{2}+\frac{Z}{2}\right) = \frac{\rho}{m_H}\left(\frac{X+1}{2}\right) \tag{65}$$

where the last equality was obtained exploiting the condition $X+Y+Z=1$.

We can therefore plug (65) into (63), obtaining

$$\bar{\kappa}_T = \kappa_T = \frac{8\pi}{3}\left(\frac{e^2}{m_ec^2}\right)^2\frac{X+1}{2m_H} \tag{66}$$

Note that in general, for higher temperatures ($T > 10^8\,\mathrm{K}$), the Compton scattering probability grows non-negligible. Compton scattering is an anisotropic, anelastic scattering mechanism

$$\lambda' - \lambda = \frac{h}{m_ec}(1-\cos\theta) \tag{67}$$

Since the effect produced by Compton scattering is small (but sometimes not negligible), people used to correct the Thomson scattering cross-section as such

$$\sigma_{T,\mathrm{corr}} = \left[1+0.35\left(\frac{T}{10^9\,\mathrm{K}}\right)^{1/2}+0.73\left(\frac{T}{10^9\,\mathrm{K}}\right)\right]^{-1}\sigma_T$$

2.7.3 Inverse Bremsstrahlung

Another channel that is particularly important when BB and BF channels are quiescent[15], is that of inverse Bremsstrahlung[16]. Particle physicists usually manage to convince us that the *differential cross-section* (read: velocity dependent) for inverse Bremsstrahlung is described by

$$\frac{\mathrm{d}\sigma_{FF}}{\mathrm{d}v_e} = \frac{4\pi}{3\sqrt{3}hc}\frac{Z^2e^6}{m_e^2}\frac{1}{v_e}\frac{1}{\nu^3}n_ef(v_e)\bar{g}_{ff}(\nu,v_e) \tag{68}$$

where here Z is the electrical charge of the ion interacting with electrons, $f(v_e)$ is some velocity distribution for electrons (typically a Maxwellian distribution) and $\bar{g}_{ff}$ is the *Gaunt factor* a fancy numerical coefficient that we will approximate as roughly constant and of order unity in the following. Note that since there isn't an energy threshold for this opacity channel, in principle we have to take into account all possible species that might be contributing.

15 Oh my, what a fancy word I've found.
16 Remember the capital "B" and the two "s", it's a German word (well, two, actually).

If we take an average value for the velocity

$$\langle \frac{1}{v_e} \rangle = \left(\frac{2m_e}{\pi k T} \right)^{1/2} \propto \left(\frac{m_e}{T} \right)^{1/2}$$

then we can consider an average cross section

$$\langle \sigma_{FF} \rangle = \text{const} \cdot \frac{Z^2 n_e}{(kT)^{1/2}} \frac{\bar{g}_{ff}}{\nu^3}$$

Recalling the definition of opacity (63)

$$\begin{aligned} \kappa_{\nu,ff}\rho &= \sum_{A=1}^{N} n(A, Z_A)\langle \sigma_{FF} \rangle \\ &= \sum \frac{X_A \rho}{A m_H} \langle \sigma_{FF} \rangle \end{aligned}$$

To the cross section only the more abundant species will participate (so Hydrogen and Helium). Thus

$$\kappa_{\nu,FF} \sim \frac{X}{m_H} \langle \sigma_{FF}(1, \nu) \rangle + \frac{Y}{4m_H} \langle \sigma_{FF}(2, \nu) \rangle$$

If we assume completely ionized matter (65) and notice that $\sigma \propto T^{-1/2}\nu^{-3} \rightarrow T^{-3.5}$, then after computing the Rosseland mean opacity, we're left with

$$\bar{\kappa}_{FF} \sim 4 \cdot 10^{22}(X + Y)(1 + X)\rho T^{-3.5} \tag{69}$$

2.7.4 Photoionization

Not surprisingly, for photoionization mechanisms to arise, it's necessary the presence of atoms with at least one bounded electron. Differently from the channels we've taken a look so far, photoionization is a threshold process. For it to happen, photons must have an energy at least equal to the ionization potential of the impinged atom. In general

$$h\nu = \chi_n + \frac{p_e^2}{2m_e}$$

with

$$\chi_n = \frac{2T^2 m_e e^4}{hn^2}(Z')^2$$

where Z' is the effective charge of the ion. Unless we're particularly happy of doing a lousy job, we should, in principle, consider all relevant energy levels of the ions, the ionization degree of the species and the number of species available. In short: There are many things you might want to keep track of.

What you would find out, were you actually to do the computations (and do not make any mistakes in the process), is that the photoionization channel has the same density and temperature dependencies as the Inverse Bremsstrahlung one, but with a greater multiplicative coefficient, thus making it usually more relevant.

Assuming a prevalence of hydrogenoid atoms[17], it is possible to prove that the cross section for photoionization is given by

$$\sigma_{bf} = \frac{64\pi^4 m_e e^{10} (Z')^4 g}{3\sqrt{3} c h^6 n^5 \nu^3} \tag{70}$$

where all symbols have their customary meaning. Said N_A the numeric density of the atomic species A

$$N_A = \frac{X_A \rho}{A m_{\mathrm{H}}}$$

we can define a numeric density expressing the average number of electrons for each atom A which are bounded to an energy state n. This quantity, which we're calling $N_{A,n}$ is given by

$$N_{A,n} = \frac{N_e n^2 h^3 \exp(\chi_n / kT)}{2(2\pi m_e kT)^{3/2}}$$

Given these definitions, we can write down the expression for the opacity

$$\kappa_{bf}(\nu)\rho = \sum_{A,n} \sigma_{bf}(\nu) \frac{X_A \rho}{A m_{\mathrm{H}}} N_{A,n}$$

in which we're going to consider only elements with $A \geq 6$, for Hydrogen and Helium are completely ionized at the temperatures we're considering and are thus unable to participate to the opacity. So, we can write the numeric density of electrons as

$$N_e \sim \frac{\rho}{m_{\mathrm{H}}} \left(\frac{X+1}{2} \right)$$

so that pluggining in all the various definitions we obtain

$$\kappa_{bf}(\nu) = \sum_{A,n} \frac{4\pi^2 e^2 g}{3\sqrt{3} c \nu^3} \left(\frac{\chi_n^2}{m_e} \right) \frac{kT}{kT} \frac{X_A \rho (1+X)}{A m_{\mathrm{H}}^2} \frac{nh \exp(\chi_n / kT)}{(2\pi m_e kT)^{3/2}}$$

where we've used the definition of the ionization potential χ_n to eliminate the effective ion charge. Summing over A gives us the metallicity Z, thus leaving us with

$$\kappa_{bf}(\nu) = \sum_{n} \frac{4\pi^2 e^2 g}{3\sqrt{3} c \nu^3} \left(\frac{\chi_n^2}{m_e} \right) \frac{kT}{kT} \frac{Z \rho (1+X)}{A m_{\mathrm{H}}^2} \frac{nh \exp(\chi_n / kT)}{(2\pi m_e kT)^{3/2}}$$

If we consider only the energy level $n = 1$ we may find that the opacity, leaving behind all numerical factors, is a function of

$$\kappa_{bf}(\nu) \propto \left[\frac{\chi_n}{kT} \exp(\chi_n (kT) \left(\frac{kT}{h\nu} \right)^3) \right] Z(1+X) \rho T^{-3.5}$$

and the term within squared parentheses is approximately of order 1, since terms with $h\nu \sim kT$ will be weighed more when performing the Rosseland mean. If we restore all numerical factors and perform the aforementioned Rosseland mean, you'd find

$$\bar{\kappa}_{bf} \sim 4 \cdot 10^{25} \, Z(1+X) \rho T^{-3.5} \tag{71}$$

which, as anticipated, has the same temperature and density dependencies as the free-free contribution.

17 That is atoms of given Z that through processes we're not even interested in have been left with just one electron.

Level Jump

With *level jump* we refer to the absorption of radiation by an electron bound to an ion that causes the electron to move from one bound state to a more energetic one. After the transition the electron will have to move down to the original bound state, for the ion has to return in equilibrium with the surroundings. The photon emitted during this de-excitation will have the same energy of the absorbed one, but it is re-emitted in a random direction.

This process necessarily involves photons of certain frequencies, corresponding to the energy differences between the initial and final bound state of the electron. Bound-bound processes can become important contributors to the total opacity for temperatures below $\sim 10^6$ K.

Since a formal discussion of bound-bound opacitive mechanisms would have to make use of Einstein's coefficients (and we would end up not finding a nice analitical solution, but rather statistics for the process) we'll skip right to the next opacitive channel.

2.7.5 Electron Conduction

As we might expect from its name, electron conduction is a *diffusive process*, in the sense that it can be described by a diffusive equation of the form

$$\partial_t n = D_{\text{cond}} \partial_r^2 n \quad D_{\text{cond}} = \frac{\lambda_e \langle v_e \rangle}{3}$$

If electrons are non-degenerate, the cross section for conductive mechanisms is much larger than, say, the photoionization cross section. In fact, the mean free path of electrons is much smaller than that of photons

$$\lambda_{\text{cond}} \propto \frac{1}{n_i \sigma_{\text{cond}}} \ll \lambda_\gamma$$

But things change rather dramatically if the gas becomes degenerate: The cross section for the process decreases and thus the mean free path for electrons raises considerably. For the purposes of our discussion, let us recall the Fermi momentum

$$p_F = \left(\frac{3n_e}{8\pi}\right)^{1/3} h \quad n_e = \frac{\rho}{\mu_e m_{\text{H}}} \tag{72}$$

If diffusion is isotropic, we can assume the generalized Fick law to hold, and so

$$F(r) = -\frac{1}{3}\langle v_e \rangle \lambda_e \frac{\mathrm{d}U(r)}{\mathrm{d}r} = -\frac{1}{3}\langle v_e \rangle \lambda_e \frac{\mathrm{d}U}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{1}{3}\langle v_e \rangle \lambda_e c_V \frac{\mathrm{d}T}{\mathrm{d}r}\bigg|_{\text{amb}} \tag{73}$$

where we've introduced c_V the constant volume specific heat. We'll call

$$k_{\text{cond}} = \frac{1}{3}\langle v_e \rangle \lambda_e c_V \tag{74}$$

the thermal conduction coefficient. To proceed further we have to determine the Coulombian cross section

$$m_e v_e^2 \approx \frac{Z_C e^2}{b}$$

where b is the impact parameter. Inverting last equation and defining a conduction cross section $\sigma_{\text{cond}} \approx \pi b^2$ yields

$$\sigma_{\text{cond}} \propto \frac{\langle Z_C^2 \rangle}{\langle v_e^4 \rangle}$$

Since the gas is degenerate, the electrons that will have the larger contribution are those with momentum approximately equal to the Fermi momentum (72), and thus

$$v_e \approx \frac{p_F}{m_e} \propto \left(\frac{\rho_e}{\mu_e} \right)^{1/3}$$

hence

$$\sigma_{\text{cond}} \propto \left(\frac{\mu_e}{\rho_e} \right)^{4/3}$$

Now we can finally calculate the mean electronic free path

$$\lambda_e = \frac{1}{n_i \sigma_{\text{cond}}} = \frac{\mu_i m_{\text{H}}}{\rho} \left(\frac{\rho_e}{\mu_e} \right)^{4/3} \frac{1}{\langle Z_C^2 \rangle}$$

Moreover, we're in the need of an expression for the specific heat

$$c_V = \frac{8\pi^3 m_e k^2 T p_F}{3h^3} \propto \left(\frac{\rho_e}{\mu_e} \right)^{1/3} T$$

At last we've finally found an expression for the thermal conduction coefficient

$$k_{\text{cond}} = \rho \frac{\mu_i}{\mu_e^2} \frac{T}{\langle Z_C^2 \rangle}$$

From this we can then define an opacity coefficient for thermal electronic conduction

$$\kappa_{\text{cond}} := \frac{4acT^3}{3k_{\text{cond}}\rho} \approx 4 \cdot 10^{-8} \left(\frac{\mu_e^2}{\mu_i} \right) \left(\frac{T}{\rho} \right)^2 \langle Z_C^2 \rangle \tag{75}$$

Note that with this we can define a *total transported flux*

$$F_{\text{tot}} = F_{\text{cond}} + F_{\text{rad}} = -\frac{4acT^3}{3\rho} \left(\frac{1}{\bar{\kappa}} + \frac{1}{\bar{\kappa}_{\text{cond}}} \right) = -\frac{4acT^3}{3\rho \bar{\kappa}_{\text{tot}}}$$

The opacitive coefficient within parentheses is the one that is customarily found in professional opacitive tables.

3 STELLAR MODELS

3.1 INTRODUCTION

In the last chapter we've managed to write down the four (five, if you consider the EoS) equations through which we're able to describe stellar structures. For the sake of completeness (and not to make you flip your pages back and forth), I'll write them all down below

$$\begin{aligned}
\frac{\mathrm{d}M_r}{\mathrm{d}r} &= 4\pi r^2 \rho(r) \\
\frac{\mathrm{d}P_r}{\mathrm{d}r} &= -\frac{GM_r}{r^2} \\
F(r) &= -\frac{16}{3}\frac{\sigma_B T^3}{\rho\bar{\kappa}}\frac{\mathrm{d}T}{\mathrm{d}r} \\
\frac{\mathrm{d}L(r)}{\mathrm{d}r} &= 4\pi r^2 \rho(r)\epsilon(r) \\
p &= p(\rho, T, X, Y, X_i)
\end{aligned}$$

Those who are used to working with stellar structure equations are often solving them (numerically, of course) using different independent variables in the stead of the radial distance r.

For example, in the inner regions of stars, where the mass isn't saturated, we may use the mass M_r and express all the derivatives consequently. When the mass is saturated we have to switch to another varible, for example the pressure p (the associated region is the *sub-atmosphere*). As we move to the outskirts of the star, all relevant quantities will be saturated to their boundary value, and we have to switch to the *optical depth* to parametrize the stellar structure equations (this is the case for the *photosphere*).

The interested reader may refer to [5], §3.3, to have an idea of how the boundary conditions are chosen and linked together.

Let us now take a closer look at how the deed is done.

3.2 FITTING METHOD FOR STELLAR STRUCTURES INTEGRATION

In principle, you could use an atmosphere model to describe more or less accurately the atmosphere of a star. However, it may also happen that you *don't want* to use an atmosphere model but rather use an approximate model. In the present section, we're going to discuss a little bit of this can be done. Clearly, these rough calculation will hold, if approximately, only in small main sequence stars, which have "simpler" atmospheres.

The procedure here described follows closely the discussion by (blank, at the moment I've yet to find someone talking about this).

Consider the equation for hydrostatic equilibrium. In the atmosphere, we're going to use as independent variable the optical depth τ, so that

$$\frac{\mathrm{d}p}{\mathrm{d}r} = \frac{\mathrm{d}p}{\mathrm{d}r} \cdot \frac{\mathrm{d}\tau}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2}$$

But recall the definition of optical depth (110) and absorption coefficient (8), so that

$$\frac{\mathrm{d}\tau}{\mathrm{d}r} = (-)\rho\bar{\kappa}$$

with $\bar{\kappa}$ a suitably averaged opacity coefficient. Since in the atmosphere the mass of the star is saturated $M_r = M_{\text{tot}}$ and so is the radius, the equation for hydrostatic equilibrium becomes

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{g}{\bar{\kappa}}$$

where $g = GM_{\text{tot}}/R_*^2$ is the surface gravity of the star. What about the other equations? The mass is saturated, so the 1st equation is irrelevant; for the same reason, also the 4th equation is irrelevant, for there is no energy production in the atmosphere[1].

So we're left with only the equation for energy transport and the EoS. Note that, in general, we cannot write an expression for radiative transfer as in (109), because we're not assured LTE to hold.

So what we're going to do is writing down an approximate relation for the temperature, that is going to depend on the optical depth and the effective temperature

$$T = T(\tau, T_{\text{eff}})$$

One possible relation is the celebrated equation for the *grey atmosphere*

$$T(\tau) = \frac{3}{4}T_{\text{eff}}^4\left(\tau + \frac{2}{3}\right) \tag{76}$$

which assumes LTE to hold and negligible frequency-dependence on the absorption coefficient (and thus for the opacity and optical depth as well). According to this equation, the photosphere, characterized by $T = T_{\text{eff}}$, is to be found at an optical depth $\tau = 2/3$.

This approximation is sometimes used when we're interested in observing differential effects and we can be a little loose on having an accurate estimate of the effective temperature.

Another expression, that is still quite in use, is that found by [10], which is essentially a grey atmosphere modified so to include the absorption of some elements

$$T^4(\tau) = \frac{3}{4}T_{\text{eff}}^4\left(\tau - 1.017 - 0.30e^{-2.54\tau} - 0.291e^{-30\tau}\right) \tag{77}$$

where the coefficients are the product of a fitting procedure.

It's clear that those who create hydrodynamic atmosphere models try to take into account every possible phenomenon that may occur (and be relevant) in the atmosphere, such as turbulence. Technically, especially in larger

1 Note that although in the atmosphere there is no energy produced by nuclear reaction, we may still have to consider the contribution of energy generation by the means of gravitational collapse. However, densities are very low in the atmosphere, making ϵ_g negligible.

stars with convective envelopes, it would be sensible to consider the contribution of *turbulent pressure* to the total balance

$$p = p_{\text{gas}} + p_{\text{rad}} + p_{\text{turb}}$$

As you should know if you've attended the Fluid Dynamics or the Astrophysical Processes course, turbulence is one hell of a beast to deal with. To give you an idea of how much complex it is to even build a half-functioning model of turbulence, here's Werner Heisenberg's take on the subject[2]:

> "*When I meet God, I am going to ask him two questions: why relativity? And why turbulence? I really believe he will have an answer for the first.*"

If we really have to include turbulence in the picture, what's often done is invoking the Boussinesq's approximation and write a barotropic expression for the turbulent pressure

$$p_{\text{turb}} = \rho \bar{v}_{\text{turb}}^2 \quad \bar{v}_{\text{turb}}^2 = \beta c_s^2$$

where $\beta \lesssim 1$ is a free parameter that has to be estimated to find an accord. But is this worth the trouble? Probably not.

On top of not knowing how to deal with turbulence, we are not able, as yet, to give a precise description of the ambient temperature gradient in the atmosphere, which already enters our equations with a free parameter.

If you really want to make a meticulous job, don't bother stuffing your equations with all this stuff and get for yourselves a three-dimensional atmosphere model.

3.2.1 Starting the Integration

Our external boundary conditions are the star luminosity L_* and the effective temperature T_{eff}. Note, however, that the former doesn't help us much at this stage, since it still doesn't enter our equations in the atmosphere.

Let's assume we're starting with a very small optical depth $\tau = \tau_0 \ll 1$, so that the pressure (and the opacity) is small $p = \epsilon \ll 1$. So, assuming we're given an expression of the form $T = T(\tau, T_{\text{eff}})$, we can calculate T plugging in our values for $\tau = \tau_0$ and T_{eff}.

From the EoS, since we already knew $p = \epsilon$ and T we've just calculated, we obtain the density ρ. At this point, using all of this and the chemical composition, we can calculate the mean opacity $\bar{\kappa}$ and put everything in the equation of hydrostatic equilibrium and check if our solution is self-consistent. If not, we re-iterate the process (changing our starting values for τ and p) until the solution satisfies hydrostatic equilibrium within a given precision.

Now that we have all our varibles determined on the external boundary, we can start integrating towards the inner regions. To do so, derivatives in our equations become incremental ratios and we solve mesh after mesh until we reach the base of the atmosphere (the boundary with the sub-atmosphere).

2 Yes, this is the same Werner Heisenberg that laid the foundations of Quantum Mechanics.

Integrating through the sub-atmosphere, we have to raccord the values of the n-th mesh of the region with the most external one of the stellar interior. So we'll have to match p_N, ρ_N, T_N, L_N and R_N.

As anticipated, in the stellar interior we're going to be integrating with respect to the mass M_r, and we'll be using as boundary conditions at the center of the star ρ_C, p_C, $L_r \approx 0$, $M_r \approx 0$.

Consider the equivalent formulation for the equation of hydrostatic equilibrium

$$\frac{\mathrm{d}p}{\mathrm{d}M_r} = -\frac{GM_r}{4\pi r^4}$$

In a given mesh we know the value of the right hand side of the equation, meaning we also know the value of the derivative. We can therefore linearize the relation for the pressure and (moving outwards) find

$$p(M_r + \Delta M_r) = p(M_r) + \frac{\mathrm{d}p}{\mathrm{d}M_r}\Delta M_r$$

As mentioned above, the derivative is known from the hydrostatic equilibrium. In principle we're not assured that the variation is exactly linear, so we're going to inevitably incurr into errors. What is usually done (automatically) is halving the mass step and check how the solution changes, eventually stopping when the variation gets under a given threshold.

Typically, the integration stops not in the center of the star (which would be non-sensical, computationally and physically speaking), but rather in a mesh nearby the center called *fitting mesh*. Once this special mesh is defined, we start integrating outwards from the center of the star with the aforementioned boundary conditions, stopping only when the fitting mesh is reached.

This leaves us with two sets of values at the fitting mesh, one obtained from fitting from the exterior inwards and one from the inside outwards. Clearly, there's no reason these two sets must turn out to be equal (computationally speaking), thus obtaining differences

$$X_i - X_e \neq 0$$

where X_i and X_e represent some physical property of the star as computed from, respectively, integrating outwards and inwards. Physically speaking, however, since the values are describing the same mesh, they must be coinciding.

To decide whether or not the difference is acceptable, we can set an upper bound to the maximum relative difference we're accepting

$$\frac{X_i - X_e}{X_i} \leq 10^{-4} - 10^{-3}$$

If that condition is not met, we change both the inner and outer boundary conditions and re-iterate the process, thus modifying the two sets of solution at the fitting mesh. The procedure will then be stopping when the difference between the solutions is negligible.

Moreover, we can check the dependence of our solutions from the boundary conditions we've chosen, calculating partial derivatives. For example, we may want to check the dependence of the inner solution for the pressure on the central temperature

$$\frac{\partial p_i}{\partial T_C} \sim \frac{p_i|_{T_C+\delta T_C,\, p_C} - p_i|_{T_C,\, p_C}}{T_C}$$

so that the difference between the inner pressure solution and the external pressure solution is

$$\Delta(p_i - p_e) = \left(\frac{\partial p_i}{\partial T_C}\right)\Delta T_C + \left(\frac{\partial p_i}{\partial p_c}\right)\Delta p_C - \left(\frac{\partial p_e}{\partial L}\right)\Delta L + \left(\frac{\partial p_e}{\partial T_{\text{eff}}}\right)\Delta T_{\text{eff}}$$

This variation must be able to cancel out the difference that we originally had in the fitting mesh. We're going to write similar expressions for all our variables. Sooner or later the procedure will hopefully converge and we'll find out the right boundary solutions.

The procedure we've just described is able to solve stellar structure equations *but* is not able to follow the "fine structure" of thinner structures within the star, for example a thin layer where Hydrogen is burned. This method serves just to build a first model of the star, paving the way for more complex and accurate methods, such as the one that is going to be described in the next section.

3.3 HENYEY METHOD

As mentioned in the passing at the end of last section, it is customary to use the approximate solution found with the fitting method at time t as a test solution for the *Henyey method* [8] [4] at time $t + \mathrm{d}t$.

The fitting method provides us with test-values for the k-th mesh; what we're going to do is checking mesh by mesh if the stellar structure equations are verified. We can expand derivatives as incremental ratios and write an expression, for each equation, with the form

$$f_{i,k} = \delta_{i,k} \neq 0$$

where $f_{i,k}$ is just a fancy way to indicate the i-th stellar equation in the k-th mesh and $\delta_{i,k}$ is the, hopefully small, result of the i-th equation in the k-th mesh. Clearly, at the first iteraction, it is going to be hard that it is vanishing as we'd like it to be. In fact, in general $\delta_{i,k}$ is a function of all the physical properties of the star in the k-th mesh and of the $(k+1)$-th mesh. We can express it as

$$\delta_{i,k} = \mathcal{F}(p_k, T_k, R_k, L_k, p_{k+1}, T_{k+1}, R_{k+1}, L_{k+1})$$

What we're going to do is slightly changing the test-values for each mesh until $\delta_{i,k}$ is found within the desired accuracy. The change in $\delta_{i,k}$ can be expressed as the sum of all the variation induced by the little changes in parameters Said X_i one of the physical properties, we can write

$$\Delta\delta_{i,k} = \sum_j \frac{\partial \delta_{i,k}}{\partial X_{j,k}}\Delta X_{j,k} + \sum_j \frac{\partial \delta_{i,k}}{\partial X_{j,k+1}}\Delta X_{j,k+1}$$

We have have 8 unknown variables, but four of them are always in common between two adjacent meshes. Since $i = 1, ..., 4$ and $k = 1, ..., N-1$, we thus have $4n - 4$ equations and $4n$ variables to determine. We're in the need of adding boundary conditions.

The most obvious conditions to add are $L_C \sim 0$, $R_C \sim 0$ in the center of the star, so that $\Delta R_1 \sim 0$, $\Delta L_1 \sim 0$, and at the base of the sub-atmosphere $p_N = f_1(L, T_e)$, $T_N = f_2(L, T_e)$, $R_N = f_3(L, T_e)$, $L_N = f_4(L, T_e)$. Given these conditions, there are plenty of (rigorous) mathematical theorems that grants that the solution we're going to find this way is, both locally and globally, *unique* (within the requested accuracy, that should go without saying).

4 NUCLEAR REACTIONS

4.1 INTRODUCTION

Most observed stars, including the Sun, are powered by thermonuclear fusion. This term denotes nuclear reactions induced by the thermal motion of the ions, wherby lighter nuclei comnine to form a heavier one.

If the combined mass of the interacting nuclei is larger than the mass of the produced nucleus, the mass difference is converted into energy according to the famous (to the point of being obnoxious) Einstein's relation.

The mass difference between the interacting protons and the product of their interaction stems for the properties of the nuclear binding energy E_B, which is the energy required to separate nucleons against their mutual attraction due to nuclear forces, and is essentially a measure of the stability of the nucleus.

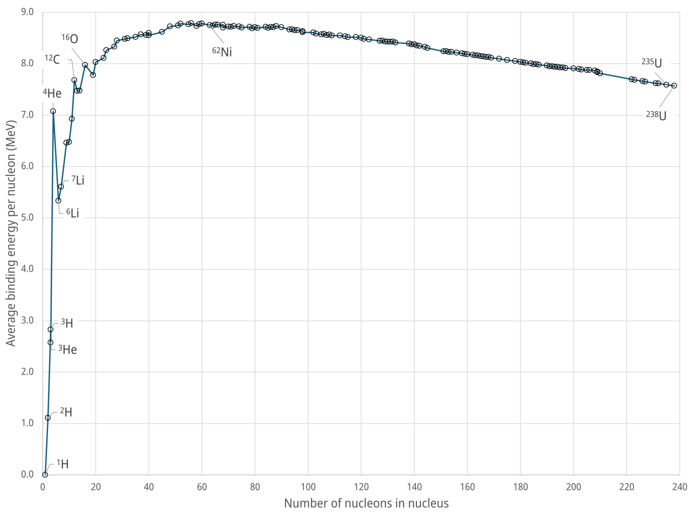

Figure 9: Binding energy per nucleon for a selection of nuclides. The nuclide with the highest value, ^{62}Ni, is noted.

The binding energy per nucleon, shown in Fig.9 is a most interesting quantitiy; its value increases with A first steeply, then more gently, until it reaches a maximum of 8.79 MeV around $A = 56$, which corresponds to ^{56}Fe, and then drops slowly for heavier nucleons.

Most people claim, for reasons we'll soon find out, that Iron nuclei are the most stable ones, while it's false. It's actually ^{62}Ni nuclei that claim this figurative top position.

Elements up to Iron (read: Nichel) will therefore produce energy by thermonuclear fusion, whereas for higher values of A it is the splitting of heavier nuclei into ligther ones (*nuclear fission*) that does the deed.

We'll now focus on the determination of the nuclear energy generation coefficient.

4.2 ENERGY PRODUCTION RATE

Consider a reaction of the type

$$\mathrm{A} + \mathrm{b} \rightarrow \mathrm{C} + \mathrm{d}$$

in which the target nucleus A interacts with the particle b and forms nucleus C plus particle d. Three (or more) body interactions have a much smaller probability of happening and can therefore be neglected to a first approximation[1].

We define a cross section σ as the number of reactions per target nucleus per unit time, divided by the number of incident particles per unit area per unit time. By denoting with v the relative velocity of species A and b, the number of reactions r is given by $r = v\sigma(v)N_A N_b$, but since particles in stellar gas have a non-negligible velocity distribution, we'll have to integrate over the whole allowed range of velocities

$$r = N_A N_b \int_0^\infty v\sigma(v) f(v)\,\mathrm{d}v = \frac{N_A N_b}{1+\delta_{Ab}} \langle \sigma v \rangle \tag{78}$$

where δ_{Ab} is a Kronecker delta. By recalling that the mass fraction of a generic element k is given by

$$X_k = \frac{N_k m_{\mathrm{H}A_k}}{\rho}$$

we get the following expression for the reaction rate per unit mass

$$R_{Ab} = \rho \frac{X_A X_b}{m_{\mathrm{H}}^2 A_A A_b} \frac{\langle \sigma v \rangle_{Ab}}{1+\delta_{Ab}} \tag{79}$$

and the nuclear energy generation coefficient can be therefore written as

$$\epsilon_N = R_{Ab} Q_{Ab} \tag{80}$$

with Q_{Ab} being the amount of energy released by a single reaction. The value of Q_{Ab} can be determined from the difference between the sum of the masses of the interacting particles and the sum of the masses of the products

$$Q = \sum_r m_r(c^2) - \sum_p m_p(c^2) \tag{81}$$

Often nuclear masses are expressed in atomic mass units, and [15] points out that is worth recalling that 931.494 MeV is the amount of energy associated with one mass unit ($1.6605 \cdot 10^{-24}$ g). Moreover, if a positron is produced by the nuclear reaction, it is customary to add to Q_{Ab} its annihilation energy, whereas, in the case neutrinos are produced, their energy has to be subtracted from the total energy production budget.

In order to determine ϵ_N we still have to discuss how $\langle \sigma v \rangle$ can be determined. Matter involved in nuclear reactions is well approximated by a fully ionize perfect gas, and the particle velocities and kinetic energies follow the well-known Maxwellian distribution.

1 We're going to have to consider at least one of those if we want to produce heavier elements.

If the individual particles have a maxwell velocity distribution, it is possible to show that their relative velocities v will also follow the same distribution. Since the corresponding kinetic energy is $E = \mu v^2/2$, where μ is the reduced mass of the interacting particles, the expression for $\langle \sigma v \rangle$ thus becomes

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E e^{-E/kT} \, \mathrm{d}E$$

but in order to complete the calculation we need further information about the dependence of $\sigma(E)$ on the energy E. This, of course, can be done by studying the process of thermonuclear fusion in more detail.

Whenever a pair of particles A and b interacts (that is, whenerver they come close enough that the attractive nuclear force tends to fuse them together) a compound nucleus C′ is formed, with atomic and mass number that is the sum of the respective numbers of its "progenitors".

Clearly, the nucleus C′ has no memory of how it was formed.

This compound nucleus is in a particular excited state for a given time τ, after which it'll break up in many possible ways, producing, among the various possibilities, the particles C and d.

The probability that our generic reaction $\mathrm{A} + \mathrm{b} \rightarrow \mathrm{C} + \mathrm{d}$ is really happening, depends on the product of the probability that the interacting particles come close enough to experience the effect of the attractive nuclear force, and the probability that this interaction produces the particles C and d.

In order to obtain the fusion of charged particles, they have to be able to get close enough that the short range forces dominate over the long range repulsive Coulombian forces. Nuclear attraction dominates for distances that are of the order of the nuclear size, so $\sim 10^{-13}$ cm. For larger distances, the repulsive electrostatic potential prevails

$$V_{\text{Coulomb}} = \frac{Z_A Z_b e^2}{r}$$

where Z_A, Z_b are the electric charges of A and b. Clearly, the electrostatic potential barrier can be overcome only if particles have a kinetic energy of the same order $kT \sim V_{\text{Coulomb}}$, which, for all but few cases, it's physically impossible.

Then how come nuclear reactions are indeed happening in the cores of stars?

As was investigated by G. Gamow [7], there is a small but quite finite probability of 'tunnelling' through the Coulombian potential barrier, even for particles of energy lower than V_{Coulomb}. It can be shown that the probability of tunnelling, in terms of the energy E of the relative motion of the colliding particles is

$$P_G = p_0 e^{-b/E^{1/2}} \quad b = \left(\frac{\mu}{2} \right)^{1/2} \left(\frac{2\pi Z_A Z_b e^2}{\hbar} \right) \tag{82}$$

Through this miracle, nuclear fusion can happen at temperatures much lower than predicted by classical physics.

At a given E the probability decreases for increasing Z_A, Z_b (the potential barrier gets higher), and progressively higher temperatures are needed for nuclear 'burnings' involving heavier nuclei.

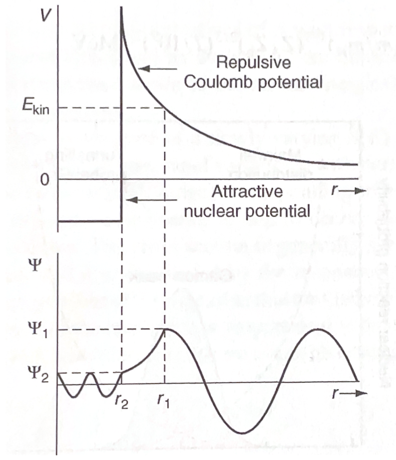

Figure 10: Illustration of the potential seen by a particle b when approaching a particle A with a kinetic energy E_{kin} and the corresponding wavefunction Ψ. Credits: [15].

In astrophysics it is customary to define the quantity

$$S(E) = \sigma(E) E e^{+b/E^{1/2}} \tag{83}$$

called *astrophysical factor*, that is mainly sensitive to the nuclear contribution to the cross section. We may now rewrite the equation for $\langle \sigma v \rangle$ by including $S(E)$

$$\langle \sigma v \rangle = \left(\frac{8}{\mu \pi} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{kT} - \frac{b}{\sqrt{E}} \right) \mathrm{d}E \tag{84}$$

In many cases of astrophysical relevance, the so-called non-resonant reactions, $S(E)$ is a slowly varying function of E in the range of interest. Note how in (84) the first term of the exponential represents the high energy wing of the Maxwellian distribution, while the second one represents the energy dependence of the tunnelling probability. Sometimes a scale energy E_{Gamow} is defined, so that $b^2 = E_{\text{Gamow}}$.

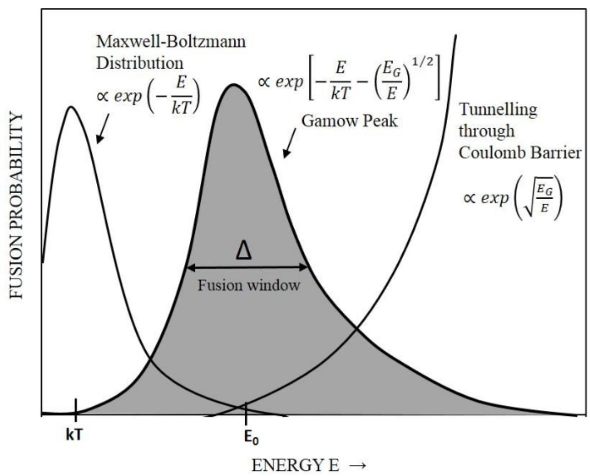

Figure 11: Illustration of the location and shape of the Gamow peak (not to scale) as compared with $e^{-E/kT}$ and $e^{-b/\sqrt{E}}$.

Their product, shown in Fig.11, produces a strongly peaked curve, with a maximum at the so-called *Gamow peak*, which also represents the most efficience energy for the nuclear reaction to occur.

The energy E_0 of the peak is

$$E_0 = \left(\frac{bkT}{2}\right)^{2/3} \approx 0.122 \left(\frac{\mu}{m_\mathrm{H}}\right)^{1/3} (Z_A Z_b)^{2/3} \left(\frac{T}{10^9\,\mathrm{K}}\right)^{2/3} \mathrm{MeV} \tag{85}$$

and the FWHM ΔE_0 is given by

$$\Delta E_0 \sim 0.237 \left(\frac{Z_A^2 Z_b^2 \mu}{m_\mathrm{H}}\right)^{1/6} \left(\frac{T}{10^9\,\mathrm{K}}\right)^{5/6} \mathrm{MeV} \tag{86}$$

It is in the Gamow peak energy range that S(E) needs to be known in order to derive the reaction rate. Unfortunately, this energy range is at least a factor of 10 lower than the energies at which stellar reactions can be produced in laboratories; therefore, all we're able to do is extrapolate the measured $S(E)$ values to much lower energies if an empirical determination of $S(E)$ is sought.

As mentioned somewhere in the above, the astrophysical factor is often a slowly varying function of energy, and (84) is usually solved considering a second-order Taylor expansion for S and $e^{-b/\sqrt{E}}$ around E_0. If we approximate the exponential term as a Gaussian function[2]

$$I_\mathrm{max} \approx \exp\left(-\frac{E}{kT} - \sqrt{\frac{E_G}{E}}\right) \implies I_\mathrm{max} \approx I(E_0)$$

meaning

$$I_\mathrm{max} = \exp\left(-\frac{3E_0}{kT}\right) = \exp(-\tau)$$

This way we can write

$$I_\mathrm{max} = e^{-\tau} \exp\left[-\left(\frac{E-E_0}{\Delta/2}\right)^2\right] \quad \Delta = \mathrm{FWHM}$$

Once plugged in (84), this provides

$$\langle\sigma(v)v\rangle = \sqrt{\frac{2}{\mu}} \frac{\Delta}{(kT)^{3/2}} S_\mathrm{eff}\, e^{-3E_0/kT} \tag{87}$$

with

$$\frac{S_\mathrm{eff}}{S(E_0)} = 1 + \frac{5kT}{36E_0} + \frac{S'(E_0)}{S(E_0)}\left(E_0 + \frac{35}{36}kT\right) + \frac{1}{2}\frac{S''(E_0)}{S(E_0)}\left(E_0^2 + \frac{89}{36}kTE_0\right) \tag{88}$$

The values for the astrophysical factor and its derivaties are obtained either from laboratory measurements (with the aforementioned limitations) or from theoretical quantum mechanical computations.

2 It is actually not a bad approximation, but the curve is slightly skewed, so we'll have to make some corrections afterwards.

Let's see how can we deduce the (approximate) temperature dependence. Say T_0 a reference temperature, which may be the temperature for which the burning of a particular element is possible.

We've shown that in general, the energy at Gamow's peak is dependent on the temperature

$$E_0 = \left(\frac{bkT}{2}\right)^{2/3} \qquad b = \frac{(2\mu)^{1/2}\pi e^2 Z_A Z_b}{\hbar}$$

Let us expand (84) around our reference temperature in the way presented below

$$\langle \sigma v\rangle_T \approx \langle \sigma v\rangle_{T_0}\left(\frac{T}{T_0}\right)^{\nu}$$

Taking the logarithms of both memebers

$$\ln(\langle \sigma v\rangle_T) \approx \ln(\langle \sigma v\rangle_{T_0}) + \nu \ln\left(\frac{T}{T_0}\right)$$

hence

$$\nu \approx \frac{\ln(\langle \sigma v\rangle_T) - \ln(\langle \sigma v\rangle_{T_0})}{\ln T - \ln T_0} = \frac{\mathrm{d}\ln\langle \sigma v\rangle}{\mathrm{d}\ln T}\Big|_{T=T_0}$$

Let us express this in terms of the variable $\tau = 3E_0/kT$ we've used earlier when expanding the exponential term in (84).

You can show that

$$\tau \propto T^{-1/3} \implies T \propto \tau^{-3} \implies \ln T = C - 3\ln\tau$$

while for $\langle \sigma v\rangle_T$ we use (87). Let's explicit the various dependencies

$$\langle \sigma v\rangle_T \propto \frac{\Delta}{T^{3/2}}e^{-\tau} \qquad \Delta \propto T^{5/6}$$

which, after putting all together, implies

$$\langle \sigma v\rangle_T \propto T^{-2/3}e^{-\tau} \propto \tau^2 e^{-\tau} \implies \ln\langle \sigma v\rangle_T \propto D + 2\ln\tau - \tau$$

Now we can finally plug everything in the definition of ν. Calculating

$$\mathrm{d}\ln\langle \sigma v\rangle_T = \mathrm{d}\tau\left(\frac{2}{\tau} - 1\right) \qquad \mathrm{d}\ln T = -\frac{3}{\tau}\mathrm{d}\tau$$

$$\nu = \frac{\mathrm{d}\ln\langle \sigma v\rangle}{\mathrm{d}\ln T}\Big|_{T=T_0} = -\frac{2}{3} + \frac{\tau}{3}$$

meaning that

$$\boxed{\langle \sigma v\rangle \propto T^{-\frac{1}{3}(\tau_0 - 2)}} \tag{89}$$

For example, consider a reaction from the ppI chain

$$p + p \to D + e^+ + \bar{\nu}_e \implies \langle \sigma v\rangle \propto T^4$$

4.2.1 Resonant Reactions

So far we've been focusing on reactions for which the astrophysical factor (83) was slowly varying with energy, the so-called *non-resonant reactions*. We've mentioned before that given two reactants, multiple products are possible, depending on the nature of the interaction (elastic, anelastic, etc).

$$\begin{aligned} \mathrm{A}+\mathrm{b} \to \mathrm{C}^* &\to \mathrm{A}+\mathrm{b} \text{ (elastic scattering)} \\ &\to \mathrm{A}+\mathrm{b}^* \text{ (anelastic scattering)} \\ &\to \mathrm{D}+\mathrm{e} \\ &\to \mathrm{C}+\gamma \text{ (radiative capture)} \end{aligned}$$

Note that if the energy of the center of mass added to the Q-value of the specific reaction is "close enough" to one of the energetic level of the compound nucleus C^*, then the reaction is *resonant*. We'll consider only *narrow* resonant reactions, for which the width of the resonance is much smaller than the resonant energy

$$\Gamma \ll E_R \tag{90}$$

It should come as no surprise that, around resonance, the interaction cross section becomes extremely larger, and it's no more the one we've found in (84), but it's rather the one found by Breit and Wigner

$$\sigma_{BW}(E) = \frac{\lambda^2}{4\pi} \frac{\omega\Gamma_1\Gamma_2(1+\delta_{Ab})}{(E-E_R)^2 + \left(\frac{\Gamma}{2}\right)^2} \tag{91}$$

where λ is the De-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2\mu E}} \tag{92}$$

and Γ_1 is the probability amplitude of forming the compound nucleus, which is also equal to its decay probability; Γ_2 is the probability amplitude of decaying in the products of the reaction (say D, e), while Γ is the width of the excited level responsible of the resonance. Note that Γ is proportional to the total lifetime τ of the excited state according to

$$\Gamma = \frac{h}{2\pi\tau} \tag{93}$$

and Γ_1, Γ_2 have a similar definition, but considering the lifetime for the decay of the compound nucleus into the original interacting particles A+b, and into the products D+e, respectively. In teh case of heavy compound nuclei many resonances are present in the range of stellar energies.

On the other hand, the factor ω is related to the spin of the reactants and to the (total) angular momentum of the compound nucleus

$$\omega = \frac{2J+1}{(2J_A+1)(2J_b+1)}$$

At this point we can write anew the reaction rate (78), but with a few caveats. First of all, now we're in the presence of a narrow resonance, meaning that if the Gamow peak E_0 is far from resonance E_R, we won't be able to observe the resonance.

On the other hand, if $E_R \sim E_0$, the resonance overrides all the other possible decay products, and thus we have to use (91) rather than the tunnelling probability.

We still have to average over the velocity distribution

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{BW}(E) E e^{-E/kT} \, \mathrm{d}E \tag{94}$$

But since the resonance is narrow around E_R and, for extension, around E_0, we can bring the Maxwellian contribution out of the integral, evaluating them in E_R

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} E_R e^{-E_R/kT} \int_0^\infty \sigma_{BW}(E) \, \mathrm{d}E$$

This integral can be calculated, thus finding

$$\langle \sigma v \rangle = \left(\frac{2\pi}{\mu k T} \right)^{3/2} \hbar^2 \omega \gamma e^{-E_R/kT} \tag{95}$$

where $\gamma = \Gamma_1 \Gamma_2 / \Gamma$. Unlike narrow resonances, *broad resonances* are more troublesome, since they have broad tails that can influence the interaction cross section even when the reaction is well out of resonance conditions.

4.2.2 Electronic Shielding

It is important to notice that in our previous discussion and the cross sections provided in the literature refer to reactions *happening in vacuum*. Instead, as I'm sure the sharpest among my four readers had been itching to point out for a while, ions in stellar interiors are surrounded by a sea of free electrons that tend to cluster near the nuclei and reduce their Coulomb potential.

Therefore an approaching particle will feel a Coulombian potential different from the case of an isolated positive charge. This shielding effect of the electrons increases the reaction rate $\langle \sigma v \rangle$ by a so-called *screening factor* f that will necessarily be a function of the matter density.

Consider the upper cartoon of Fig.10, and assume that the impinging particle on our target nucleus has an energy much lower than the Coulombian potential. The poor thing would not be able to interact effectively with the target nucleus and will "stop" at a distance known as *classical turning point*, which is easily computed

$$r_{TP} = \frac{Z_A Z_b e^2}{E_b} \tag{96}$$

In reality, however, the target nucleus has some electrons, that will be located at a certain distance from the nucleus. Within the electronic cloud, the Coulombian potential vanishes.

Then, if our impinging projectile had an energy such that $r_{TP} > r_A$, where r_A is the atomic radius, the resulting effect would be incredible since, because of the electronic shielding, the effective electrostatic potential would be identically zero.

Let us define an energy

$$U_e = \frac{Z_A Z_b e^2}{r_A}$$

so that if $E_b > U_e$, the projectile can break into the electronic cloud. This is indeed what happens, since U_e is rather small ($\sim 0.03\,\text{keV}$). If that is the case, then necessarily $r_{TP} < r_A$.

So how does the electrostatic potential changes? We can imagine the cloud as a symmetric source of potential around Z_A, so that

$$\Phi_A = -\frac{Z_A e}{r_A}$$

Therefore the total potential within the cloud decreases a bit

$$\Phi_{\text{tot}} = \frac{Z_A e}{r} - \frac{Z_A e}{r_A}$$

but we're interested in evaluating it at distances comparable to the range of nuclear forces, where our nuclear fusion interactions can take place.

This has the effect of lowering the Coulombian barrier

$$V_{\text{eff}} = \frac{Z_A Z_b e^2}{r_N} - \frac{Z_A Z_b e^2}{r_A}$$

The correction we've introduced is rather small, since $r_N/r_A \sim 10^{-5}$. In a sense, we could consider this effect as an increased energy of the impinging projectile $E_b \to E_b + U_e$. This clearly induces a slight increasing of the reaction rate by a factor f.

The shielding factor f is defined as

$$f = \frac{\sigma(E_b + U_e)}{\sigma(E_b)} \tag{97}$$

Albeit small, this effect is not at all negligible. From the definition (97) we can substitute the expressions for the cross sections (I'm dropping the b-subscript)

$$\frac{\sigma(E+U_e)}{\sigma(E)} = \frac{E}{E+U_e}\frac{S(E+U_e)}{S(e)}\exp\left(-\frac{2\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2(E+U_e)}} + \frac{2\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2E}}\right)$$

which, since $U_e \ll E$, we can Taylor-expand. First of all we can notice that the first two ratios are, at linear order, of order unity. Then, expanding the square root, we obtain

$$f \approx \exp\left(\frac{\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2E}}\frac{U_e}{E}\right)$$

which becomes more and more relevant for lower energies of the center of mass, which are those of astrophysical relevance, since the astrophysical factor starts to climb up back again.

To the reaction rates we've been calculating so far we have to add a correction to address the effect of electronic shielding in the stellar plasma. The idea is that, although the stellar plasma is globally neutral, whenever a impinging projectile comes close to an ionized nucleus, the two charged nuclei start repelling each other, whilst electrons tend to be attracted.

If that's the case, then we can imagine our stellar plasma as made of a myriad of charged nuclei that are getting "shielded" by electronic clouds at a certain distance from them, so that when a nucleus interact with another nucleus, the electronic cloud, which is just an electron over-density, trails behind it.

Say that n_i is the numeric density of the ions

$$n_i \approx \frac{\rho}{\mu_i m_\mathrm{H}} \approx \frac{1}{\frac{4\pi}{3}a^3}$$

with a the average distance between ions. We can invert the latter expression

$$a \sim \frac{1}{n_i^{1/3}} \sim \left(\frac{\mu_i n_\mathrm{H}}{\rho}\right)^{1/3}$$

In the inner regions of the stellar plasma, the density is quite large, meaning that the distance between two ions gets smaller.

We can define *strong shielding* the condition for which

$$\frac{Z_1 Z_2 e^2}{akT} \gg 1 \tag{98}$$

In such a scenario we can imagine that ions are so far away from each other that there's effectively only one ion per electronic cloud. This is a condition that can be verifying only at very high densities, which are not those typical of the regions where nuclear reactions are active.

In stars, strong shielding is typical of regions where electrons are degenerate, but generally it's much more probable that, in regions of active fusion, the shielding is *weak* (which is exactly the opposite of the strong shielding we mentioned before) or an intermediate scenario that is known as *intermediate shielding*, which can be addressed only through numerical simulations.

Consider a *weak shielding* scenario, where the positions of ions is only slightly skewed with respect to the global absence of electric fields. Generally, the distance between the electronic cloud and a bare nucleus R is much larger than the distance among ions a.

In the surrounding of the target ion A with positive charge Z_A, there's going to be a non-zero electric field, which is going to induce a different distribution of ions and electrons from that that would be giving a vanishing net charge. In general we can write

$$\bar{n}_i \approx \frac{\rho X_i}{A_i m_\mathrm{H}} \rightarrow \bar{n}_e = \sum_i Z_i \bar{n}_i$$

assuming, for simiplicity, complete ionization in a region of active fusion. On average we know that

$$\Phi = 0 \quad \bar{\rho} = \sum_i Z_i e \bar{n}_i - e\bar{n}_e = 0$$

Instead, in the surrounding of each charged nucleus, there is a small, but non vanishing, electrostatic potential. Statistical mechanics tells us that

$$n_i = \bar{n}_i \exp\left(-\frac{Z_i e\Phi}{kT}\right) \approx \bar{n}_i \left(1 - \frac{Z_i e\Phi}{kT}\right)$$
$$n_e = \bar{n}_e \exp\left(\frac{e\Phi}{kT}\right) \approx \bar{n}_e \left(1 + \frac{e\Phi}{kT}\right)$$

Note that, despite the deviation from neutrality, the charge displacement is very small, so that the new numeric density of charges is not that much different from the one that grants neutrality.

The total electronic density is now

$$\rho = \sum_i Z_i e\bar{n}_i - e\bar{n}_e = \sum_i \left(-\frac{Z_i^2 e^2 \Phi \bar{n}_i}{kT} - \frac{e^2 \Phi \bar{n}_e}{kT}\right)$$

where we've used the global neutrality condition. Let's manipulate this last expression a bit so to extract the distance of the electronic cloud (*Debye radius*)

$$n = \sum_i (\bar{n}_e + \bar{n}_i) \quad \text{at } \Phi = 0$$

and we introduce the quantity χ classically defined as

$$\chi = \frac{\sum_i Z_i^2 \bar{n}_i + n_e}{n}$$

Substituting what we've just found we obtain

$$\rho = -\frac{\chi e^2 \Phi}{kT} n$$

Now we'd like to understand how the electrostatic potential changes nearby the charged ion, so to determine how the electronic cloud is distributed. Recall Poisson's equation

$$\nabla^2 \Phi = -4\pi\rho \implies \frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = -4\pi\rho$$

We can plug in the charge density ρ

$$\frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = 4\pi\frac{\chi e^2 \Phi}{kT} n$$

At last, we can define the *Debye radius*

$$r_D = \left(\frac{kT}{4\pi\chi n e^2}\right)^{1/2} \tag{99}$$

so that our spherically symmetric Poisson's equation becomes

$$\frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = \frac{\Phi}{r_D}$$

which can be solved for the electrostatic potential Φ

$$\Phi = \frac{Z_i e}{r} \exp\left(-\frac{r}{r_D}\right)$$

If we expand it for $r/r_D \ll 1$ we find out that

$$\Phi \approx \frac{Z_i e}{r} - \frac{Z_i e}{r - D} \tag{100}$$

which can be interpreted as the potential generated by a charged ion corrected for the presence of an electronic charge at distance r_D. We can also define a *Debye energy*

$$E_D = \frac{Z_i Z_j e^2}{r_D} \tag{101}$$

which is the limit energy for which the impinging projectile wouldn't be able to penetrate into the electronic cloud. Note that since the Debye radius is quite large, the associated energy is $E_D \ll E_0$. In "normal" stars[3] the Debye radius is much larger than

$$E_D = \frac{Z_i Z_j e^2}{r_D} \ll \frac{Z_i Z_j e^2}{r_0} := E_0$$

For example, for the Sun, at its center, the Debye radius is $r_D \approx 10^{-9} - 10^{-8}$ cm, while the ratio $r_D/r_0 \sim 50 - 100$.

As we've done before, we have to calculate the shielding factor and the associated $\langle \sigma v \rangle_S$

$$\langle \sigma v \rangle_S = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{ns}(E + E_D) E e^{-E/kT} \, \mathrm{d}E$$

which we can integrate changing variable to $E' = E + E_D$

$$\langle \sigma v \rangle_S = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{ns}(E')(E' - E_D) e^{-(E' - E_D)/kT} \, \mathrm{d}E'$$

Now, since $E_D \ll E_0$, we can neglect its contribution coming from the factor $E' - E_D$, thus leaving us with

$$\langle \sigma v \rangle_S \approx \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} e^{+E_D/kT} \int_0^\infty \sigma_{ns}(E') E' e^{-E'/kT} \, \mathrm{d}E'$$

where we can easily recognize the shielding factor f, which comprises essentially of the exponential

$$f = e^{E_D(kT)} \sim 1 \tag{102}$$

so that the shielded cross section is just a small correction on the non-shielded cross section

$$\langle \sigma v \rangle_S = f \langle \sigma v \rangle_{NS}$$

For typical solar values, given the Debye radius we've found earlier, we'd obtain a shielding factor for the pp reaction of order $f \sim 1.03$.

---

3 Read: non-strongly degenerate stars as white dwarves or neutron stars.

4.3 FUSION REACTIONS

Before we even begin looking into what happens when a star enters the longest stage of their evolution cycle, the main sequence, we'll consider briefly the reactions occurring in the phase known as *pre-main sequence*.

As you should already know a star, whose nuclear reactions have yet to start, will balance the energy loss through their intrinsic luminosity depleting gravitational energy. This means that, as the star shrinks, it heats up, thus radiating away more energy and cooling down, ready to repeat the cycle over and over.

When temperature gets high enough ($\sim 10^6\,\mathrm{K}$), the combustion of the so-called *light elements* starts. The first that starts burning is Deuterion, which is already scarce ($X^{\text{in}}_{D,\odot} \sim 2 - 4 \cdot 10^{-5}$) and gets burned quickly

$$ {}^2_1\mathrm{D} + ^1_1 \mathrm{p} \rightarrow ^3_2 \mathrm{He} + \gamma $$

For temperatures just a little higher (respectively $2 \cdot 10^6\,\mathrm{K}$ and $2.5 \cdot 10^6\,\mathrm{K}$), two other reactions occur

$$ \begin{aligned} &{}^6_3\mathrm{Li} + ^1_1 \mathrm{p} \rightarrow ^3_2 \mathrm{He} + ^4_2 \mathrm{He} \\ &{}^7_3\mathrm{Li} + ^1_1 \mathrm{p} \rightarrow 2\,^4_2\mathrm{He} \end{aligned} $$

Clearly, the ignition temperatures we've given are only approximative since they actually depend on the element abundance and density of the star. One thing in particular is curious about ${}^7_3\mathrm{Li}$. Only 1% of the abundance we observe in a Sun-like star (which is already not a lot, $X_{^7\mathrm{Li}} \sim 10^{-8}$) is produced during the BBN (*Big Bang Nucleosynthesis*). On average it is only burned in stars, while it's produced through *spallation on nuclei by cosmic rays*

$$ {}^4_2\mathrm{He} + ^{12}_6 \mathrm{C} \rightarrow ^7_3 \mathrm{Li} + 2\,^4_2\mathrm{He} + ^1_1 \mathrm{p} $$

On top of this other reactions involving Berillium and Boron can be occurring, if temperatures get high enough. For Berillium, assuming a temperature of at least $T \sim 3.5 \cdot 10^6\,\mathrm{K}$

$$ \begin{aligned} {}^9_4\mathrm{Be} + ^1_1 \mathrm{p} &\rightarrow ^4_2 \mathrm{He} + ^6_3 \mathrm{Li} \\ &\rightarrow ^8_4 \mathrm{Be} + ^2_1 \mathrm{D} \rightarrow 2\,^4_2\mathrm{He} + ^2_1 \mathrm{D} \end{aligned} $$

the last reaction in particular is strongly unstable, and ${}^8_4\mathrm{Be}$ immediately decays ($\tau \sim 10^{-16}\,\mathrm{s}$) with a double-$\alpha$ decay. Notice that normally ${}^9_4\mathrm{Be}$ is produced, for example, from reactions involving heavier elements

$$ {}^1_1\mathrm{p} + ^{14}_7 \mathrm{N} \rightarrow ^9_4 \mathrm{Be} + ^4_2 \mathrm{He} + 2\,^1_1\mathrm{p} $$

and has an overall initial abundance of $X_{^9\mathrm{Be}} \sim 10^{-10}$.

We've got reactions for the isotopes of Boron as well ($T \sim 4.5 \cdot 10^6\,\mathrm{K}$, $X_{^{11}\mathrm{B}} \sim 5 \cdot 10^{-9}$)

$$ \begin{aligned} &{}^{11}_5\mathrm{B} + ^1_1 \mathrm{p} \rightarrow 3\,^4_2\mathrm{He} \\ &{}^{10}_5\mathrm{B} + ^1_1 \mathrm{p} \rightarrow ^4_2 \mathrm{He} + ^7_4 \mathrm{Be} \end{aligned} $$

It's curious to know that the abundance of ${}^{11}\mathrm{B}$ is four larger than ${}^{10}\mathrm{B}$.

During the star's evolution along the Hayashi track (see Chapter 5), the star increases its temperature due to the virial theorem, a radiative core will form at the center and grow in mass as the star evolves. When the radiative core appears, the star is no longer fully convective; this happens when

$$\nabla_{\text{rad}} < \nabla_{\text{amb}}$$

and the star will move out of the Hayashi track. The reactions we've presented are not relevant from the point of view of energy production, not even in this stage, but reduce practically to zero the abundances the abundances of the light elements involved (and do not appreciably affect the abundances of the elements they produce).

If the stars were fully convective during this stage, we would have a complete depletion of Lithium and Deuterium at the surface, since matter in a convective star is always well mixed. However, as we've mentioned en passant, the convective region will start to shrink, eventually; if the convective region is confined to the external and cooler region before all the ***insert light element here*** reservoir bas been destroyed in fully convective phase, some remaining nuclei will still be present in the convective envelope.

This means that the resulting surface abundances of these light elements depend on the interplay between the increase of the stellar temperature, the resulting efficiency of the nuclear reactions involed and the timescale for the shrinking of the convective envelope.

As a rule of thumb, we can say that, once the chemical composition is fixed, larger masses will have lower surface depletion of light elements, and the depletion increases along with the star's metallicity.

If this is the case, then why ${}^{6}_{3}\text{Li}$ is only 1% as abundant as theory predicts for Sun-like stars? Good question, we don't know. Apparently, it just continues to get burned also when the star enters the main sequence.

4.3.1 Hydrogen Burning: pp-chain

The H-burning mechanism is essentially the nuclear fusion of four protons into one ${}^{4}_{2}\text{He}$ nucleus, garnished with two positrons and two electronic neutrinos.

The energy production for the whole set of reaction we'll be going through in a moment can be shown to amount to 27.7 MeV, only decreasing to 26.1 MeV if we consider the energy subtracted by neutrinos. In any case, this amount of energy is still 10 times larger than that produced in any other nuclear reaction process occurring in stars.

Assuming burning of all protons into Helium you'd produce (in a Sun-like star) $E \sim 6.5 \cdot 10^{57}\,\text{MeV}$. Since the luminosity of the Sun is $L_{\odot} \sim 2.5 \cdot 10^{39}\,\text{MeV}\,\text{s}^{-1}$, you'd obtain a timescale for nuclear burning of $\tau_{\text{nuc}} \sim 10^{11}\,\text{yr}$. Accounting for the fact that only 10% of the total Hydrogen is burned in the center, you get a more realistic timescale of $\tau_{\text{nuc}} \sim 10^{10}\,\text{yr}$.

The main channel of energy production in this phase is (for less-massive stars) the *p-p chain*.

The reaction networks involved in the so-called p-p chain are the following:

ppI ($T \sim 8 \cdot 10^6\,\mathrm{K}$)
$^1_1\mathrm{p} +^1_1 \mathrm{p} \rightarrow^2_1 \mathrm{D} + e^+ + \nu_e$
$^2_1\mathrm{D} +^1_1 \mathrm{p} \rightarrow^3_2 \mathrm{He} + \gamma$
$^3_2\mathrm{He} +^3_2 \mathrm{He} \rightarrow^4_2 \mathrm{He} + 2\,^1_1\mathrm{p}$

ppII ($T \sim 15 \cdot 10^6\,\mathrm{K}$)
$^3_2\mathrm{He} +^4_2 \mathrm{He} \rightarrow^7_4 \mathrm{Be} + \gamma$
$^7_4\mathrm{Be} + e^- \rightarrow^7_3 \mathrm{Li} + \nu_e$
$^7_3\mathrm{Li} +^1_1 \mathrm{p} \rightarrow 2\,^4_2\mathrm{He}$

ppIII ($T \sim 20 \cdot 10^6\,\mathrm{K}$)
$^3_2\mathrm{He} +^4_2 \mathrm{He} \rightarrow^7_4 \mathrm{Be} + \gamma$
$^7_4\mathrm{Be} +^1_1 \mathrm{p} \rightarrow^8_5 \mathrm{B} + \gamma$
$^8_5\mathrm{B} \rightarrow^8_4 \mathrm{Be} + e^+ + \nu_e \rightarrow 2\,^4_2\mathrm{He} + e^+ + \nu_e$

He-p
$^3_2\mathrm{He} +^1_1 \mathrm{p} \rightarrow^4_2 \mathrm{He} + e^+ + \nu_e$

Although it seems rather hard to digest at first sight, it's actually much simpler than it looks.

First of all, we know that these reactions can't possibly involve free neutrons, for there aren't any: Those few that had remained after the BBN have now decayed (10^{10} yr is much larger than $\sim 10\,\mathrm{min}$, isn't it?), so the only possiblity for our star to produce Deuterium is through Hydrogen burning, with a branching fraction $\Gamma_{\mathrm{pp}} \sim 99.8\%$[4].

The four reactions that we've presented are the four main ways $^3_2\mathrm{He}$ can be used to eventually produce $^4_2\mathrm{He}$. The first reaction in the ppI chain requires that protons experience a β^+ decay, and can only occur if the protons are brought together by a nuclear collision. This process is governed by *weak* interactions and therefore has a low probability of occurring: Its nuclear cross section is in fact quite low ($\sigma_{\mathrm{pp}} \sim 10^{-23}\,\mathrm{b}$).

Until a higher temperature, roughly $8 \cdot 10^6\,\mathrm{K}$, is reached, the reactions producing $^3_2\mathrm{He}$ are more frequent than those consuming it, and as a consequence the abundance of $^3_2\mathrm{He}$ increases, only decreasing when successive reactions become effective. In a short time this element reaches an "equilibrium" abundance, which can be as high as earning it the title of *pseudo-primary element*.

The relative frequency of the ppII and ppIII depends strongly on temperature; in particular, the $^3_2\mathrm{He} +^4_2 \mathrm{He}$ reaction becomes competitive with respect to $^3_2\mathrm{He} +^3_2 \mathrm{He}$ for $T \approx 15 \cdot 10^6\,\mathrm{K}$. The reason for it lays mainly in (82) and its dependence on the reduced mass μ: the reduced mass of $^3_2\mathrm{He} +^4_2 \mathrm{He}$ is larger

4 Actually, there's another reaction able to produce Deuterium, which is $2\,^1_1\mathrm{p} + e^- \rightarrow^2_1 \mathrm{D} + \nu_e$, which, you'll reckon, is quite unlikely to happen, since it's a three-body reaction on top of being mediated by electroweak interactions. The associated branching fraction is $\Gamma_{\mathrm{pep}} \sim 0.2\%$

than that of ${}^3_2\mathrm{He} + {}^3_2\mathrm{He}$, meaning that the probability of tunnelling through the potential barrier is higher for the latter.

As a rule of thumb, you may say that with increasing temperature the importance of the ppII and ppIII increases with respect to the ppI if there is a sufficient concentration of ${}^4_2\mathrm{He}$.

Key informations about our favorite energy production chains are summarized in Table 1.

	ppI	ppII	ppIII	He-p
Γ	$\sim 85\%$	$\sim 15\%$	$\sim 0.1\%$	$\sim 0.0001\%$
$T\,[K]$	$\sim 8 \cdot 10^6$	$\sim 15 \cdot 10^6\,\mathrm{K}$	$\sim 20 \cdot 10^6\,\mathrm{K}$	-
K.R.	Electroweak	Electronic capture	2α	Electroweak

Table 1: Table reporting key informations about the p-p energy production chain. "K.R." stands for Key Reaction, not the most significant, but rather the one that differentiate each chain from the others.

As per what we were saying in the above, the branching ratios for each reaction will be extremely sensitive to temperature, so the values reported in the table are to be considered for Sun-like stars currently burning Hydrogen in their core.

Note that neutrinos produced by different reactions will have, naturally, different energies, meaning different spectra will be produced. We can try to calculate an average flux of neutrinos in the following way

$$n_\nu = 2\frac{L_\odot}{26.1\,\mathrm{MeV}} \approx 1.9 \cdot 10^{38}\,\mathrm{s}^{-1}$$

which we can convert in a flux we can measure on Earth simply dividing by an appropriate power of the Earth-Sun distance (2)

$$\phi_\nu \sim \frac{1.9 \cdot 10^{38}}{4\pi D^2} \approx 6.7 \cdot 10^{10}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

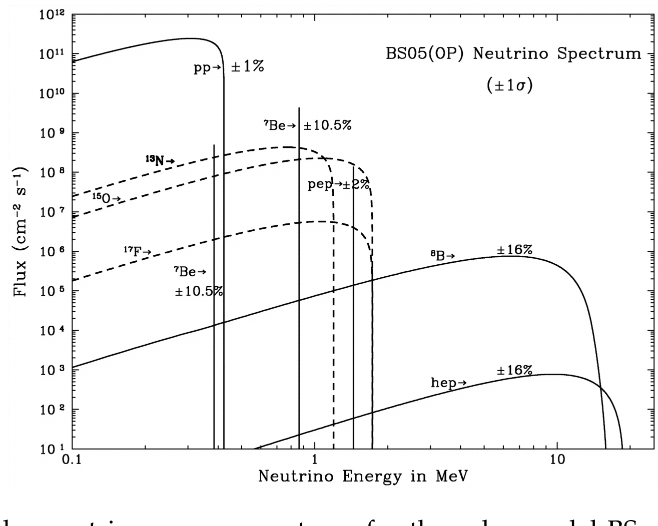

Figure 12: Solar neutrino energy spectrum for the solar model BS05(OP). The uncertainties are taken from Table 8 of [3].

The sharper amongst my readers will have surely noticed that there are some elements which are simultaneously involved in destruction and production processes (e.g., ${}^{2}_{1}\mathrm{D}$ or ${}^{3}_{2}\mathrm{He}$). These elements are usually called *secondary elements*.

It should come as no surprise that the variation of the abundance of element i will depend on all the creation and destruction reactions involving that element[5]. If you recall how we've defined the reactoin rate (78), we can then say that the variation over time of the abundance (or number of particles) of a certain species will be

$$\dot{N}_i = r_i^C - r_i^D$$

Let us consider for example, the equilibrium equation for Deuterion

$$\dot{N}_{\mathrm{D}} = \frac{N_p N_p}{2} \langle \sigma v \rangle_{pp} - N_{\mathrm{D}} N_p \langle \sigma v \rangle_{\mathrm{D}p}$$

Equilibrium will be thus reached when $\dot{N}_{\mathrm{D}} = 0$, so when

$$\frac{N_{\mathrm{D}}}{N_p} = \frac{\langle \sigma v \rangle_{pp}}{2 \langle \sigma v \rangle_{\mathrm{D}p}}$$

What do we learn from this? The equilibrium abundance is *directly proportional* to the creation rate and *inversely proportional* to the destruction rate.

It is evendent that if the abundance of ${}^{2}_{1}\mathrm{D}$ is larger than its abundance at equilibrium, then the destruction process prevails over the production one, and the chemical abundance will tend to evolve quickly towards its equilibrium configuration.

Under the assumption that $N_{\mathrm{D}} \gg N_{\mathrm{D,eq}}$ we can determine a timescale for the process, since the destruction process will be dominating

$$\frac{\mathrm{d}N_{\mathrm{D}}}{\mathrm{d}t} = -N_{\mathrm{D}} N_p \langle \sigma v \rangle_{\mathrm{D}p} \tag{103}$$

whose solution is of the type $N_{\mathrm{D}} = N_{\mathrm{D},0} \exp(-t/\tau)$, with

$$\tau = \frac{1}{N_p \langle \sigma v \rangle_{\mathrm{D}p}} \tag{104}$$

For typical temperatures at which the p-p chain is fully efficient, the value of τ is of the order of 1 s, meaning the equilibrium configuration is reached quite hastily.

From this we're learning also something else really important: With given initial conditions, the reaction chain can reach and (virtually) remain in equilibrium, but the reaction rate will be led by the slower reaction which (104) tells us is that with the smaller cross section.

In the p-p chain, the reaction telling the time to the others is the very first one, which has, as we've seen, an abysmally small cross section

$$\sigma_{\mathrm{pp}} \sim 10^{-23}\,\mathrm{b} \quad S(0)_{pp} \sim 4.1 \cdot 10^{-22}\,\mathrm{keV\,b^{-1}}$$

which, for example, we can compare to the very following reaction, the one involving deuterion, that has $S(0)_{\mathrm{D}p} \sim 2.1 \cdot 10^{-4}\,\mathrm{keV\,b^{-1}}$.

So there are 18 orders of magnitude of difference.

5 The general equation is quite long and convoluted, and I see no point into copying it here. The interested reader can refer to [15], §3.17, equation 3.39.

4.3.2 Hydrogen Burning: CN-NO

The CN-NO (bi)cycle is the combination of two independent cycles: The CN cycle and the NO cycle. The presence of some isotopes of Carbon, Nitrogen and Oxygen are necessary for either cycle to begin. Being both produced and destroyed during these cycles, those elements act as catalysts. The reaction networks involved are the following:

$$
\begin{aligned}
&\textbf{CN cycle}\\
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{13}_{7}\mathrm{N} + \gamma\\
&{}^{13}_{7}\mathrm{N} \rightarrow {}^{13}_{6}\mathrm{C} + e^{+} + \nu_e\\
&{}^{13}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{14}_{7}\mathrm{N} + \gamma\\
&{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma\\
&{}^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e\\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{16}_{8}\mathrm{O}^{*} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} \quad (\Gamma \sim 99\%)\\
&\qquad\qquad\qquad\qquad\ \rightarrow {}^{16}_{8}\mathrm{O} + \gamma \qquad (\Gamma \sim 1\%)
\end{aligned}
$$

This branch of the cycle has to follow the timescale of ${}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma$, which has the smaller cross section, followed by the first reaction of the cycle.

In the CN cycle the isotopes of C and N act as catalysts, and so behave as secondary elements. As a consequence, the cycle can start almost from any reaction if the involved isotope is present, and during a complete loop around the cycle the isotope is consumed and then produced again.

This, however, doesn't mean that the concentrations of the different isotopes will be unchanged as the final abundances depend strongly on the relative rates of the nuclear reactions in the cycle. Only at a high enough temperature ($T \approx 15 \cdot 10^6$ K) will the isotopes achieve their equilibrium abundance.

The branching ratio between the proton captures on ^{15}N, producing respectively ^{16}O and ^{12}C, is of the order of 10^{-4}, thus the amount of ^{16}O produced is practically negligible. Nevertheless, for slightly higher temperatures ($T \approx 20 \cdot 10^6$ K)the small production of ^{16}O unlocks a new branch for the cycle, one which transforms Oxygen into Nitrogen.

$$
\begin{aligned}
&\textbf{NO cycle}\\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{16}_{8}\mathrm{O}^{*} \rightarrow {}^{16}_{8}\mathrm{O} + \gamma \quad (\Gamma \sim 1\%)\\
&{}^{16}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{17}_{9}\mathrm{F} + \gamma\\
&{}^{17}_{9}\mathrm{F} \rightarrow {}^{17}_{8}\mathrm{O} + e^{+} + \nu_e\\
&{}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma\\
&{}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{18}_{9}\mathrm{F}^{*} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \quad (\Gamma \sim 99\%)\\
&\qquad\qquad\qquad\qquad\ \rightarrow {}^{18}_{9}\mathrm{F} + \gamma \qquad (\Gamma \sim 1\%)
\end{aligned}
$$

It should be noticed that for a typical PopI chemical composition (i.e., about solar) the global amount of CNO elements is of order of 75% of the total metallicity

$$X_{\mathrm{CNO}} \approx 0.75Z \quad \frac{X_{\mathrm{O}}}{X_{\mathrm{C}}} \approx 0.7$$

We know, however, that the total abundance has to be conserved (unless particular kinds of burnings are involved, that is)

$$\sum_i N_i(t=0) = \sum_i N_i(t)$$

meaning only the relative abundances between elements can be changing. In particular, we know that the slowest reaction of the CNO cycle is the $^{14}\mathrm{N}(p,\gamma)\,^{15}\mathrm{O}$, so the most abundant element in the CNO cycle processed material is indeed ^{14}N. As a rule of thumb, the change of the CNO element caused by the different burning channels is

- CN cycle processed matter: C$\downarrow$ N$\uparrow$
- CNO cycle processed matter: C$\downarrow$ N$\uparrow$ O$\downarrow$

Once the full CNO is operating, more than 95% of the cycle's elements are converted into ^{14}N, which also enables a new chain for unbound neutrons production.

At even higher temperatures, a new chain of reactions for processing Fluorine is unlocked

$$\begin{aligned}
&{}^{18}_{9}\mathrm{F} \rightarrow {}^{18}_{8}\,\mathrm{O} + e^{+}\nu_e \\
&{}^{18}_{8}\mathrm{O} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{19}_{9}\,\mathrm{F}^{*} \rightarrow {}^{15}_{7}\,\mathrm{N} + {}^{4}_{2}\,\mathrm{He} \\
&\qquad\qquad\qquad\qquad\quad \rightarrow {}^{19}_{9}\,\mathrm{F} + \gamma \\
&{}^{19}_{9}\mathrm{F} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{16}_{8}\,\mathrm{O} + {}^{4}_{2}\,\mathrm{He} \\
&\qquad\qquad\quad \rightarrow {}^{20}_{10}\,\mathrm{Ne} + \gamma
\end{aligned}$$

Until temperature raises even higher ($T \approx 25 \cdot 10^6\,\mathrm{K}$) there is a *leakage* on un-burned Neon from the star. Only in very massive MS-stars and during the AGB[6] does a star suffice the necessary conditions to activate the Neon-Sodium cycle

Ne-Na cycle

$$\begin{aligned}
&{}^{20}_{10}\mathrm{Ne} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{21}_{11}\,\mathrm{Na} + \gamma \\
&{}^{21}_{11}\mathrm{Na} \rightarrow {}^{21}_{10}\,\mathrm{Ne} + e^{+} + \nu_e \\
&{}^{21}_{10}\mathrm{Ne} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{22}_{11}\,\mathrm{Na} + \gamma \\
&{}^{22}_{11}\mathrm{Na} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{23}_{12}\,\mathrm{Mg} + \gamma \\
&{}^{23}_{12}\mathrm{Mg} \rightarrow {}^{23}_{11}\,\mathrm{Na} + e^{+} + \nu_e \\
&{}^{23}_{11}\mathrm{Na} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{24}_{13}\,\mathrm{Mg}^{*} \rightarrow {}^{20}_{10}\,\mathrm{Ne} + {}^{4}_{2}\,\mathrm{He} \\
&\qquad\qquad\qquad\qquad\qquad\quad \rightarrow {}^{24}_{12}\,\mathrm{Mg} + \gamma
\end{aligned}$$

which reflects into an increase of the abundance of Sodium and a decrease in the abundance of Oxygen, the so-called *Oxygen-Sodium anticorrelation*, which

6 *Asymptotic Giant Branch.*

is a distinctive characteristic of most globular clusters. At $T \approx 50 \cdot 10^6$ K, Magnesium is ignited, forming Aluminium

Mg-Al cycle

$$
\begin{aligned}
&{}^{24}_{12}\mathrm{Mg} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{25}_{13}\mathrm{Al} + \gamma \\
&{}^{25}_{13}\mathrm{Al} \rightarrow {}^{25}_{12}\mathrm{Mg} + e^{+} + \nu_e \\
&{}^{25}_{12}\mathrm{Mg} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{26}_{13}\mathrm{Al} + \gamma \\
&{}^{26}_{13}\mathrm{Al} \rightarrow {}^{26}_{12}\mathrm{Mg} + e^{+} + \nu_e \\
&{}^{26}_{12}\mathrm{Mg} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{27}_{13}\mathrm{Al} + \gamma
\end{aligned}
$$

Note, however, that when temperature raises over a given threshold, protonic captures and α-captures are competitive to β^+ decays. Two more protonic captures are given for isotopes of Aluminium

$$
\begin{aligned}
&{}^{26}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{27}_{14}\mathrm{Si} + \gamma \\
&{}^{27}_{14}\mathrm{Si} \rightarrow {}^{27}_{13}\mathrm{Al} + e^{+} + \nu_e \\
&{}^{27}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{28}_{14}\mathrm{Si}^{*} \rightarrow {}^{24}_{12}\mathrm{Mg} + {}^{4}_{2}\mathrm{He} \\
&\qquad\qquad\qquad\qquad\quad \rightarrow {}^{28}_{14}\mathrm{Si} + \gamma
\end{aligned}
$$

which reflects the so-called *Magnesium-Aluminium anticorrelation*. This property we're not able to see in all clusters, from which we'd able to infer, in principle, the presence of a contamination in the cluster.

This is a good point to stop for a second (and take a deep breath) and ask ourselves who's burning Hydrogen how.

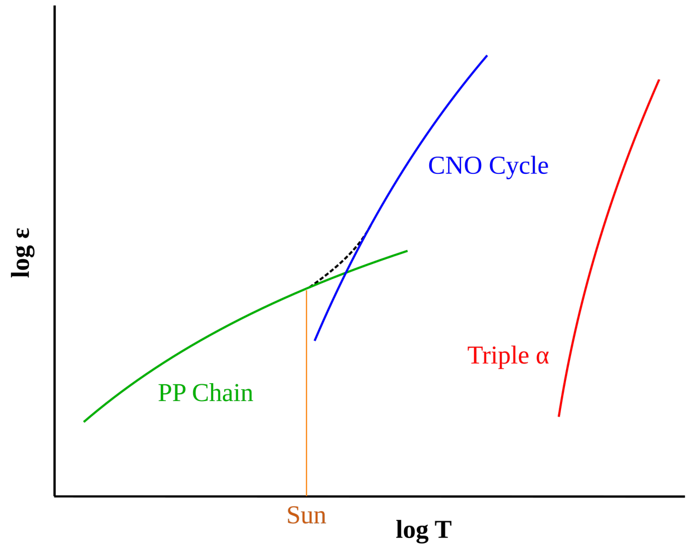

Figure 13: Logarithm of the relative energy output (ϵ) of proton-proton (p-p), CNO, and triple-α fusion processes at different temperatures. The dashed line shows the combined energy generation of the p-p and CNO processes within a star. Credits: Wikipedia.

It's possible to show that the energy production rate scales with $\sim T^4$ for the p-p chain, and with $\sim T^{17}$ for the CNO bi-cycle. When temperature gets over $\sim 20 \cdot 10^6$ K, the CNO takes over the p-p, as shown in Fig.13. At the end of the day, it all comes down to the mass of the star.

Lower Main Sequence stars (LMS), that have about solar chemical composition and are roughly in the range $M \lesssim 1.2 M_\odot$, will burn Hydrogen through the p-p chain, as will do *Very Low Mass Stars* (VLMS), which have masses $M \lesssim 0.3 M_\odot$ and are completely convective and very "cold".

Upper Main Sequence stars (UMS), that have masses $M \gtrsim 1.2 M_\odot$ will be burning Hydrogen through the CNO bi-cycle.

Note that as temperature climbs up to other heights[7], under conditions such as those found in novae and X-ray bursts, the rate of proton captures can exceed the rate of beta-decay, pushing the burning to the proton drip line.

The essential idea is that a radioactive species will capture a proton before it can beta decay, opening new nuclear burning pathways that are otherwise inaccessible. Because of the higher temperatures involved, these catalytic cycles are typically referred to as the *hot CNO cycles*. For example

HCNO I

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{13}_{7}\mathrm{N} + \gamma \\
&{}^{13}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{14}_{8}\mathrm{O} + \gamma \\
&{}^{14}_{8}\mathrm{O} \rightarrow {}^{14}_{7}\mathrm{N} + e^{+} + \nu_e \\
&{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma \\
&{}^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e \\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}
\end{aligned}
$$

4.3.3 Helium Burning

All stars with initial total mass larger than $\sim 0.5 M_\odot$ are able to attain, in their interiors, the thermal conditions required for the onset of Helium burning. Unlike during the Hydrogen burning, the physical process at work in stars of different masses during the main core He-burning stage are quite similar.

The first and fundamental reaction in the He-burning process is the production of $^{12}\mathrm{C}$ from the fusion of three Helium nuclei. This reaction is called the 3α reaction. Since it's a three-body encounter, it has very low probability, and this reaction usually occurs in two separate steps

$$
\begin{aligned}
&{}^{4}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{8}_{4}\mathrm{Be} \\
&{}^{8}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{12}_{6}\mathrm{C} + \gamma
\end{aligned}
$$

Also, temperature is now high enough to allow α-captures on Carbon nuclei, producing Oxygen

$$
{}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{16}_{8}\mathrm{O} + \gamma
$$

In a short time ($\sim 10^{-16}$ s), Berillium decays back into two α particles, thus making the probability of the second reaction extremely low. However, as the interior temperature climbs up, the probability of the second reaction increases, due to these combined effects:

7 Poetical delirium.

- the number of $\alpha + \alpha$ reactions increases, while the decay time for Berillium stands unchanged, thus significantly increasing the abundance of Berillium;
- the nuclear cross section of the second reaction strongly increases.

A temperature of about $\sim 100 \cdot 10^6$ K is necessary before 3α reactions start producing a sizeable amount of energy. Also, any increase in the density favors incredibly this reaction for

$$\epsilon_{3\alpha} \propto \rho^2 T^{41}$$

Even considering all of this, the measured abundances of Carbon and Oxygen in the Universe wouldn't be matching the results the reactions we've written down would bear, because they're still strongly inefficient reactions.

In 1954, Fred Hoyle [9] predicts that in order to achieve the observed abundances of Carbon and Oxygen, the 3α reaction must be *resonant*.

That was indeed the case: the α-capture on Berillium nuclei has a broad resonance at approximately 300 keV (roughly to $\approx 2 \cdot 10^8$ K), meaning that the reaction gets influenced from the "tail" of the resonance, which would also explain how so much Oxygen is produced[8]

$$\frac{\mathrm{C}}{\mathrm{Z}} \sim 0.20 \quad \frac{\mathrm{O}}{\mathrm{Z}} \sim 0.47 \quad \frac{\mathrm{N}}{\mathrm{Z}} \sim 0.06$$

Since small stars which can achieve Helium burning can continue to do so for millions of years, we've now found a way to circumvent the instability hidden into elements with $A = 5, 8$, which prevented the formation of heavier nuclei.

Note that since the luminosity is fixed only by the *mass* of the star and from its *evolutionary stage*, various reactions might be competitive in order to balance the total luminosity of the star. So, for example, the contribution to luminosity might be distributed between the 3α and the $^{12}\mathrm{C}(\alpha, \gamma)\,^{16}\mathrm{O}$

$$L_{\mathrm{tot}} = L_{3\alpha} + L_{\mathrm{C}-\alpha}$$

with the latter being slightly favored because it has a lower Helium consumption rate (only one α instead of 3), which allows the star to live longer.

Production of free neutrons

Other important reactions are given

$$\begin{aligned} {}^{16}_{8}\mathrm{O} + {}^{4}_{2}\mathrm{He} &\rightarrow {}^{20}_{10}\mathrm{Ne} + \gamma \\ {}^{20}_{10}\mathrm{Ne} + {}^{4}_{2}\mathrm{He} &\rightarrow {}^{24}_{12}\mathrm{Mg} + \gamma \\ {}^{24}_{12}\mathrm{Mg} + {}^{4}_{2}\mathrm{He} &\rightarrow {}^{28}_{14}\mathrm{Si} + \gamma \end{aligned}$$

8 Technically, Oxygen production is influenced by *three* resonant reactions.

but the most notable feature of Helium burning is that the star is now able to produce *free neutrons*. Consider for example

$$
\begin{aligned}
&{}^{14}_{7}\mathrm{N} + {}^{4}_{2}\,\mathrm{He} \rightarrow^{18}_{9}\,\mathrm{F} + \gamma\\
&{}^{18}_{9}\mathrm{F} \rightarrow^{18}_{8}\,\mathrm{O} + e^{+} + \nu_e\\
&{}^{18}_{8}\mathrm{O} + {}^{4}_{2}\,\mathrm{He} \rightarrow^{22}_{10}\,\mathrm{Ne} + \gamma\\
&{}^{22}_{10}\mathrm{Ne} + {}^{4}_{2}\,\mathrm{He} \rightarrow^{25}_{12}\,\mathrm{Mg} + +{}^{1}_{0}\mathrm{n}
\end{aligned}
$$

For stars of solar metallicity, in which $Z_{\mathrm{CNO}} \sim 0.01 Z_{\odot}$, we'd have, for example, a flux of $7 \cdot 10^{20}$neutrons g^{-1}, but temperatures of order $300 - 350 \cdot 10^{6}\,\mathrm{K}$ are needed. So, during the AGB phase, this chain of neutrons production is possible for $M \gtrsim 3M_{\odot}$ (and dominant for $M \gtrsim 5M_{\odot}$). Smaller stars produce free neutrons with a reaction involving Carbon

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\,\mathrm{p} \rightarrow^{13}_{7}\,\mathrm{N} + \gamma\\
&{}^{13}_{7}\mathrm{N} \rightarrow^{13}_{6}\,\mathrm{C} + e^{+} + \nu_e\\
&{}^{13}_{7}\mathrm{C} + {}^{4}_{2}\,\mathrm{He} \rightarrow^{17}_{O}\,\mathrm{O}^{*} \rightarrow^{16}_{8}\,\mathrm{O} + +{}^{1}_{0}\mathrm{n}
\end{aligned}
$$

that is completed only at temperatures of order $\approx 80 \cdot 10^{6}\,\mathrm{K}$, but not much higher since Helium burning would soon be starting.

Paradoxically, these reactions aren't easy to do not even during the AGB phase: Stars have the necessary temperatures, but, in those same regions, they don't have protons to spare. Other possible reactions involve the burning of heavier nuclei

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{12}_{6}\,\mathrm{C} \rightarrow^{23}_{12}\,\mathrm{Mg} + {}^{1}_{0}\,\mathrm{n}\\
&{}^{16}_{8}\mathrm{O} + {}^{16}_{8}\,\mathrm{O} \rightarrow^{31}_{16}\,\mathrm{S} + {}^{1}_{0}\,\mathrm{n}\\
&{}^{56}_{26}\mathrm{Fe} + \gamma \rightarrow 13\,{}^{4}_{2}\mathrm{He} + 4\,{}^{1}_{0}\mathrm{n}
\end{aligned}
$$

where the latter is only active moments before a star goes supernova and photons are able to disintegrate Iron nuclei.

4.3.4 Heavier Elements Burning

Moving on to even higher temperatures, it is possible to ignite the combustion of heavier nuclei, such as Carbon. Since a temperature of order $T \sim 700 - 800 \cdot 10^{6}\,\mathrm{K}$ is needed, many stars will never even get that far. The minimum mass to ignite Carbon is said M_{up} and is defined as such

$$
M_{\mathrm{up}} = \begin{cases} \sim 7M_{\odot} & \text{metal poor halo stars, } Z \sim 10^{-4}\\ \sim 7M_{\odot} & \text{metallic disk stars, } Z \sim 0.015\\ \sim 4M_{\odot} & \text{intermediate metallicity stars, } Z \sim \cdot 10^{-3} \end{cases}
$$

Not surprisingly, as progressively heavier nuclei are burned, more reactions products are possible. For Carbon

Carbon Burning

$$
\begin{aligned}
{}^{12}_{6}\mathrm{C} + {}^{12}_{6}\,\mathrm{C} \rightarrow^{24}_{12}\,\mathrm{Mg}^{*} &\rightarrow^{20}_{10}\,\mathrm{Ne} + {}^{4}_{2}\,\mathrm{He} \quad (\Gamma \sim 50\%)\\
&\rightarrow^{23}_{11}\,\mathrm{Na} + {}^{1}_{1}\,\mathrm{p} \quad (\Gamma \sim 50\%)\\
&\rightarrow^{23}_{12}\,\mathrm{Mg} + {}^{1}_{0}\,\mathrm{n} \quad (\text{rare})\\
&\rightarrow^{24}_{12}\,\mathrm{Mg} + \gamma \quad (\text{very rare})
\end{aligned}
$$

Before reaching the temperature for Oxygen burning ($T \sim 1 \cdot 10^9$ K), photons get energetic enough to photodisintegrate Neon

$$^{20}_{10}\text{Ne} + \gamma \leftrightarrows\ ^{16}_{8}\text{O} +^{4}_{2}\text{He}$$

with the right hand side (so, Oxygen) being more favored. At these temperatures, α-captures are also possible

$$\begin{aligned}
^{20}_{10}\text{Ne} +^{4}_{2}\text{He} &\to^{24}_{12}\text{Mg} + \gamma \\
^{24}_{12}\text{Mg} +^{4}_{2}\text{He} &\to^{28}_{14}\text{Si} + \gamma
\end{aligned}$$

Finally we reach the temperature needed for Oxygen burning ($T \sim 1.4 \cdot 10^9$ K, $M \gtrsim 10 M_\odot$). As expected, multiple channels are available this time as well

Oxygen Burning

$$\begin{aligned}
^{16}_{8}\text{O} +^{16}_{8}\text{O} \to^{32}_{16}\text{S}^* &\to^{28}_{14}\text{Si} +^{4}_{2}\text{He} \quad (\Gamma \sim 45\%) \\
&\to^{31}_{15}\text{P} +^{1}_{1}\text{p} \quad (\Gamma \sim 45\%) \\
&\to^{31}_{16}\text{S} +^{1}_{0}\text{n} \quad (\Gamma \sim 10\%) \\
&\to^{32}_{16}\text{S} + \gamma \quad (\text{very rare})
\end{aligned}$$

Clearly, all other combinations of heavy nuclei are also possible, such as

$$\begin{aligned}
^{12}_{6}\text{C} +^{16}_{8}\text{O} &\to^{24}_{12}\text{Mg} +^{4}_{2}\text{He} \\
^{12}_{6}\text{C} +^{16}_{8}\text{O} &\to^{27}_{13}\text{Al} +^{1}_{1}\text{p} \\
^{12}_{6}\text{C} +^{20}_{10}\text{Ne} &\to 2\,^{16}_{8}\text{O}
\end{aligned}$$

Now, the next in line after Oxygen would be Silicon. But there's a problem. The ignition temperature for Silicon burning is $T \sim 4.5 \cdot 10^9$ K, while at $T \sim 3 \cdot 10^9$ K Silicon gets photodisintegrated, as well as do other less bound nuclei, for example

$$\begin{aligned}
^{28}_{14}\text{Si} + \gamma \leftrightarrows &^{27}_{13}\text{Al} +^{1}_{1}\text{p} \\
&^{24}_{12}\text{Mg} +^{4}_{2}\text{He}
\end{aligned}$$

This allows for a redistribution of chemical elements, favoring an over-abundance of more tightly bound elements. At this temperatures are more probable α-captures up until the very most bound element (take a deep breath)

$$\begin{aligned}
&^{28}_{14}\text{Si} +^{4}_{2}\text{He} \leftrightarrows\ ^{32}_{16}\text{S} + \gamma \\
&^{32}_{16}\text{S} +^{4}_{2}\text{He} \leftrightarrows\ ^{36}_{18}\text{Ar} + \gamma \\
&^{36}_{18}\text{Ar} +^{4}_{2}\text{He} \leftrightarrows\ ^{40}_{20}\text{Ca} + \gamma \\
&^{40}_{20}\text{Ca} +^{4}_{2}\text{He} \leftrightarrows\ ^{44}_{22}\text{Ti} + \gamma \\
&^{44}_{22}\text{Ti} +^{4}_{2}\text{He} \leftrightarrows\ ^{48}_{24}\text{Cr} + \gamma \\
&^{48}_{24}\text{Cr} +^{4}_{2}\text{He} \leftrightarrows\ ^{52}_{26}\text{Fe} + \gamma \\
&^{52}_{26}\text{Fe} +^{4}_{2}\text{He} \leftrightarrows\ ^{56}_{28}\text{Ni} + \gamma \\
&^{56}_{28}\text{Ni} \to^{56}_{27}\text{Co} + e^+ + \nu_e \to^{56}_{26}\text{Fe} + 2(e^+ + \nu_e)
\end{aligned}$$

The formation of $^{62}_{28}$Ni, which would be the most tightly bound element, is greatly disfavored by the scarce number of reactions producing it. That's the reason why this crazed run for the most bound element stops only at $^{56}_{26}$Fe.

4.3.5 Netronic Captures

Up until now we've seen how elements up to Iron are produced all throughout stellar nucleosynthesis, and maybe you've began to wonder how even heavier elements like Silver, Gold and Platinum (if you're into riches) or Technetium, Uranium and Radium (if you're into radioactive stuff) are produced.

The answer to all your questions was anticlimactically revealed in the title of this section: *Neutronic captures*.

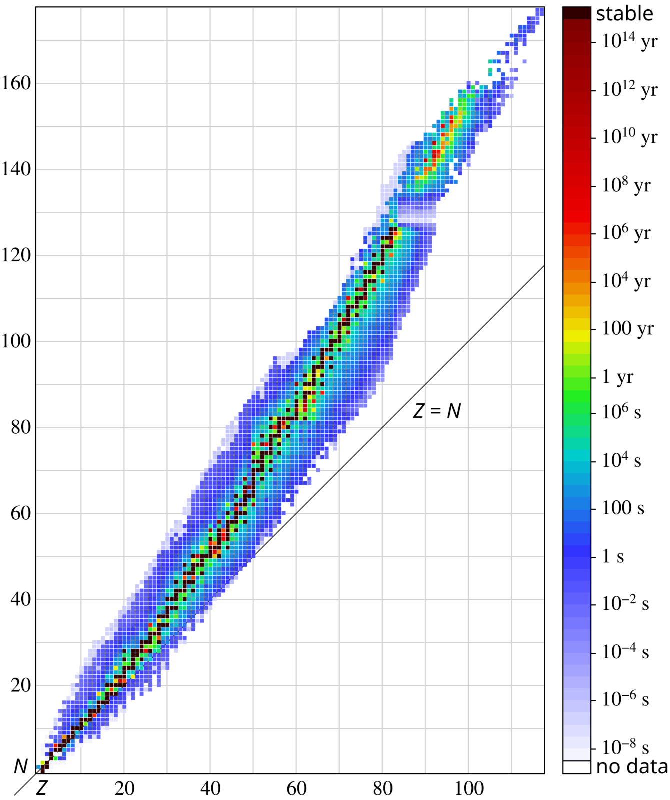

Figure 14: Isotope half-lives. The darker more stable isotope region departs from the line of protons $Z = N$ (neutrons) as the element number Z grows. Credits: Wikipedia.

As far as our understanding goes, there are two types of neutronic captures: *slow* (s) and *rapid* (r) captures. The distinction is essentially based on the comparison between the β^+-decay half-life and the average timescale for neutronic capture in the case of an unstable element being produced.

Neutronic captures are usually happening during the AGB phase, which is millions of years-long and neutrons are supplied quite steadily all throughout the process. Other sources are typically SNe, during which we have an incredible amount of free neutrons available but that is more of a short injection rather than a constant supply.

Since neutrons are, guess what, neutral, we don't have to worry about getting through the Coulombian potential as we've done earlier. If nuclei and neutrons are both thermalized (which is likely, if not certain) we can write

$$v_T = \sqrt{\frac{2kT}{\mu}} \quad \sigma \propto \frac{1}{v} \propto \frac{1}{\sqrt{E}}$$

As we've done earlier, we now have to calculate (84) but considering only the Maxwellian contribution

$$\langle \sigma(v)v \rangle = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E \exp(-E/kT) \mathrm{d}E$$

but since neutrons are neutral, the maximum reactivity is at $E_0 = kT$. Thus we can write an expression for the average interaction cross section of the form

$$\langle \sigma \rangle = \sigma(v_T) = \sigma_T$$

and

$$\langle \sigma(v)v \rangle = \text{const.} = \sigma_T v_T$$

Let us now calculate the average lifetime of a nuclide of mass A, so that we can compare it with the average timescales for neutronic captures.

The reaction we're interested in considering is

$$A(Z, N) + n \rightarrow A + 1(Z, N+1)$$

We know that we can approximate $\dot{N}_A = -N_A(t)/\tau$ and, following its definition, we also have

$$\dot{N}_A = -N_n N_A \langle \sigma(v)v \rangle_{A+n}$$

Equating the two and solving for τ

$$\tau_n = \frac{1}{N_n \langle \sigma(v)v \rangle_{A+n}} \tag{105}$$

Note that in general the cross section $\langle \sigma(v)v \rangle_{A+n}$ is not constant with respect to A because of the presence of stable nuclides, in correspondence to which there is a "depression".

For s-processes, temperatures are those of Helium burning in shells, which are in the range $1 - 6 \cdot 10^8$ K.

As a reference nucleus we'll be taking $^{134}_{56}$Ba, which has a termal velocity $v_T \sim 3 \cdot 10^8\,\mathrm{cm\,s^{-1}}$ and a cross section $\sigma_T \sim 0.1\,\mathrm{b}$ at a temperature $T \sim 4 \cdot 10^8$ K ($kT = 30$ keV). Thus

$$\sigma_T v_T \sim 3 \cdot 10^{-7}\,\mathrm{cm^3\,s^{-1}}$$

If we want to make sure that a capture is indeed a s-capture, then we have to require $\tau_n \gg \tau_\beta$. Since the average timescale for β-decays in these conditions is $\tau_\beta \sim 3 \cdot 10^8$ s, the necessary neutron density is

$$N_n < 1 \cdot 10^8\,\mathrm{cm^{-3}}$$

On the other hand, to make sure captures are of the r type, we have to require $\tau_\beta \gg \tau_n$, with an average value of $1 \cdot 10^{-3}$ s, hence

$$N_n > 10^{20}\,\mathrm{cm^{-3}}$$

5 STELLAR EVOLUTION

5.1 INTRODUCTION

6 | STELLAR CLUSTERS

7 STANDARD CANDLES AND VARIABILITY MECHANISMS

7.1 INTRODUCTION

Part I

APPENDIX

8 APPENDIX A: RADIATIVE TRANSFER

Most of our knowledge about the Universe is based on the electromagnetic radiation that reaches us from far far away. EM radiation is obviusly not the only way we can probe the Universe we live in but, in respect to neutrinos, cosmic rays or even gravitational waves, it's not a long stretch to claim it is by far the most understood.

It is most important then that an astrophysicist worthy of his (or her) name has a good grasp of the theory of radiative transfer and of its applications.

Apart from a few more key differences, I'll follow the description of radiative transfer of [5] and [14], but I won't fail to emphasize whenever I'll be doing otherwise.

Radiative transfer equation

In the presence of matter, it is not immediately obvious what changes may occur in the specific intensity as we move along a ray path. The aim of this section will be to eviscerate the matter.

Let's consider the following geometric construction

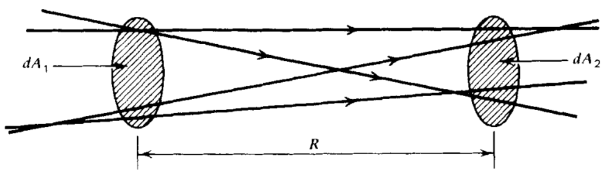

Figure 15: Geometrical construction for ray paths propagating in empty space. Credits: G. Rybicki, A. Lightman.

It won't take a lot of effort to convince yourself that in empty space the monochromatic intensity I_ν is actually conserved. In fact, from simply writing down the definitions and imposing the conservation of energy

$$I_{\nu_2}\,\mathrm{d}A_2\,\mathrm{d}t\,\mathrm{d}\Omega_2\,\mathrm{d}\nu = I_{\nu_1}\,\mathrm{d}A_1\,\mathrm{d}t\,\mathrm{d}\Omega_1\,\mathrm{d}\nu$$

the conclusion follows observing that $\mathrm{d}A_2\,\mathrm{d}\Omega_2 = \mathrm{d}A_1\,\mathrm{d}\Omega_1$.

If we consider an affine parameter of the form $\vec{x} = \vec{x}_0 + \hat{k}s$, we may as well write the previous results in a more familiar fashion

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = 0 \implies (\hat{k}\cdot\nabla)I_\nu = 0 \tag{106}$$

What changes if matter is present along the ray path? Clearly it will no longer be true that $(\hat{k}\cdot\nabla)I_\nu = 0$, but we're not that far off. All that we need is some little work on both terms.

How the right hand side of the equation should change is obvious: It needs to keep track of the "creation" and "destruction" of photons in the considered volume of spacetime.

The left hand side of the equation requires a little more care. Consider infinitesimal time and space displacements along the ray path, respectively $\mathrm{d}t$ and $\mathrm{d}\vec{x}$

$$\Delta E_\nu \,\mathrm{d}\nu = \left(I_\nu(\vec{x}+\mathrm{d}\vec{x}, t+\mathrm{d}t, \hat{k}) - I_\nu(\vec{x}, t, \hat{k})\right) \mathrm{d}t \,\mathrm{d}\Omega \,\mathrm{d}A \,\mathrm{d}\nu$$

Taking a first order expansion in respect to the affine parameter s along the ray path yields

$$\left(\frac{1}{c}\partial_t I_\nu + \partial_s I_\nu\right) \mathrm{d}t \,\mathrm{d}s \,\mathrm{d}\Omega \,\mathrm{d}A \,\mathrm{d}\nu = \text{photon addition} - \text{photon removal}$$

This equation is a generalization of eq.106 for non-stationary radiative transport and in the presence of matter. It's about time we get to know what "lives" in the right hand side of the equation.

Monochromatic emission coefficient

For the moment, we'll define the *spontaneous* monochromatic emission coefficient j_ν as

$$\mathrm{d}E_\nu \,\mathrm{d}\nu = j_\nu \,\mathrm{d}V \,\mathrm{d}t \,\mathrm{d}\Omega \,\mathrm{d}\nu \tag{107}$$

which in general has a non-zero dependence on the emission direction. Sometimes the spontaneous emission coefficient is defined by the *emissivity* ϵ_ν (**please note** that often the two names are used almost interchangeably), which is the energy emitted spontaneously per unit frequency per unit time per unit mass

$$j_\nu = \frac{\epsilon_\nu \rho}{4\pi}$$

where ρ is the mass density of the emitting medium.

If we perform the decomposition $\mathrm{d}V = \mathrm{d}A \,\mathrm{d}s$, the contribution of spontaneous emission to the specific intensity is

$$\mathrm{d}I_\nu = j_\nu \,\mathrm{d}s$$

Absorption coefficient

Similarly, we can consider the energy that is absorbed from the radiation when passing through a medium. There exists various definitions; I'll use the one we gave in class and that is incidentally the one used in [5] and [14] as well.

We define the *absorption coefficient* α_ν through the following expression

$$\mathrm{d}I_\nu = -\alpha_\nu I_\nu \,\mathrm{d}s \tag{108}$$

If we use a microscopic model, then the absorption coefficient can be understood as particles with numeric density n presenting an effective absorbing area, the *cross section* σ_ν. The coefficient α_ν can thus be rewritten in terms of

$$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$$

where κ_ν is called the mass absorption coefficient or the *mass-weighed opacity coefficient*.

Note that in eq.108, we consider "absorption" to include both "true absorption" and stimulated emission, because both are proportional to the intensity of the incoming beam. Depending on the entity of the contribution, the α_ν coefficient may be positive or even negative, giving raise to curious phenomena.

Making full use of what we've just defined, we can finally present the celebrated *equation of radiative transfer* (although in the notable absence of scattering)

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu \tag{109}$$

which is actually fairly easy to solve when one of the two coefficients vanishes.

Emission only

We set $\alpha_\nu = 0$ and the equation may be solved by direct integration

$$I_\nu(s) = I_\nu(s_0) + \int_{s_0}^{s} j_\nu(s')\, ds'$$

the result is not that interesting per se.

Absorption only

This time we set $j_\nu = 0$. The equation is easily solved this time as well

$$I_\nu(s) = I_\nu(s_0) \exp\left(-\int_{s_0}^{s} \alpha_\nu(s')\, ds'\right)$$

In this case, it's rather common to write down the equation in terms of a new variable, the *optical depth* τ_ν

$$d\tau_\nu = \alpha_\nu\, ds \tag{110}$$

Given this definition we'll say that if

- $\tau_\nu \gg 1$: the medium is *optically thick or opaque*
- $\tau_\nu \ll 1$: the medium is *optically thin or transparent*

This has some crucial implications we'll be going through in a moment.

In the stationary limit, the equation of radiative transport may be written as

$$(\hat{k} \cdot \nabla) I_\nu(\hat{k}, \vec{x}) = j_\nu(\vec{x}) - \alpha_\nu(\vec{x}) I_\nu(\hat{k}, \vec{x})$$

In terms of the *source function* $S_\nu = j_\nu / \alpha_\nu$ it becomes

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu \tag{111}$$

which can be integrated to yield the formal solution

$$I_\nu(\hat{k}, \tau_\nu) = I_\nu(\tau_{\nu,0}) \exp(-\tau_\nu) + \int_{\tau_{\nu,0}}^{\tau_\nu} d\tau'_\nu\, S_\nu \exp(-(\tau_\nu - \tau'_\nu))$$

Assume for the moment that the matter through which radiation is passing has constant properties and has no background source. Then the source function S_ν is constant and the formal equation becomes

$$I_\nu = S_\nu(1 - e^{-\tau_\nu})$$

If the medium and is optically thin, then the equation is reduced to

$$I_\nu = S_\nu \tau_\nu = j_\nu L \tag{112}$$

by taking the Taylor expansion of the exponential term and calling L some typical length of the medium.

If, on the other hand, the medium is optically thick, we can neglect the exponential $e^{-\tau_\nu}$ to obtain

$$I_\nu = S_\nu \tag{113}$$

kirchhoff's law and lte

The most notable implication of eq.113 is if we consider the specific intensity coming out of a small hole on a box kept in thermodynamic equilibrium. We know that what's going to come out of there is the blackbody radiation

$$I_\nu = B_\nu(T)$$

but what if we were to put an optically thick object just behind the hole?

If the object is in thermodynamic equilibrium with the surroundings (and it *will* be, given an appropriate amount of time), then the radiation coming out of the hole will still be blackbody radiation. But eq.113 tells us that the source function will tend to be equal to the specific intensity, hence

$$S_\nu = B_\nu(T) \tag{114}$$

which actually puts a constraint on the possible values of the emission coefficient in terms of the absorption coefficient. This is exactly what is expressed in Kirchhoff's law

$$j_\nu = \alpha_\nu B_\nu \tag{115}$$

Let us briefly consider what we have just derived. Matter often tends to emit and absorb at specific frequencies corresponding to what are commonly called *spectral lines*. We would expect then both j_ν and α_ν to have peaks (or depression) around these lines. But Kirchhoff's law forces their ratio to be equal to a smooth blackbody profile.

Thus we can expect to observe two very different scenarios if the medium is optically thin rather than optically thick. In the former, the radiation emerging from the medium is essentially determined by its emission coefficient; since j_ν is expected to present peaks, so will the radiation spectrum, which will be appear in spectral lines, as shown in Fig.16 and Fig.17.

On the other hand, the intensity coming out of an optically thick body is its source function, which must be equal to the blackbody function. Hence we expect the medium to emit in a continuum, like a blackbody.

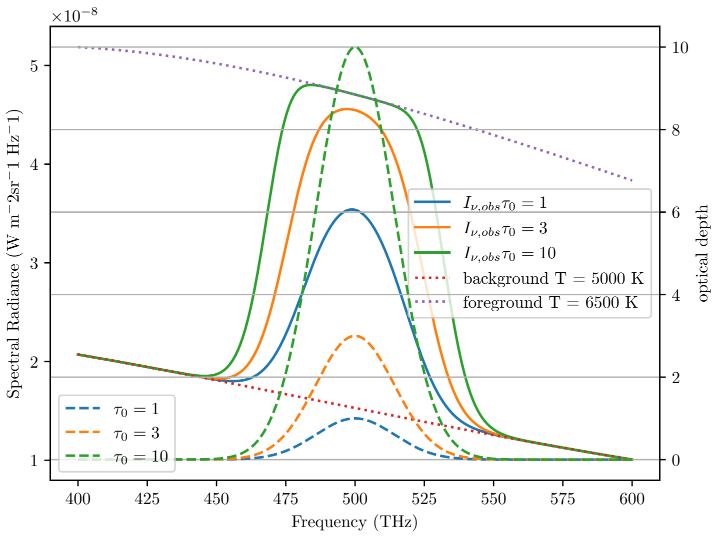

Figure 16: An example of emission features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

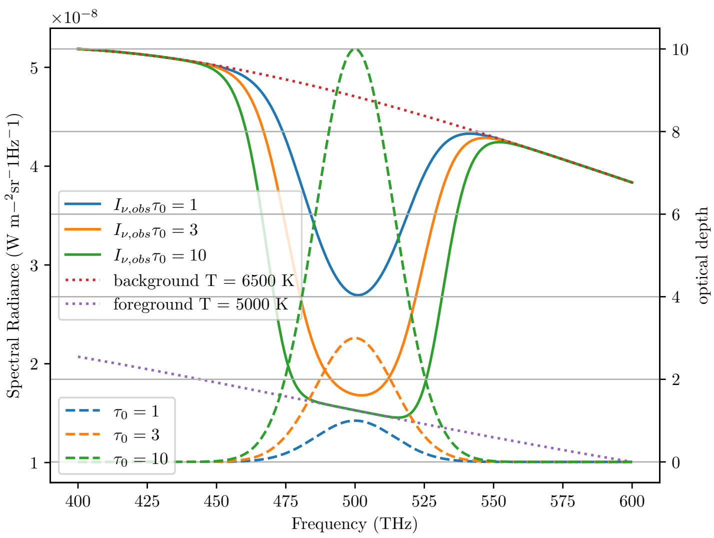

Figure 17: An example of absorption features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

All throughout this description, we've been assuming the medium to have constant properties, which has the perk of being a good approximation for many objects of interest, but still turns out to be a really poor one for many other objects. Stars, for example.

Ingenuously, we may expect stars to emit radiation like blackbodies, but they're not. Actually, stars present absorption lines, many, even, depending on the class of star. What we cannot assume in stars is them having constant properties, starting from temperature.

In fact, we could take a guess and claim that stars are in *strict* thermodynamic equilibrium. It would be a very bad guess indeed.

Local Thermodynamic Equilibrium (LTE)

Let's be honest: In a realistic situation, we *rarely* have strict thermodynamic equilibrium. If a body is in thermodynamic equilibrium, we can assume a number of important physical principles to hold, like the Maxwellian distribution

$$dn_v = 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right) dv \tag{116}$$

where n is the total number of particles per unit volume and m is the mass of each particle. Similarly, we can expect certain laws to hold, like Boltzmann's law for occupation numbers

$$\frac{n_E}{n_0} = \frac{g_E}{g_0} \exp\left(-\frac{E - E_0}{kT}\right) \tag{117}$$

and Saha's equation

$$\frac{N_{j+1} n_e}{N_j} = 2\frac{Z_{j+1}(T)}{Z_j(T)} \left(\frac{2\pi mkT}{h^2}\right)^{3/2} \exp\left(-\frac{\chi_{j,j+1}}{kT}\right) \tag{118}$$

where n_e is the density of electrons and $\chi_{j,j+1}$ is the ionization potential. Saha's equation in particular is expected to be crucial in interpreting the effect that ionization has on the emission/absorption spectrum.

The proverbial "one-million-dollar-question" then is: When can we expect a system to be in thermodynamic equilibrium and when can we expect the previous principles to hold?

Even if the system initially does not obey the, say, Maxwellian distribution, it will eventually relax to it after undergoing some *collisions*.

Collisions are crucial in establishing thermodynamic equilibrium.

When collisions are frequent, the mean free path of particles will be small, and particles will interact more effectively. When this happens, we can expect the principles aforementioned to hold. Since we're physicists, vague sentences like "*the mean free path of particles will be small*" are destined to elicit a deep sense of unease and distress. How small does the free path have to be? One meter? Two micrometers? Below the Planck lengthscale?

When we've defined the absorption coefficient α_ν, the sharpest among my four readers total may have noticed that α_ν has the dimension of the inverse of a length. It is safe to assume that α_ν^{-1} may define some distance over which a significant fraction of the radiation would get absorbed by matter.

Such a "mean-distance" is defined in a homogeneous medium as

$$< \tau_\nu >= \alpha_\nu l_\nu = 1$$

Thus, if l_ν is sufficiently small such that the temperature can be taken as a constant over such distance, we can safely say that the useful relations we have defined earlier still hold, although only locally.

In such a fortunate scenario, known as *Local Thermodynamic Equilibrium* (LTE), all the important laws requiring thermodynamic equilibrium are expected to hold, provided that we use the local temperature $T(\vec{x})$.

In the interiors of stars, for example, LTE will prove to be a very good approximation, that will get progressively worse as we approach the "surface" of the star.

One useful approximation that may be worthwhile to dedicate some of our time to is the *plane parallel atmosphere*, that will allow us to obtain notable results for describing how radiation travels through, say, the inner regions of the stellar atmosphere.

In the following, we're going to neglect the curvature of the stellar atmosphere and assume the various thermodynamic quantities to be constant over horizontal planes.

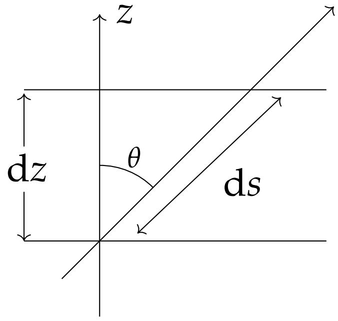

Figure 18: A ray path through a plane parallel atmosphere.

Using Fig.18 as a reference, we see that

$$\mathrm{d}s = \frac{\mathrm{d}z}{\cos\theta} = \frac{\mathrm{d}z}{\mu}$$

where we used the customary notation in astrophysics $\mu = \cos\theta$.

We shall consider a scattering free, stationary situation for the equation of radiative transport (109). Hence, due to planar symmetry, we expect the specific intensity to depend only on z and μ. For the sake of the current discussion, we perform a slight modification to the definition of optical depth, so that

$$\mathrm{d}\tau_\nu = -\alpha_\nu \,\mathrm{d}z$$

This way the equation of radiative transfer may be cast in the following form

$$\mu\partial_{\tau_\nu} I_\nu(\tau_\nu, \mu) = I_\nu - S_\nu$$

which has a formal solution easily computed

$$I_\nu \exp\left(-\frac{t_\nu}{\mu}\right)\Big|_{\tau_{\nu,0}}^{\tau_\nu} = -\int_{\tau_{\nu,0}}^{\tau_\nu} \frac{S_\nu}{\mu} \exp\left(-\frac{t_\nu}{\mu}\right) \mathrm{d}t_\nu \tag{119}$$

This is customarily solved considering two distinct intervals for μ: (I) $\mu \in [0,1]$ and (II) $\mu \in [-1,0]$. In case (I) we can assume the ray path to begin from a great depth inside the star, so that $\tau_{\nu,0} \to \infty$, while in case (II) we assume the ray to receive contributions beginning from the top of the atmosphere, where $\tau_{\nu,0} \approx 0$. For case (II), we're also assuming no radiation to be coming from *outside the star*[1].

Now we can assume LTE throughout the stellar atmosphere so that eq.115 is verified. The source function at some optical depth shall then be equal to

1 Please note that this condition may be not valid at all in close binary systems.

$B_\nu(T(\tau_\nu))$. For the source function at a nearby optical depth we can simply compute a Taylor expansion around the optical depth τ_ν

$$S(t_\nu) = B_\nu(\tau_\nu) - (t_\nu - \tau_\nu)\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu} + \mathrm{o}(t_\nu^2)$$

We can use this to solve (119), finding for both positive and negative values of μ a very important equation

$$I_\nu(\tau_\nu, \mu) = B_\nu(\tau_\nu) + \mu\,\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu} \tag{120}$$

provided the point considered is sufficiently inside the atmosphere so that $\tau_\nu \gg 1$[2]. Using this simple result, we can compute the three momenta of the equation of transport

$$U_\nu = \frac{4\pi}{c} B_\nu(\tau_\nu) \tag{121}$$

$$F_\nu = \frac{4\pi}{3}\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu} \tag{122}$$

$$P_\nu = \frac{4\pi}{3c} B_\nu(\tau_\nu) \tag{123}$$

The Grey Atmosphere

If we consider the absorption coefficient α_ν constant over all frequencies, then the atmosphere is called a "grey atmosphere". This implies that the value of the optical depth at some physical depth is constant for all frequencies. Under this assumption, we could solve

$$\mu\frac{\partial I}{\partial \tau} = I - S$$

I'll skip the explicit calculation (which is actually fairly easy for once) and present just the final result. Two more assumptions are to be made, however: The first is to assume *radiative equilibrium*, which roughly translates into requiring that there are no sources nor sinks of energy in the atmosphere, thus $\partial_\tau F = 0$; as a further semplification, we assume the *Eddington approximation* to hold everywhere in the atmosphere[3], so that

$$P = \frac{1}{3}U$$

It should be evident that this last equation is verified automatically in presence of an isotropic source of radiation, also in its frequency-dependent form.

We then come to the following conclusion

$$I_{obs}(\tau = 0, \mu) = \frac{3F}{4\pi}\left(\mu + \frac{2}{3}\right) \tag{124}$$

2 You can see [5], §2.4.1 for more detailed calculations.

3 I find most intriguing that Eddington's approximation essentially assumes that $T^\mu{}_\mu = 0$, with $T^{\mu\nu}$ the electromagnetic energy-momentum tensor if we are to treat electromagnetic radiation as a perfect fluid.

from which we deduce the equation for the *limb darkening*

$$\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\left(\mu + \frac{2}{3}\right)$$

which roughly translates into saying that the radiation that we observe at the surface is the one equivalent at a source function S evaluated at $\tau = 2/3$. This is known as the *Eddington-Barbier estimation*.

A somewhat more general way to solve the problem is assuming the following functional relation for the specific intensity

$$I_\nu(\tau,\mu) = a_\nu(\tau_\nu) + b_\nu(\tau_\nu)\mu$$

and compute the three momenta proper

$$J_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \,\mathrm{d}\mu = a_\nu \tag{125}$$

$$H_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \mu \,\mathrm{d}\mu = \frac{b_\nu}{3} \tag{126}$$

$$K_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \mu^2 \,\mathrm{d}\mu = \frac{a_\nu}{3} \tag{127}$$

Now we assume a stronger version of the *Eddington approximation* $K_\nu = J_\nu/3$ so that the two following expressions can be written

$$\frac{\partial H_\nu}{\partial \tau_\nu} = J_\nu - S_\nu \tag{128}$$

$$\frac{\partial K_\nu}{\partial \tau_\nu} = H_\nu = \frac{1}{3}\frac{\partial J_\nu}{\partial \tau_\nu} \tag{129}$$

The mixing of the two gives us a second order PDE that it's still (approximately) valid even in the outer regions of the stellar atmosphere.

RADIATIVE DIFFUSION APPROXIMATION

This section would probably be clearer if you take a brief detour to Chapter 3 to build some groundwork for scattering processes, but it still suits better the main subject of this chapter.

For the sake of coherence (and personal laziness) I'm going to talk about the radiative diffusion approximation right here, also because it makes use of the plane parallel approximation we just went through.

In Chapter 3 we will use random walk arguments to show that S_ν approaches B_ν at large *effective optical depths* in a homogeneous medium. Real media are seldom homogeneous, but often, as in the interiors of stars, there is a high degree of local homogeneity.

The equation of radiative transport in presence of scatteringz may be cast in a slightly different form

$$I_\nu = S_\nu - \frac{\mu}{\alpha_\nu + \sigma_\nu}\partial_z I_\nu$$

We shall then assume that over a distance l_* (the thermalization length) I_ν is constant and at zero-th order is $I_\nu^{(0)} = S_\nu^{(0)} = B_\nu$. Plugging this in the equation of radiative transfer gives us $I_\nu^{(1)}$ by a simple iterative procedure

$$I_\nu^{(1)} = B_\nu - \frac{\mu}{\alpha_\nu + \sigma_\nu}\partial_z B_\nu \tag{130}$$

With the simple redefining $\mathrm{d}\tau_\nu = -(\alpha_\nu + \sigma_\nu)\,\mathrm{d}z$, we can put eq.(130) in a form functionally equal to (120).

Let us now compute the flux F_ν using the above form for the intensity

$$F_\nu(z) = 2\pi \int_{-1}^{+1} I_\nu^{(1)} \mu \,\mathrm{d}\mu = -\frac{4\pi}{3}\frac{\partial_z B_\nu}{\alpha_\nu + \sigma_\nu} = -\frac{4\pi}{3}\frac{\partial_T B_\nu}{\alpha_\nu + \sigma_\nu}\partial_z T$$

Recalling the result

$$\partial_T \int_0^{+\infty} B_\nu \,\mathrm{d}\nu = \frac{4\sigma_{SB} T^3}{\pi}$$

we can define a *mean absorption coefficient* using the *Rosseland approximation for radiative diffusion*

$$\frac{1}{\alpha_R} := \frac{\int_0^{+\infty} \frac{1}{\alpha_\nu + \sigma_\nu}\,\partial_T B_\nu \,\mathrm{d}\nu}{\int_0^{+\infty} \partial_T B_\nu \,\mathrm{d}\nu} \tag{131}$$

If we integrate the monochromatic flux over the frequencies and make use of the Rosseland mean, we find a useful expression used in stellar structure models

$$F(z) = -\frac{16\sigma_{SB} T^3}{3\alpha_R}\partial_z T \tag{132}$$

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Batchelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

Figure 19: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my collegues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

BIBLIOGRAPHY

[1] M. Asplund, A. M. Amarsi, and N. Grevesse. The chemical make-up of the Sun: A 2020 vision. *Astronomy and Astrophysics*, 653:A141, September 2021.

[2] Erik Aver, Evan D. Skillman, Richard W. Pogge, Noah S. J. Rogers, Miqaela K. Weller, Keith A. Olive, Danielle A. Berg, John J. Salzer, John H. Miller Jr., and José Eduardo Méndez-Delgado. The lbt y_p project iv: A new value of the primordial helium abundance, 2026.

[3] John N. Bahcall, Aldo M. Serenelli, and Sarbani Basu. New Solar Opacities, Abundances, Helioseismology, and Neutrino Fluxes. *Astrophysical Journal Letters*, 621(1):L85–L88, March 2005.

[4] Kent G. Budge. An Improved Henyey Method for Calculating Stellar Models. *Astrophysical Journal*, 312:217, January 1987.

[5] A. R. Choudhuri. *Astrophysics for Physicists*. Astrophysics-Textbooks. Cambridge University Press, 2010.

[6] J. P. Cox and R. T. Giuli. *Principles of stellar structure*. 1968.

[7] G. GAMOW. The Quantum Theory of Nuclear Disintegration. *Nature*, 122:805–806, November 1928.

[8] L. G. Henyey, J. E. Forbes, and N. L. Gould. A New Method of Automatic Computation of Stellar Evolution. *Astrophysical Journal*, 139:306, January 1964.

[9] F. Hoyle. On Nuclear Reactions Occuring in Very Hot STARS.I. the Synthesis of Elements from Carbon to Nickel. *Astrophysical Journal Supplement*, 1:121, September 1954.

[10] K. S. Krishna Swamy. Convection in Normal Stars and Metal-Deficient Stars. *Astrophysics and Space Science*, 3(4):552–563, April 1969.

[11] Pavel Kroupa, Carsten Weidner, Jan Pflamm-Altenburg, Ingo Thies, Jörg Dabringhausen, Michael Marks, and Thomas Maschberger. The Stellar and Sub-Stellar Initial Mass Function of Simple and Composite Populations. In Terry D. Oswalt and Gerard Gilmore, editors, *Planets, Stars and Stellar Systems. Volume 5: Galactic Structure and Stellar Populations*, volume 5, page 115. 2013.

[12] Ekaterina Magg, Maria Bergemann, Aldo Serenelli, Manuel Bautista, Bertrand Plez, Ulrike Heiter, Jeffrey M. Gerber, Hans-Günter Ludwig, Sarbani Basu, Jason W. Ferguson, Helena Carvajal Gallego, Sébastien Gamrath, Patrick Palmeri, and Pascal Quinet. Observational constraints on the origin of the elements. IV. Standard composition of the Sun. *Astronomy and Astrophysics*, 661:A140, May 2022.

[13] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.

[14] George B. Rybicki and Alan P. Lightman. *Radiative Processes in Astrophysics*. 1986.

[15] Maurizio Salaris and Santi Cassisi. *Evolution of Stars and Stellar Populations*. Wiley, 2005.

[16] Edwin E. Salpeter. The Luminosity Function and Stellar Evolution. *Astrophysical Journal*, 121:161, January 1955.